

# MATHEMATICS GRADE 10: TRIGONOMETRY I

*CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.1 (10 Lessons)*

| SUB-STRAND OVERVIEW |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| Grade Level         | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Subject             | Mathematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Strand              | Strand 3.0: Geometry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Sub-Strand          | Sub-Strand 3.1: Trigonometry I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Total Duration      | 10 lessons × 40 minutes = 400 minutes total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Sub-Strand Content  | <ul style="list-style-type: none"> <li>– Trigonometric ratios of acute angles (sine, cosine, tangent defined from right-angled triangles)</li> <li>– Sine of an acute angle — definition, values, and use of mathematical tables</li> <li>– Cosine of an acute angle — definition, values, and use of mathematical tables</li> <li>– Tangent of an acute angle — definition, values, and use of mathematical tables</li> <li>– Finding trigonometric ratios using calculators</li> <li>– Relationship between sine, cosine and tangent of acute angles (<math>\tan \theta = \sin \theta / \cos \theta</math>)</li> <li>– Trigonometric ratios of complementary angles (<math>\sin \theta = \cos(90^\circ - \theta)</math>)</li> <li>– Special angles (<math>0^\circ</math>, <math>30^\circ</math>, <math>45^\circ</math>, <math>60^\circ</math>, <math>90^\circ</math>) — exact values without tables</li> <li>– Angles of elevation and depression</li> <li>– Application of trigonometry in real-life situations (heights, distances, surveying)</li> </ul>                                                                                                                                                                                               |
| Learning Outcomes   | <p>By the end of the sub-strand, the learner should be able to:</p> <ol style="list-style-type: none"> <li>define sine, cosine and tangent of an acute angle using measurements from right-angled triangles drawn by hand,</li> <li>determine trigonometric ratios of acute angles using mathematical tables,</li> <li>determine trigonometric ratios of acute angles using a calculator,</li> <li>establish the relationship between sine, cosine and tangent of acute angles (<math>\tan \theta = \sin \theta / \cos \theta</math>),</li> <li>identify and apply the relationship between trigonometric ratios of complementary angles (<math>\sin \theta = \cos(90^\circ - \theta)</math>),</li> <li>derive and recall exact trigonometric ratios for special angles (<math>0^\circ</math>, <math>30^\circ</math>, <math>45^\circ</math>, <math>60^\circ</math>, <math>90^\circ</math>),</li> <li>distinguish between angles of elevation and angles of depression in diagrams and real contexts,</li> <li>apply trigonometric ratios to solve problems involving heights and distances in real-life Kenyan contexts,</li> <li>appreciate the relevance of trigonometry in architecture, surveying, navigation, and everyday problem-solving.</li> </ol> |
| Core Competencies   | <ul style="list-style-type: none"> <li>– Communication and Collaboration: learners work in groups to draw right-angled triangles of chosen measurements, measure angles with a protractor, share computed ratios, and discuss patterns observed across different triangles.</li> <li>– Critical Thinking and Problem Solving: learners analyse why opposite/hypotenuse always gives the same ratio for a fixed angle regardless of triangle size, and use this insight to solve height-and-distance problems set in Kenyan contexts (e.g., estimating the height of a flagpole at a Nairobi school).</li> <li>– Digital Literacy: learners use scientific calculators and, where available, digital devices to determine trigonometric ratios and</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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|                                                   | <p>verify table values.</p> <ul style="list-style-type: none"> <li>– Learning to Learn: learners derive exact values for special angles independently using equilateral and isosceles right-angled triangles, building self-directed inquiry habits.</li> <li>– Self-efficacy: learners gain confidence in applying abstract angle ratios to concrete, visible real-world situations such as measuring the height of a building or a hill in the Rift Valley.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Core Values</b>                                | <ul style="list-style-type: none"> <li>– Responsibility: learners handle protractors, rulers, calculators, and mathematical tables carefully during hands-on triangle-drawing activities, returning equipment in good condition.</li> <li>– Integrity: learners record actual measured values honestly, even when results differ slightly from expected ratios, and reflect on sources of measurement error.</li> <li>– Unity: learners collaborate in diverse groups during the triangle-drawing phenomenon activity and the special-angles derivation task, valuing each member's contribution.</li> <li>– Respect: learners listen to peers' explanations during sensemaking discussions and appreciate different approaches to solving trigonometric problems.</li> <li>– Social Justice: learners recognise how trigonometry empowers communities — e.g., local surveyors and engineers in Kisumu or Mombasa use these tools — making mathematics relevant beyond the classroom.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Science &amp; Engineering Practices</b>        | <ul style="list-style-type: none"> <li>– Planning and Carrying Out Investigations: learners design and conduct a hands-on investigation by drawing right-angled triangles of self-chosen dimensions, measuring angles with a protractor, computing ratios of sides, and recording results systematically in a table.</li> <li>– Analysing and Interpreting Data: learners compare sine, cosine, and tangent values obtained from their own triangles with table/calculator values, identifying patterns (e.g., sin and cos of complementary angles are equal) and resolving discrepancies.</li> <li>– Using Mathematics and Computational Thinking: learners use ratio, proportion, and algebraic manipulation (<math>\tan \theta = \sin \theta / \cos \theta</math>) to connect the three trigonometric functions, and apply these to calculate unknown heights and distances.</li> <li>– Constructing Explanations: learners articulate in writing and speech why the ratio of sides remains constant for a given angle regardless of triangle size, grounding their explanation in the concept of similar triangles.</li> <li>– Obtaining, Evaluating, and Communicating Information: learners read articles and watch videos on complementary angles, special angles, and angles of elevation/depression, then synthesise information into annotated diagrams and model sketches.</li> </ul> |
| <b>Pertinent &amp; Contemporary Issues (PCIs)</b> | <ul style="list-style-type: none"> <li>– Life Skills (Critical Thinking and Problem Solving): learners apply step-by-step logical reasoning to set up and solve angle-of-elevation and angle-of-depression problems, building real-world decision-making skills.</li> <li>– Life Skills (Self-Esteem): learners gain confidence as they successfully derive exact special-angle values using geometric reasoning rather than memorisation, reinforcing belief in their own mathematical ability.</li> <li>– Career Guidance: learners explore careers — surveying, architecture, engineering, aviation, and game-park tourism (e.g., tracking animal positions using bearings at Maasai Mara) — where trigonometry is directly applied.</li> <li>– Environmental Education: learners consider how trigonometry supports responsible land surveying and forest-boundary mapping in Kenya, linking mathematics to sustainable land use in areas such as the Aberdare Range.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Career Connections</b>                         | <ul style="list-style-type: none"> <li>– Land Surveyor: uses trigonometric ratios daily to measure distances and heights across Kenyan terrain — from Rift Valley escarpments to coastal plains near Mombasa.</li> <li>– Civil/Structural Engineer: applies sine, cosine, and tangent when designing bridges, road gradients, and building foundations for infrastructure projects such as those along the SGR corridor.</li> <li>– Architect: uses angles of elevation and trigonometric ratios to calculate roof pitches, ramp gradients, and shadow lengths for building designs in Nairobi and other urban centres.</li> <li>– Aviation Pilot / Air Traffic Controller: uses angles of depression and elevation for approach paths and altitude calculations at airports including JKIA and Kisumu International Airport.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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|                                                 | <ul style="list-style-type: none"> <li>– Geologist / Mining Engineer: applies trigonometry when mapping rock strata angles and calculating depths at mining sites in Taita Hills or the Lake Victoria basin.</li> <li>– Game-Park Tourism Guide / Wildlife Researcher: uses angular measurements and distance estimation — grounded in trigonometric principles — when tracking and documenting animal movements in Amboseli or Tsavo.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Focus for Lessons</b>                        | <p>This unit introduces Grade 10 learners to the foundational concepts of trigonometry through a hands-on, phenomenon-driven approach: learners begin by physically drawing right-angled triangles of their own chosen dimensions, measuring angles with a protractor, and computing the ratios of sides — discovering for themselves that these ratios depend only on the angle, not the size of the triangle. The unit progresses from defining sine, cosine, and tangent, through the use of mathematical tables and calculators, to elegant patterns in complementary and special angles, culminating in the powerful real-life applications of angles of elevation and depression in Kenyan contexts such as estimating the height of a water tower or a hill in the Rift Valley. Throughout, the storyline returns repeatedly to the anchoring phenomenon, with each new concept deepening students' ability to explain and model what they first observed.</p>                                                                                                                                                                                                                                                                                                                                  |
| <b>Driving Question / Key Inquiry Questions</b> | <p>DRIVING QUESTION: How can the angle a sunbeam makes with the ground allow us to calculate the height of any tree, building, or hill in Kenya without measuring it directly?</p> <p>KICD KEY INQUIRY QUESTIONS:</p> <ol style="list-style-type: none"> <li>1. How do we apply trigonometric ratios in day-to-day activities?</li> <li>2. How are the trigonometric ratios of acute angles related to each other?</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Anchoring Phenomenon</b>                     | <p>ANCHORING PHENOMENON — "Why does my shadow tell me how tall the tree is?"</p> <p>A student standing in the school compound in Kisumu on a sunny afternoon notices that her 1.6 m shadow is exactly 1.2 m long, while the shadow of a nearby mango tree is 9.0 m long. She wonders: can she figure out the height of the tree without climbing it? A classmate suggests they also measure the angle the sun's rays make with the ground using a simple clinometer made from a protractor and a straw. When they do, they get an angle of approximately <math>53^\circ</math>. The class is struck by the puzzle: how can an angle — something you measure with a protractor on paper — tell you the height of a real tree? This phenomenon is puzzling because it implies a hidden, consistent relationship between angles and the ratios of lengths in right-angled triangles, a relationship that seems to hold no matter how large or small the triangle. Students are asked to predict how this could be possible before any instruction begins, activating prior knowledge of ratio and similar triangles and generating genuine curiosity about trigonometric ratios.</p>                                                                                                                      |
| <b>Storyline Thread</b>                         | <p>Lesson 1 — Predict &amp; Anchor Phenomenon: Students observe the mango-tree shadow scenario (anchoring phenomenon). In groups they predict how an angle could reveal a height, record predictions on sticky notes, and post them on the Driving Question Board (DQB). Teacher introduces the activity: each group draws three right-angled triangles of different sizes but with the same chosen acute angle, measures sides, and computes opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent ratios — discovering these are constant. Sine, cosine, and tangent are formally defined.</p> <p>Lesson 2 — Sine of an Acute Angle: Learners watch a short video on the sine of an acute angle. They use mathematical tables to read sine values for various angles and verify against their Lesson 1 triangle measurements. Teacher conducts cold-call questioning (minimum 6 students) on how to locate values in tables. Students annotate their DQB: "Sine depends only on the angle, not the triangle's size."</p> <p>Lesson 3 — Cosine of an Acute Angle: Guided reading on cosine of an acute angle. Learners read cosine values from tables, compare with sine values, and begin noticing a complementary pattern. Pairs solve five structured problems finding unknown sides</p> |

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|                      | <p>of right-angled triangles in Kenyan contexts (e.g., a ramp at a Nakuru hospital). DQB updated with new question: "Is there a link between sine and cosine?"</p> <p>Lesson 4 — Tangent and the Sine–Cosine–Tangent Relationship: Learners derive <math>\tan \theta = \sin \theta / \cos \theta</math> algebraically and verify numerically using table values for <math>30^\circ</math>, <math>40^\circ</math>, <math>60^\circ</math>, <math>70^\circ</math>. Teacher facilitates a whole-class sensemaking discussion. Students revisit the mango-tree phenomenon and use tangent to calculate the tree's height from the <math>53^\circ</math> sun angle and the 9.0 m shadow. DQB updated: the driving question is partially answered for the first time.</p> <p>Lesson 5 — Trigonometric Ratios Using Calculators: Learners use scientific calculators to find sine, cosine, and tangent values, and to find angles given a ratio (inverse functions). Activity: finding trigonometric ratios of acute angles using calculators with a structured worksheet. Students check their Lesson 1 hand-drawn triangle results against calculator values and compute percentage error, linking to integrity value. DQB updated with precision note.</p> <p>Lesson 6 — Complementary Angles: Learners read an article and watch a short video on complementary angles. They discover <math>\sin \theta = \cos(90^\circ - \theta)</math> through a table-comparison task, then prove it geometrically using a right-angled triangle diagram. Cold-call challenge: teacher asks 8 students to state a complementary-angle pair and write the corresponding identity. Supporting phenomenon (tile-cutter's <math>45^\circ</math>) is introduced to motivate the next lesson.</p> <p>Lesson 7 — Special Angles (Derivation): Learners derive exact values for <math>0^\circ</math>, <math>30^\circ</math>, <math>45^\circ</math>, <math>60^\circ</math>, <math>90^\circ</math> using two geometric constructions: an equilateral triangle (side 2) for <math>30^\circ/60^\circ</math> and an isosceles right-angled triangle (legs of 1) for <math>45^\circ</math>. Video and reading resources on special angles are used to consolidate. Each group creates a "special angles passport card" — a pocket-sized reference they design themselves. DQB updated with exact-value table.</p> <p>Lesson 8 — Special Angles (Application &amp; Model Building): Learners use exact special-angle values (no calculator) to solve problems: e.g., a builder in Eldoret needs the length of a roof rafter for a <math>60^\circ</math> pitched roof of 5 m span. Reading and article on special angles applied to real structures. Groups revise their cumulative model (annotated right-angled triangle diagram) to include the special-angle values and the <math>\tan = \sin/\cos</math> relationship.</p> <p>Lesson 9 — Angles of Elevation and Depression: Learners watch a video and complete a reading on angles of elevation and depression. Using the airport glide-path supporting phenomenon, students calculate horizontal distances and heights. Outdoor mini-activity (if space allows): students use a clinometer (protractor + straw + plumb-line) to measure the angle of elevation to the top of the school's flagpole or water tank and calculate its height using a measured baseline distance. Results are compared across groups and discrepancies discussed.</p> <p>Lesson 10 — Real-Life Applications, DQB Closure &amp; Final Model Revision: Learners tackle an integrated problem set drawn from Kenyan contexts — surveying a plot boundary in Kisumu, calculating the height of a telecommunications mast in Kericho, and verifying the glide path at Wilson Airport. Groups present solutions. Full DQB review: every original prediction and question is revisited and resolved. Each learner submits a final annotated model diagram showing all unit concepts connected back to the anchoring mango-tree phenomenon, demonstrating how trigonometric ratios of acute angles answer the driving question.</p> |
| Supporting Phenomena | <p>– SUPPORTING PHENOMENON 1 — "The leaning phone mast": Learners watch a short video of a telecommunications mast on a hillside near Kericho that leans at an angle. Engineers on site use only a tape measure and an angle-reading instrument to decide whether the mast is safe. Students are asked: what mathematical relationship allows two measurements to reveal a structural risk? This supports the introduction of sine and cosine of acute angles.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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|  | <ul style="list-style-type: none"> <li>– SUPPORTING PHENOMENON 2 — "Reading the mountain": Learners examine a photograph taken from the Nairobi–Nakuru highway showing Mount Longonot. A road sign gives the horizontal distance to the crater rim as 4.2 km. Students measure the apparent angle of elevation from the photograph using a printed protractor overlay and calculate the crater height — connecting angles of elevation to real geography.</li> <li>– SUPPORTING PHENOMENON 3 — "The tile cutter's 45°": A short article describes how a tile-fitter in Mombasa cutting square tiles diagonally always uses a 45° guide. Learners investigate why <math>\sin 45^\circ = \cos 45^\circ</math> and why the diagonal of a unit square is <math>\sqrt{2}</math>, naturally motivating the special-angles lesson.</li> <li>– SUPPORTING PHENOMENON 4 — "Airport glide path": Learners read a brief excerpt explaining that aircraft approaching Wilson Airport in Nairobi follow a standard 3° glide-path angle of depression. They calculate how far from the runway threshold a plane is when it is 300 m above ground, making angles of depression concrete and consequential.</li> </ul> |
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### **ATTRIBUTION AND LICENSING**

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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## LESSON 1 (40 minutes): Predict and Anchor: Why Does My Shadow Tell Me How Tall the Tree Is?

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>Purpose</b>              | To launch the sub-strand by anchoring learning in a real Kenyan phenomenon and generating genuine curiosity about the hidden relationship between angles and the ratios of sides in right-angled triangles.                                                                                                                                                                                                                                  |
| <b>Knowledge</b>            | Learners will recall properties of right-angled triangles and the concept of ratio from prior learning, and will be introduced to the idea that a fixed acute angle produces a constant ratio of sides regardless of the size of the triangle.                                                                                                                                                                                               |
| <b>Skills</b>               | Learners will draw right-angled triangles of different sizes sharing the same acute angle, measure sides with a ruler, compute three ratios (opposite over hypotenuse, adjacent over hypotenuse, opposite over adjacent), and record results in a structured table.                                                                                                                                                                          |
| <b>Attitudes</b>            | Learners will demonstrate curiosity by making genuine predictions before instruction, and will show integrity by recording their measured values honestly even when results are unexpected.                                                                                                                                                                                                                                                  |
| <b>Key Inquiry Question</b> | How do we apply trigonometric ratios in day-to-day activities?                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Purpose in Storyline</b> | This is the opening lesson that launches the entire unit phenomenon. Learners encounter the mango-tree shadow puzzle from Kisumu, make unaided predictions, and then discover through hands-on triangle drawing that angle alone fixes the ratio of sides — the foundational insight that makes all subsequent lessons meaningful. The Driving Question Board is opened here and will be updated in every subsequent lesson until Lesson 10. |
| <b>Safety Notes</b>         | Ensure rulers and pencils are used safely during the drawing activity. If the lesson is taken outdoors to observe shadows, learners should not stare directly at the sun. No significant hazards are anticipated for an indoor lesson.                                                                                                                                                                                                       |

### B. LESSON OVERVIEW

The teacher presents the mango-tree shadow scenario set in Kisumu: a student of height 1.6 m casts a 1.2 m shadow while a nearby mango tree casts a 9.0 m shadow, and a classmate measures the sun-ray angle as approximately 53 degrees using a homemade clinometer. Learners first predict in groups how an angle measured with a protractor could reveal the height of a real tree, posting sticky-note predictions on the newly opened Driving Question Board. Groups then draw three right-angled triangles of different sizes but sharing one chosen acute angle, measure all three sides, and compute the three side ratios, discovering the ratios are constant across all triangle sizes. The teacher formally names these ratios sine, cosine, and tangent and connects them back to the phenomenon, leaving the full calculation for Lesson 4.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                        | Learner Experience                                                                                                                                                                                  | Teacher Moves                                                                                                                                                                               | Sensemaking Strategy                                                                                                                                                                                     | Formative Assessment Strategy                                                                                                                                                                                 |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Finding reciprocal trig ratios</a><br><small>Source: Khan Academy (English)</small> | Learners sit in groups of four. The teacher reads aloud the mango-tree scenario slowly and deliberately, pausing to let learners visualise the compound in Kisumu, the student, her shadow, and the | Teacher displays or sketches the scenario on the board: a stick figure of height 1.6 m with a shadow of 1.2 m beside a mango tree with a shadow of 9.0 m, labelling the sun-ray angle as 53 | Predict-before-instruction activates prior knowledge of ratio and similar triangles from Grade 8 and 9 and surfaces alternative conceptions. The deliberate 2-minute silence ensures all learners engage | Teacher scans sticky notes during circulation to identify the range of prior knowledge: learners who mention similar triangles or ratio correctly, and learners who predict unrelated ideas such as measuring |

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| <p>- US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                                     | <p>tree shadow. Each learner individually writes one prediction on a sticky note in response to the prompt: How could knowing only the angle of the sun help you find the height of the mango tree without climbing it? After two minutes of individual silent thinking, group members share their sticky notes with each other, discuss similarities and differences, and agree on one group prediction to post on the Driving Question Board. Individual sticky notes are also retained for personal portfolios. Predicted ideas typically include: the tree and student form similar triangles, the ratio of height to shadow stays the same, or the angle controls some number relationship.</p> | <p>degrees. Teacher says: I want you to look at this picture carefully and just think. Do not calculate yet. On your sticky note, write your best prediction: how could the angle of those sun rays possibly tell us the height of the tree? You have two minutes of silence to think on your own. [WAIT TIME: 2 minutes of strict silence — teacher circulates without speaking, making eye contact and nodding to encourage.] After two minutes: Now share with your group. Choose the one prediction you all find most convincing. [WAIT TIME: 1.5 minutes for group share.] Teacher then cold-calls two groups to share their group prediction aloud, saying: Group two, share your sticky note with us. What is your group thinking? After each group responds, teacher says only: Interesting. Hold that idea. We are going to test it today. Teacher does not confirm or deny any prediction at this stage.</p> | <p>individually before social influence occurs. Posting on the DQB externalises thinking and creates accountability — learners will return to these predictions in Lesson 4 and Lesson 10 to evaluate them.</p>                                                                                                                                                                                                                                                                                | <p>the shadow with a longer ruler. This informs the amount of scaffolding needed in the Explain Phase. The teacher makes a mental note of at least two groups to revisit during the Model Building Phase.</p>                                                                                                             |
| <p><b>Observe Phase</b><br/> <b>VIDEO:</b><br/> <a href="#">Intro to the trigonometric ratios</a><br/> Source: Khan Academy (Kiswahili)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p> | <p>Each group receives a plain A4 sheet of paper, a ruler, a protractor, and a pencil. The teacher gives the instruction: draw three right-angled triangles — one small, one medium, one large — each containing the same acute angle. Groups choose their own angle (most will choose 30, 45, or 60 degrees because these are familiar from geometry, but any angle is allowed). Learners use the protractor to set the chosen angle at the base-left vertex, draw the base (adjacent side), erect a perpendicular (opposite side) using the right-angle corner, and connect the hypotenuse. The three triangles differ in base length but share the identical chosen angle.</p>                    | <p>Teacher models drawing one triangle on the board, narrating each step: I am placing my protractor at the bottom-left corner and marking 40 degrees. I draw my base along the bottom. At the right end of the base I draw a perpendicular line going straight up — that is my opposite side. Then I connect the top of that line back to my starting point — that is my hypotenuse. Teacher labels the sides clearly. Teacher then says: Now it is your turn. You choose your angle. Draw all three triangles on the same sheet of paper. Measure every side carefully. Record your values in the table. You have seven minutes. [WAIT TIME: 7 minutes</p>                                                                                                                                                                                                                                                           | <p>The hands-on drawing task grounds the abstract idea of angle in a physical, manipulable object. Drawing multiple sizes of the same triangle is the critical observational move — it sets up the cognitive conflict of the next phase, because learners will expect the sides to differ but will discover the ratios do not. Choosing their own angle gives learners agency and means different groups will have different angles, allowing cross-group comparison in the Explain Phase.</p> | <p>Teacher observes whether learners correctly identify opposite, adjacent, and hypotenuse relative to the chosen angle (not the right angle). A common error is labelling sides relative to the wrong vertex. Teacher notes any group making this error for targeted correction during the Explain Phase discussion.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Learners label the right angle, the chosen angle (theta), the opposite side, the adjacent side, and the hypotenuse on each triangle, using the vocabulary from the board. They then measure all three sides of each triangle to the nearest millimetre and record the values in a prepared table with columns: triangle size, opposite, adjacent, hypotenuse.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>— teacher circulates, checking that triangles genuinely share the same angle and that labelling is correct. Teacher corrects protractor misuse quietly at individual groups without whole-class interruption.] If a group finishes early, teacher prompts: Look at your three triangles. Do you notice anything interesting about the measurements?</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Solving for a side in right triangles with trigonometry</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners add three new columns to their table: opposite divided by hypotenuse, adjacent divided by hypotenuse, and opposite divided by adjacent. They compute each ratio as a decimal to two decimal places for all three triangles. Within groups, learners compare the three values in each ratio column and record what they notice. Groups are asked to write a one-sentence observation on the back of their A4 sheet. The teacher then facilitates a whole-class discussion, calling on groups that chose different angles to share their ratio values. The class discovers that each group has three nearly identical numbers in each ratio column, and that different groups have different ratio values because they chose different angles. The teacher formally names the ratios: opposite over hypotenuse is the sine of the angle, adjacent over hypotenuse is the cosine of the angle, and opposite over adjacent is the tangent of the angle. The teacher connects this immediately back to the Kisumu scenario: if the sun makes an angle of 53 degrees with the ground, then the tangent of 53 degrees equals the height of the tree divided by the length of the shadow.</p> | <p>Teacher calls the class back to attention and asks: What do you notice about the three values in your first ratio column — opposite over hypotenuse? Cold-call to Group 1: Read out your three values. [WAIT TIME: 5 seconds for learner to respond.] Teacher writes the values on the board. Then to Group 3: Your group chose a different angle. Read out your three values for the same ratio. [WAIT TIME: 5 seconds.] Teacher writes these alongside. Teacher asks the class: These two groups chose different angles. Why are the values in each group almost the same, but different between the two groups? [WAIT TIME: 15 seconds of open silence before taking responses — this is a high-cognitive-demand question requiring genuine sensemaking.] Teacher accepts two or three learner responses without evaluating them yet, then says: The ratio depends only on the angle — not on how big the triangle is. That is the key idea of today. Let me give each ratio a name. Teacher writes on the board: sine theta equals opposite over hypotenuse, cosine theta equals adjacent over hypotenuse, tangent theta equals opposite over adjacent. Teacher says: Now look</p> | <p>The cross-group comparison is a deliberate sensemaking move: seeing that different groups with the same angle get the same ratio, but different angles give different ratios, isolates the variable (angle) and the invariant (ratio for that angle). The formal naming of sine, cosine, and tangent is delayed until after learners have constructed the pattern themselves, ensuring the vocabulary labels a concept already partially understood. Explicitly leaving the tree height unsolved preserves the productive tension of the phenomenon across the unit.</p> | <p>Teacher listens for whether learners can articulate that the ratio is constant because all three triangles are similar (same angles, proportional sides). If no group produces this explanation spontaneously, the teacher asks: What do we call triangles that have the same angles but different sizes? — connecting to Grade 9 similar triangles. Learners who can identify the correct sides relative to the labelled angle demonstrate readiness for Lesson 2; those who cannot are noted for targeted support.</p> |

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| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>The teacher directs attention to a dedicated section of the classroom wall or board reserved for the Driving Question Board, which will remain visible for all ten lessons. The large driving question is written at the top: How can the angle a sunbeam makes with the ground allow us to calculate the height of any tree, building, or hill in Kenya without measuring it directly? Each group brings their group prediction sticky note from the Predict Phase and places it on the DQB under the heading: Our Predictions. The teacher then invites every learner to write one question that arose during today's activity on a new sticky note and post it under the heading: Questions We Now Have. Learners circulate briefly to read each other's questions. The teacher reads several questions aloud, groups related ones together, and labels clusters. Typical emerging questions include: How do we find the exact value of sine or cosine for any angle? Is there a pattern between sine and cosine? Do these ratios work for angles bigger than 90 degrees? The teacher says: Every</p> | <p>Teacher says: This board belongs to you. Every prediction you made at the start is on it. Every question you have just posted is on it. At the end of every lesson you will add something new to it — something you have learned or a new question that opened up. By Lesson 10, you will come back to your own prediction from today and decide: were you right? Teacher physically attaches the group sticky notes, saying the group name aloud as each is placed, creating a sense of community ownership. Teacher does not remove or correct any prediction at this stage, including incorrect ones, explaining: We do not erase predictions — we revisit them with evidence later. Teacher writes the TO BE SOLVED note from the Explain Phase directly on the DQB near the driving question: <math>\tan 53^\circ \times 9.0 \text{ m} = \text{height of the mango tree}</math> — we will calculate this in Lesson 4.</p> | <p>The DQB makes the intellectual journey of the unit visible and persistent. It activates metacognitive awareness — learners know they are at the beginning of an inquiry and that their own questions drive the sequence. Clustering questions publicly helps learners see the coherence of the sub-strand without the teacher having to impose it. Preserving incorrect predictions models scientific integrity — ideas are tested, not assumed.</p> | <p>The teacher reads the posted questions during the brief circulation moment to gauge depth of curiosity and identify any misconceptions expressed as questions (for example, a learner might post: Does a bigger angle always give a bigger triangle? — revealing confusion between angle size and side length). These inform planning for Lesson 2 questioning sequences.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | one of these questions will be answered during our ten lessons. Watch this board grow.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Model Building Phase</b><br> <b>VIDEO:</b><br><a href="#">Trigonometric ratios in right triangles - dubbed(KY)</a><br>Source: Khan Academy (Kiswahili)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Trigonometric ratios in right triangles</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | Each learner opens a dedicated Trigonometry Model Page — a blank or lightly gridded page at the front of their exercise book reserved for the evolving unit model. For Lesson 1, learners draw one clearly labelled right-angled triangle and annotate it with the three side names (opposite, adjacent, hypotenuse) relative to a labelled acute angle theta. They write the three ratio definitions: $\sin \theta = \frac{O}{H}$ , $\cos \theta = \frac{A}{H}$ , $\tan \theta = \frac{O}{A}$ . Below the triangle, they write one sentence connecting this to the Kisumu scenario: In the mango-tree problem, the angle of 53 degrees is theta, the shadow of 9.0 m is the adjacent side, and the tree height we want to find is the opposite side. Learners initial and date this page. The teacher reminds learners that this page will be revised and extended in every lesson until it becomes a complete model of the entire sub-strand by Lesson 10. | Teacher says: At the back of your book, open a page labelled Trigonometry Model — Lesson 1. This page stays in your book for all ten lessons. Today, draw the triangle, label all the parts, write the three ratio definitions. Most importantly, write a sentence that connects the triangle to the mango tree. That sentence is the reason we are learning this. [WAIT TIME: 4 minutes for silent individual drawing and annotation.] Teacher circulates and spots-checks that side labels are relative to theta, not to the right angle. Teacher says to the class: In Lesson 2 we will discover exactly how to find the value of sine for any angle without drawing a triangle. Then in Lesson 4 we will finally calculate the height of that mango tree in Kisumu. Keep your predictions on the board — we are going to check them. | The model page is a living artefact of the learner's growing understanding, not a note-taking exercise. By starting it with the phenomenon connection required in the first entry, the teacher embeds the habit of always linking abstract content to the real-world anchor. The model page will be used in Lesson 10 as the final assessment artefact, giving it longitudinal value. | Teacher collects one or two model pages briefly at the close of the lesson (or photographs them with a phone) to check correctness of labelling before Lesson 2. Exit criterion: learner can correctly label opposite, adjacent, and hypotenuse on a right-angled triangle relative to a specified acute angle, and can write all three ratio definitions. Learners who cannot meet this criterion receive brief individual feedback at the start of Lesson 2. |

## D. TEACHER REFLECTION

After the lesson, consider these questions: Did at least 80 percent of groups independently notice that the ratios were constant across triangle sizes, or did most groups need direct prompting? Were the 15-second wait-time silences in the Explain Phase maintained, or did the teacher fill the silence too quickly? Did the DQB feel owned by the learners or did it feel like a teacher display? Were any learners unable to identify opposite, adjacent, and hypotenuse correctly — and have they been noted for Lesson 2 support? Did the lesson end with genuine unresolved curiosity about the tree height, or did the session feel closed? The most critical indicator of success is whether learners leave wanting to know the value of tangent 53 degrees — if they do, the phenomenon has done its job.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                     |                                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| What did I observe? | A student in Kisumu noticed that she could measure the angle of the sun with a clinometer and that both she and the mango tree |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|

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|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | cast shadows on the same sunny afternoon. Her shadow was 1.2 m and the tree shadow was 9.0 m, with a sun angle of approximately 53 degrees.                                                                                                                                                                                                                           |
| <b>What did I learn?</b>                     | When you draw right-angled triangles with the same acute angle but different sizes, the ratio of opposite to hypotenuse stays the same, the ratio of adjacent to hypotenuse stays the same, and the ratio of opposite to adjacent stays the same. These constant ratios are called sine, cosine, and tangent of that angle.                                           |
| <b>How does this explain the phenomenon?</b> | The ratios are constant because all right-angled triangles with the same acute angle are similar triangles — their sides are proportional. This means the angle alone determines the ratio, so if we know the angle and one side we can always find the other side. That is how the 53-degree sun angle can reveal the mango tree height from just the shadow length. |

## LESSON 2 (80 minutes): Sine of an Acute Angle: Reading the Language of Triangles

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | To develop learners' understanding of the sine ratio as a consistent, angle-dependent relationship in right-angled triangles, and to build fluency in reading sine values from mathematical tables.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Knowledge</b>            | By the end of the lesson, the learner should be able to: (a) define the sine of an acute angle as the ratio of the opposite side to the hypotenuse in a right-angled triangle; (b) state that the sine ratio depends only on the angle, not on the size of the triangle; (c) read sine values for given acute angles from mathematical tables.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Skills</b>               | By the end of the lesson, the learner should be able to: (a) use mathematical tables accurately to look up sine values for acute angles; (b) verify sine values obtained from tables against measurements made on triangles drawn in Lesson 1; (c) apply the sine ratio to estimate heights in real contexts such as the mango tree shadow problem.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Attitudes</b>            | Learners appreciate the elegance and consistency of the sine ratio as a mathematical tool that works universally — from a small triangle drawn on paper to the height of a mango tree in Kisumu — and show responsibility in the careful, accurate use of mathematical tables.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Key Inquiry Question</b> | How do we apply trigonometric ratios in day-to-day activities? How are the trigonometric ratios of acute angles related to each other?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Purpose in Storyline</b> | In Lesson 1 learners measured sides of right-angled triangles and computed the ratio $\text{opposite} \div \text{hypotenuse}$ for several angles, noticing the ratio stayed the same regardless of triangle size. In Lesson 2 they discover that this consistent ratio has a name — sine — and that mathematicians have pre-computed it for every angle in mathematical tables. By reading those tables and matching the values to their own Lesson 1 measurements, learners gather crucial evidence that $\sin \theta = \text{opposite/hypotenuse}$ is a universal, angle-only relationship. This advances the driving question: if $\sin 53^\circ$ is always 0.7986, then once you know the sun makes a $53^\circ$ angle with the ground, you can calculate the mango tree's height from nothing but the shadow length — no climbing required. |
| <b>Safety Notes</b>         | No hazardous materials are used in this lesson. Learners should handle mathematical tables carefully to avoid tearing. If the video is shown on a projected screen, ensure the room is not darkened to the point where learners cannot write notes safely.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

### B. LESSON OVERVIEW

Learners begin by revisiting the anchoring phenomenon — the student in Kisumu and her mango-tree shadow — and are challenged to predict what the ratio  $\text{opposite/hypotenuse}$  for the  $53^\circ$  sun angle should be. They then watch a short focused video introducing the sine ratio and its notation. The central activity is a guided table-reading exercise in which learners use mathematical tables to find sine values for a set of angles (including those measured in Lesson 1), then compare table values to their own calculated ratios from the previous lesson. The teacher conducts structured cold-call questioning across at least 6 students to build fluency with table navigation. Learners close by annotating the Driving Question Board with the key insight: "Sine depends only on the angle, not the triangle's size," and revising their cumulative models to incorporate the sine ratio.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                | Learner Experience            | Teacher Moves                  | Sensemaking Strategy           | Formative Assessment Strategy |
|----------------------|-------------------------------|--------------------------------|--------------------------------|-------------------------------|
| <b>Predict Phase</b> | Learners are shown the Kisumu | Display the phenomenon diagram | Think-Pair-Share with recorded | Observable indicator: Every   |

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| <p> <b>VIDEO:</b><br/> <a href="#">The trig functions &amp; right triangle trig ratios</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p>                      | <p>shadow scenario on the board: a girl of height 1.6 m casts a shadow of 1.2 m; a mango tree casts a shadow of 9.0 m; the sun's rays make an angle of <math>53^\circ</math> with the ground. Each learner individually writes a prediction in their exercise books: 'What is the value of <math>\text{opposite} \div \text{hypotenuse}</math> for the sun-ray triangle at <math>53^\circ</math>? Will this value change if the mango tree were taller or shorter?' Learners also predict whether the ratio they calculated in Lesson 1 for their drawn triangles will match a value they can look up somewhere. They then share predictions with a shoulder partner (Think-Pair) before the teacher takes 3 cold-call responses to surface a range of ideas — including likely misconceptions that the ratio changes with triangle size.</p>                            | <p>(sun ray, tree, shadow forming a right-angled triangle labelled with <math>O</math> = tree height, <math>H</math> = sun-ray length, angle = <math>53^\circ</math>). Say verbatim: 'Look at this triangle. In your book, write your best estimate of the value of <math>\text{opposite} \div \text{hypotenuse}</math> for a <math>53^\circ</math> angle. You have 90 seconds — work silently.' [Wait 90 seconds.] 'Now share with your partner for 30 seconds.' [Wait 30 seconds.] Cold-call Student 1: 'Achieng, what did you predict and why?' [Wait 10 seconds for response.] Cold-call Student 2: 'Otieno, do you agree with Achieng? What would change if the tree were twice as tall?' [Wait 12 seconds.] Cold-call Student 3: 'Wanjiru, if the angle stayed at <math>53^\circ</math> but the shadow was 18 m long instead of 9 m, would your ratio change?' [Wait 10 seconds.] Record all three predictions on the board under the heading 'Our Predictions — we will check these later.'</p> | <p>public predictions. This strategy externalises prior knowledge and creates cognitive dissonance when learners later discover the ratio is constant — making the sine concept genuinely surprising and memorable. Recording predictions publicly creates accountability and a reference point for revision at the end of the lesson.</p>                                                                                                        | <p>learner has a written numerical prediction in their exercise book before sharing begins. The teacher scans books during the 90-second window. Misconception to listen for: 'The ratio gets bigger when the triangle gets bigger' — flag this without correcting yet, saying: 'Let's keep that idea — we will test it today.'</p>                                                                                                                                          |
| <p><b>Observe Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Learners watch a 6–8 minute video introducing the sine ratio: its definition (<math>\sin \theta = \text{Opposite} / \text{Hypotenuse}</math>), its notation, and a worked example showing that two right-angled triangles with the same <math>40^\circ</math> angle but different sizes yield the same sine value (approximately 0.6428). The video includes a Kenyan context — a surveyor in the Rift Valley measuring the angle of elevation to a hill. After the video, learners immediately open their mathematical tables to the sine pages. Working in pairs, they complete a structured table-reading activity: they are given a list of 8 angles (<math>20^\circ</math>, <math>30^\circ</math>, <math>37^\circ</math>, <math>45^\circ</math>, <math>53^\circ</math>, <math>60^\circ</math>, <math>72^\circ</math>, <math>90^\circ</math>) and must record</p> | <p>Before playing the video say: 'As you watch, write down: (1) the definition of sine, (2) one thing that surprises you.' Play the video. After the video, give the instruction: 'Open your mathematical tables to the sine section. I will give you 4 minutes to find all 8 sine values on your worksheet. Start now.' [Circulate, observe table-navigation strategies.] After 4 minutes, begin cold-call table-reading drill — minimum 6 students: Cold-call Student 4: 'Kamau, what is <math>\sin 20^\circ</math>? Tell me exactly which column and row you used.' [Wait 10 seconds.] Cold-call Student 5: 'Nafula, what is <math>\sin 37^\circ</math>? Did anyone get a different value?'</p>                                                                                                                                                                                                                                                                                                     | <p>Data-to-concept bridging: Learners collect two parallel data sets — table-derived sine values and their own Lesson 1 measured ratios — and place them side by side. This concrete comparison makes the abstraction (sine is a universal constant for a given angle) visible and personally verifiable rather than teacher-asserted. The structured cold-call drill on table navigation builds procedural fluency through spaced retrieval.</p> | <p>Observable indicators: (1) Each pair's worksheet shows 8 correctly located sine values with the column/row reference noted. (2) For <math>\sin 53^\circ</math>, learners can articulate that their Lesson 1 measured ratio (<math>\approx 0.80</math>) matches the table value (0.7986) within measurement error. Listen for: any learner who says 'my ratio was different' — use this as a teachable moment about measurement precision, not as an error to dismiss.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>the sine value from the tables for each. For <math>53^\circ</math> and one angle of their choice from Lesson 1, they also retrieve the ratio they computed from their own drawn triangles and write both values side by side for comparison.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>[Wait 10 seconds.] Cold-call Student 6: 'Mwangi, what is <math>\sin 53^\circ</math>? Now look at the ratio you measured for your <math>53^\circ</math> triangle in Lesson 1 — what do you notice?' [Wait 12 seconds.] Cold-call Student 7: 'Asha, what is <math>\sin 90^\circ</math> and does that surprise you? Why?' [Wait 10 seconds.] Cold-call Student 8: 'Omondi, what is <math>\sin 45^\circ</math>? Can you express that as a fraction or a decimal?' [Wait 10 seconds.] Cold-call Student 9: 'Chebet, if I gave you <math>\sin 60^\circ</math>, could you immediately tell me which side would be the longest — opposite or adjacent? Explain.' [Wait 15 seconds.] Summarise on the board: write each angle and its sine value as learners confirm answers. Draw attention to <math>\sin 53^\circ \approx 0.7986</math> and link back to the shadow scenario.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Trigonometric ratios in right triangles - dubbed(KY)</a><br/> Source: Khan Academy (Kiswahili)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners work in groups of four to complete a sensemaking task: they are given three right-angled triangles of different sizes, all with a <math>37^\circ</math> angle, with the opposite and hypotenuse sides labelled in centimetres (small: 3 cm and 5 cm; medium: 6 cm and 10 cm; large: 9 cm and 15 cm). They calculate opposite/hypotenuse for each and compare to <math>\sin 37^\circ</math> from their tables. They then write a group explanation in their own words: 'Why does the ratio stay the same even as the triangle grows?' Groups share their explanations; the teacher guides learners to connect this to the concept of similar triangles (linked to prior learning). The teacher then models applying <math>\sin 53^\circ = 0.7986</math> to solve the mango tree problem: if the shadow is 9.0 m and the angle of elevation is <math>53^\circ</math>, what is the</p> | <p>Distribute the three-triangle handout and say: 'In your group, compute opposite divided by hypotenuse for each triangle and compare to your table value for <math>\sin 37^\circ</math>. You have 5 minutes. Nominate a scribe and a speaker.' [Circulate, prompt groups with: 'What do you notice about your three ratios? Can you explain WHY they are the same using what you know about similar triangles?'] After 5 minutes, take group reports: 'Group 2 — what did you find? Use the word SIMILAR in your explanation.' [Wait 10 seconds.] Affirm correct use of similar triangles language. Then model the mango tree calculation on the board step-by-step: '<math>\sin 53^\circ = \text{opposite} / \text{hypotenuse} \rightarrow 0.7986 = \text{tree height} / \text{sun-ray length}</math>. But wait — we know the shadow, not the sun-</p>                     | <p>Analogical reasoning with scaling: by computing the ratio for three triangles that are scalar multiples of each other (3-5, 6-10, 9-15), learners see that scaling preserves angles and ratios — connecting sine to the geometry of similar triangles they already know. This makes the 'why' of sine visible, not just the 'what'. The individual problem application immediately transfers understanding to a new Kenyan context.</p> | <p>Observable indicators: (1) Every group's written explanation includes a reference to 'similar triangles' or 'same shape, different size.' (2) In individual work, at least 80% of learners correctly write the sine equation (<math>\sin 40^\circ = \text{height} / \text{hypotenuse}</math>) before attempting to solve. (3) Kiptoo's cold-call response reveals whether learners can set up — not just solve — the equation. Misconception to watch for: learners multiplying angle <math>\times</math> side length instead of using the ratio.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>hypotenuse (sun-ray length)? From that, what is the tree height? Learners solve a parallel problem individually: a mobile phone mast in Nakuru casts a shadow of 12 m when the sun's elevation angle is <math>40^\circ</math>. Find the height of the mast using <math>\sin 40^\circ</math> from tables.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>ray. Which ratio connects opposite and adjacent? We will explore that in Lesson 3. For now, let's use what we do know.' Guide learners to set up: <math>\text{shadow/hypotenuse} = \cos 53^\circ</math> [preview only — do not teach cosine yet]. Then pivot to the Nakuru mast problem and circulate while learners solve individually. Cold-call Student 10: 'Kiptoo, walk us through your setup for the Nakuru mast. What did you write as your sine equation?' [Wait 15 seconds.]</p>                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Solving for a side in right triangles with trigonometry</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Each learner receives a sticky note (or writes in a designated DQB section of their book if physical sticky notes are unavailable). They respond to the prompt: 'What new evidence have you gathered today that helps answer our driving question: How can an angle tell us the height of a tree?' They also add a second note under the sub-heading 'New Questions I Have Now' — learners often spontaneously ask about what happens with the shadow length (cosine) or how tables were originally calculated. The class DQB is updated collectively: the teacher reads out 3–4 sticky notes, clusters them, and adds the lesson's anchor insight in a distinct colour: '<math>\sin \theta = O/H</math> and depends ONLY on <math>\theta</math> — not on the triangle's size.' The teacher explicitly links this to the Kisumu phenomenon: 'So now we know — <math>\sin 53^\circ</math> is always 0.7986. The sun doesn't change the angle just because the tree is taller. This means we are one step closer to calculating that mango tree's height.'</p> | <p>Distribute sticky notes and say: 'You have 3 minutes. On your note, write one piece of evidence from today that moves us closer to answering our driving question. Be specific — use numbers if you can.' [Wait 3 minutes, circulate.] Collect notes. Read 3 aloud, paraphrasing: 'This learner says <math>\sin 53^\circ = 0.7986</math> and it matched what they measured — that is strong evidence.' Place notes on the board. Ask the class: 'What single sentence captures what we learned today about sine?' Take 2 cold-call responses, then write the class-agreed anchor insight. Point to the driving question and say: 'What do we still NOT know that we need to answer the full question?' Accept 2–3 responses and add them to the 'Still Wondering' column of the DQB.</p> | <p>Public knowledge aggregation on the DQB: making individual insights collective and visible builds a shared scientific narrative. The 'Still Wondering' column sustains inquiry momentum and gives the teacher real-time data on what gaps remain before Lesson 3 (cosine).</p> | <p>Observable indicators: (1) Every learner's sticky note contains a specific, evidence-based claim (not a vague statement like 'I learned sine'). (2) The class-agreed anchor insight correctly states that sine depends only on the angle. (3) 'Still Wondering' questions reveal whether learners are ready for cosine and tangent — the teacher notes these for Lesson 3 planning.</p> |
| <b>Model Building</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Learners retrieve their cumulative model from Lesson 1 — a labelled</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Say: 'Take out your Lesson 1 model. Today we are going to</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Iterative model revision: learners physically modify a persistent</p>                                                                                                                                                                                                          | <p>Observable indicators: (1) The opposite side in every model is</p>                                                                                                                                                                                                                                                                                                                      |

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| <p><b>Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Finding reciprocal trig ratios</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>right-angled triangle showing the phenomenon (sun ray, tree, shadow). They now add the sine relationship to their model: they annotate the hypotenuse and opposite side, write '<math>\sin \theta = O/H</math>' inside the triangle, and add the specific value '<math>\sin 53^\circ \approx 0.7986</math>' with an arrow linking it to the mango tree scenario. They also add a 'model note box' alongside the triangle with the statement: 'This ratio is fixed for any given angle — it does not change when the triangle gets bigger or smaller (similar triangles).' Pairs peer-review each other's models against a simple 3-item checklist:</p> <ol style="list-style-type: none"> <li>(1) Is the opposite side correctly identified relative to the <math>53^\circ</math> angle?</li> <li>(2) Is <math>\sin \theta = O/H</math> written correctly?</li> <li>(3) Is the 'angle-only' insight recorded? Learners make any corrections in a different colour pen.</li> </ol> | <p>upgrade it with what we now know about sine. You have 6 minutes to annotate your model — add labels, the formula, and the specific value for <math>53^\circ</math>. Your model should tell the story of why you can calculate the mango tree's height.' [Circulate, prompt: 'Where is your opposite side relative to the <math>53^\circ</math> angle?' and 'Have you written what stays constant?'] After 6 minutes: 'Now swap with your partner and use the checklist on the board to review their model. Give them one specific, positive comment and one specific suggestion.' [Wait 4 minutes for peer review.] Close by saying: 'Your model is growing. By Lesson 10 it will contain everything needed to answer our driving question completely. What did you add today?' Take 2 final cold-call responses.</p> | <p>artefact across lessons, making their evolving understanding visible and cumulative. The peer-review checklist introduces disciplined critique — a key science and engineering practice — while keeping the focus on the phenomenon rather than abstract correctness.</p> | <p>correctly labelled relative to the stated angle (not the right angle or the other acute angle). (2) The formula <math>\sin \theta = O/H</math> appears on all models. (3) The 'angle-only' insight is recorded in the model note box. (4) Peer-review checklists are completed with specific (not generic) feedback. Models not meeting criterion (1) are flagged for teacher follow-up at the start of Lesson 3.</p> |
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## D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the cold-call table-reading drill, which learners struggled to navigate the sine tables — did they have difficulty with rows vs. columns, or with decimal place value? Plan a 3-minute re-teaching moment at the start of Lesson 3 if more than 3 learners showed this difficulty. (2) Did learners' explanations in Phase 3 explicitly reference similar triangles, or did they simply state the ratio 'happened to be the same'? If the similar-triangles connection was weak, consider adding a brief visual proof in Lesson 3's opening. (3) Review the 'Still Wondering' sticky notes from the DQB: are learners asking about the shadow-to-height ratio (which requires cosine/tangent)? Use their own questions as the opening hook for Lesson 3. (4) Were the Lesson 1 measured ratios close enough to the sine table values to build confidence? If many learners found large discrepancies, this is a measurement-precision issue — address it by discussing significant figures before Lesson 3's cosine work. (5) Did the Nakuru mast individual problem reveal any systematic errors in equation setup? Collect and review at least 10 learners' books before Lesson 3.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

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| <p><b>What did I observe?</b></p> | <p>Learners watched a video showing that two right-angled triangles with the same <math>40^\circ</math> angle but different sizes produce the same value of opposite <math>\div</math> hypotenuse. They also observed that a surveyor in the Rift Valley uses this relationship to measure hill heights without direct measurement.</p> |
| <p><b>What did I learn?</b></p>   | <p>The sine of an acute angle (<math>\sin \theta</math>) is defined as opposite <math>\div</math> hypotenuse in a right-angled triangle and is a fixed value that depends</p>                                                                                                                                                           |

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|                                              | only on the angle — not on the size of the triangle. For the mango tree scenario, $\sin 53^\circ \approx 0.7986$ , a value that can be read directly from mathematical tables.                                                                                                                                                                                                                                                                                   |
| <b>How does this explain the phenomenon?</b> | Learners explained that because triangles with the same angle are similar (same shape, scaled size), their side ratios are always equal — which is why $\sin 53^\circ$ is the same whether the triangle is drawn on paper or formed by a sun ray and a 20-metre tree in Kisumu. This means knowing just the angle and one side length is enough to find any other side — bringing us one step closer to calculating the mango tree's height without climbing it. |

## LESSON 3 (40 minutes): Cosine of an Acute Angle: The Adjacent Side Speaks

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>Purpose</b>              | By the end of this lesson, learners will understand that the cosine of an acute angle is a fixed ratio of the adjacent side to the hypotenuse, read cosine values from mathematical tables, and use cosine to find unknown sides of right-angled triangles in real Kenyan contexts.                                                                                                                                                                               |
| <b>Knowledge</b>            | The cosine of an acute angle $\theta$ is defined as $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$ ; cosine values depend only on the angle and not on the size of the triangle; cosine values can be read from mathematical tables and decrease as the angle increases from 0 degrees to 90 degrees.                                                                                                                                                  |
| <b>Skills</b>               | Reading cosine values from mathematical tables for given angles; identifying the adjacent side and hypotenuse in a right-angled triangle for a specified angle; using $\cos \theta$ to calculate unknown sides; comparing sine and cosine values to begin detecting a complementary pattern.                                                                                                                                                                      |
| <b>Attitudes</b>            | Curiosity about the hidden relationship between sine and cosine; diligence in careful table reading; collaborative respect when working in pairs on structured problems.                                                                                                                                                                                                                                                                                          |
| <b>Key Inquiry Question</b> | How are the trigonometric ratios of acute angles related to each other? — specifically, how does cosine compare with the sine values already discovered, and what pattern begins to emerge?                                                                                                                                                                                                                                                                       |
| <b>Purpose in Storyline</b> | In Lesson 2 learners established that sine captures the ratio involving the side opposite the angle and the hypotenuse. Lesson 3 introduces cosine, the ratio involving the adjacent side, giving learners a second measurement tool. Together sine and cosine will allow them to approach the mango-tree problem from multiple angles (literally) and set up the crucial relationship $\tan \theta = \frac{\sin \theta}{\cos \theta}$ to be derived in Lesson 4. |
| <b>Safety Notes</b>         | No physical hazards. Remind learners to handle mathematical tables booklets with care to avoid tearing; rulers used for underlining table rows should be kept flat on desks.                                                                                                                                                                                                                                                                                      |

### B. LESSON OVERVIEW

Learners begin with a guided reading excerpt about the cosine ratio in the context of a wheelchair ramp at a Nakuru hospital. They predict what  $\cos 53$  degrees will be relative to  $\sin 53$  degrees by sketching the mango-tree right-angled triangle and labelling adjacent and opposite sides. Through a structured table-reading activity they discover cosine values for a range of angles, compare them with sine values, and notice that  $\cos \theta$  for a given angle equals  $\sin$  of the complementary angle — a pattern they record as a question without yet proving it. Pairs then solve five structured problems involving Kenyan contexts. The Driving Question Board is updated with a new question: Is there a link between sine and cosine?

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                           | Learner Experience                                                                        | Teacher Moves                                                                                  | Sensemaking Strategy                                                                          | Formative Assessment Strategy                                                              |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Solving for a side</a> | Learners open their exercise books to the right-angled triangle diagram of the mango-tree | Teacher holds up the mango-tree diagram on the board (drawn or projected) and traces the 9.0 m | Anchoring the new ratio to the already-familiar mango-tree triangle gives cosine an immediate | Teacher listens to the two cold-called pairs and notes on a clipboard whether learners (a) |

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| <p><a href="#">in right triangles with trigonometry</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/><a href="#">Trigonometric ratios in right triangles</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES</a> for similar readings</p>                                                         | <p>scenario drawn in Lesson 1. The teacher reminds them that in Lesson 2 they used the side opposite 53 degrees (height of tree) and the hypotenuse (length of sun ray). The teacher now points to the remaining labelled side — the 9.0 m shadow on the ground — and asks: what ratio would use THAT side and the hypotenuse? Each learner individually writes a prediction in their book: (a) what they think this new ratio will be called, (b) whether they expect its value to be larger or smaller than <math>\sin 53</math> degrees (approximately 0.7986), and (c) why. After 2 minutes of silent individual writing, learners share predictions with their seat partner for 1 minute.</p>                                                                                                                                                                                          | <p>shadow with a finger, saying: In Lesson 2 we listened to the opposite side — it gave us sine. Today we listen to THIS side, the one that runs along the ground next to the angle. Without telling me the answer, write down what you predict: is the ratio of this ground-shadow side to the sun-ray hypotenuse going to be bigger than 0.7986, smaller, or the same — and why? WAIT 90 seconds of complete silence. Then: Share your prediction with the person next to you. You have 60 seconds. After sharing, cold-call two pairs using their names: ask one learner to give the prediction and the partner to give the reason.</p>                                                                                                                                              | <p>physical meaning. Requiring both a numerical prediction and a reason forces learners to reason from the geometry (the adjacent side is shorter than the hypotenuse, so the ratio must be less than 1; comparing it with sine requires thinking about which side is longer — opposite or adjacent — in a 53-degree triangle, where opposite is longer than adjacent, so cosine should be less than sine).</p>                                                                                                                                                          | <p>correctly identify that the ratio must be between 0 and 1 and (b) reason about whether it will be less than or greater than <math>\sin 53</math> degrees. This reveals whether learners have a geometric sense of the triangle rather than just memorised the sine formula.</p>                                                                                                                                                            |
| <p><b>Observe Phase</b><br/> <b>VIDEO:</b><br/><a href="#">Solving for a side in right triangles with trigonometry</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/><a href="#">Relate ratios in right triangles</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES</a> for similar readings</p> | <p>Learners complete a guided reading of a one-page excerpt (prepared by the teacher or taken from the KICD-approved textbook) about a ramp built at a Nakuru County Referral Hospital to allow wheelchair access. The ramp rises 0.9 m over a horizontal distance of 1.2 m and the ramp surface (hypotenuse) is 1.5 m. The excerpt defines the cosine of the angle of inclination as adjacent divided by hypotenuse, i.e., 1.2 divided by 1.5 = 0.8, and names the angle theta. After reading, learners open mathematical tables to the cosine pages and complete a structured table-reading worksheet: they find cos values for 10, 20, 30, 37, 45, 53, 60, 70, 80 degrees and record them alongside the sine values they already recorded in Lesson 2. They then answer two observation questions on the worksheet: (1) What happens to cosine as the angle increases? (2) For which</p> | <p>Teacher circulates and reads the excerpt aloud once with the class to model fluent reading of mathematical text. Then says: Now open your mathematical tables to the cosine section. The layout is the same as the sine section — I want complete silence for 5 minutes while you find each value and write it in your table. WAIT 5 minutes, circulating to check table-reading technique. If a learner is struggling, teacher points to the correct row and column silently. After 5 minutes: Compare your cosine column with your sine column. Answer the two observation questions on your worksheet. You have 3 minutes. WAIT 3 minutes. Cold-call three learners: What did you notice as the angle gets bigger? What angle makes sine and cosine equal? Why might that be?</p> | <p>Placing cosine values directly beside sine values in the same table is a deliberate side-by-side comparison strategy. Learners are not told about the complementary relationship — they discover that <math>\cos 53</math> approximately equals <math>\sin 37</math>, <math>\cos 30</math> approximately equals <math>\sin 60</math>, etc. The Nakuru hospital ramp provides a concrete low-technology context that is familiar (disability access ramps are present in many Kenyan public buildings) and makes the adjacent-over-hypotenuse definition tangible.</p> | <p>Teacher scans completed worksheets during circulation. Key checkpoints: Is cos listed as decreasing (from close to 1 at small angles to close to 0 at large angles)? Has the learner identified 45 degrees as the angle where sine equals cosine? Misconception to flag: some learners may read the wrong column in the cosine table (confusing mean-difference columns); teacher corrects silently by pointing to the column headers.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | angle is $\cos \theta$ equal to $\sin \theta$ ?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Solving for a side in right triangles with trigonometry</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Teacher formally defines cosine and then sets pairs to work on five structured problems. Problem contexts: (1) A wheelchair ramp at Nakuru County Referral Hospital makes an angle of 12 degrees with the ground. The ramp surface is 4.0 m long. Find the horizontal distance (<math>\cos 12</math> degrees times 4.0). (2) A telecommunications cable runs from the top of a 10 m vertical pole in Kisumu to a point on the ground, making an angle of 40 degrees with the ground. Find the length of the cable (the hypotenuse) using the adjacent side (10 m) and <math>\cos 40</math> degrees — learners must recognise that 10 m is the adjacent side to the 40-degree angle at the base. Wait — teacher pauses class to confirm the triangle orientation before continuing. (3) A solar panel on a roof in Eldoret is tilted at 25 degrees to the horizontal. If the panel is 2.4 m long (hypotenuse), find the horizontal width of the panel on the roof (adjacent). (4) A footpath in Kakamega Forest runs at 35 degrees to the horizontal for 500 m. Find the horizontal displacement. (5) A ladybird walks along a ramp of length 30 cm inclined at 48 degrees. How far does it travel horizontally? For each problem, pairs must: draw and label the right-angled triangle, identify adjacent and hypotenuse, write the cosine equation, substitute table values, and solve.</p> | <p>Teacher writes on the board: <math>\cos \theta = \text{adjacent} / \text{hypotenuse}</math>, and says: Cosine is the ratio of the side that RUNS ALONGSIDE the angle — the adjacent side — to the hypotenuse. It is constant for a given angle regardless of triangle size, exactly as sine was. Now with your partner, tackle the five problems on your worksheet. Draw the triangle FIRST — do not write an equation until you have drawn and labelled. WAIT 10 minutes for pair work, circulating actively. At the 5-minute mark, pause the class: Cold-call one pair to draw Problem 2 triangle on the board. Ask the class: Is the 10 m side the opposite or adjacent to the 40-degree angle? WAIT 20 seconds. Resolve any disagreement. Resume pair work for 5 more minutes. Teacher selects one pair to present Problem 4 solution to the class and facilitates a 2-minute whole-class check.</p> | <p>Requiring learners to draw and label the triangle before writing equations is a sensemaking protocol that prevents ratio-swapping errors (the most common mistake at this stage). The deliberate mid-activity pause on Problem 2 addresses a known difficulty: when the vertical side is given and the angle is at the base, learners often misidentify adjacent and opposite. Presenting a Kenyan forest footpath, a solar panel, and a hospital ramp in the same lesson emphasises that cosine is a universal tool used in everyday Kenyan infrastructure.</p> | <p>Teacher uses a clipboard checklist: for each pair, note whether (a) triangles are drawn and correctly labelled before equations are written, (b) the cosine equation is correctly set up (<math>\cos \theta = \text{adj}/\text{hyp}</math>), (c) table values are read correctly to 4 decimal places, and (d) arithmetic is carried through correctly. Common error to watch for: inverting the equation to <math>\text{hyp} = \text{adj} / \cos \theta</math> — teacher asks the pair to re-examine their triangle and confirm which quantity is unknown.</p> |
| <b>Driving Question Board (DQB) Creation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Each pair writes one new sticky note to add to the class Driving Question Board. The sticky note must contain: (a) a statement of                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Teacher distributes sticky notes (yellow for new questions, green for new evidence — a colour code established in Lesson 1). Says:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | The two-colour sticky-note system makes the epistemological distinction between evidence (what we observed) and questions (what                                                                                                                                                                                                                                                                                                                                                                                                                                     | Teacher reads all posted sticky notes after class. Quality indicator: the green evidence note should correctly state that cosine equals                                                                                                                                                                                                                                                                                                                                                                                                                           |

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| <p> <b>VIDEO:</b><br/> <a href="#">Solving for a side in right triangles with trigonometry</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p>                                          | <p>what cosine measures in the mango-tree scenario (the ratio of the 9.0 m shadow to the sun-ray length), and (b) the new question arising from their observation table: Is there a link between sine and cosine? Why does <math>\cos 53</math> degrees look similar to <math>\sin 37</math> degrees? Pairs post their sticky notes in a new column on the DQB labelled Cosine Evidence. The teacher then reads aloud the cumulative DQB narrative: In Lesson 1 we found that ratios in right-angled triangles are constant for a given angle. In Lesson 2 sine gave us a ratio using the opposite side. Today cosine gives us the adjacent side ratio. Something strange is appearing in our tables. That strange pattern is where we are heading next.</p>                                                                                                                                                 | <p>Write your cosine statement in green and your new question in yellow. You have 2 minutes. WAIT 2 minutes. Invite two pairs to read their sticky notes before posting. After posting, teacher stands at the DQB and explicitly connects today to the storyline: Our shadow question is getting closer to an answer. We now have TWO ratios — sine uses opposite and hypotenuse, cosine uses adjacent and hypotenuse. In Lesson 4 we will find the third ratio — tangent — and we will discover it connects sine and cosine in a beautiful way that will finally let us calculate the mango-tree height directly from the sun-angle alone.</p>                                      | <p>we still wonder) visible on the board. Narrating the cumulative storyline aloud reinforces that each lesson is a step in a coherent investigation, not isolated content, and primes learners for the tangent derivation in Lesson 4.</p>                                                                                                                                                                                                                                                         | <p>adjacent divided by hypotenuse and should give a specific numerical example from the lesson (e.g., <math>\cos 53</math> degrees is approximately 0.6018 in the mango-tree triangle). If sticky notes are vague or incorrect, teacher uses the first 2 minutes of Lesson 4 to clarify.</p>                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Solving for a side in right triangles with trigonometry</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners return to the cumulative annotated right-angled triangle diagram begun in Lesson 1 and extended in Lesson 2. In their exercise books they add the following to the mango-tree triangle diagram: (1) A label on the adjacent side (9.0 m shadow) with the notation adj. (2) The cosine definition box: <math>\cos \theta = \frac{\text{adj}}{\text{hyp}}</math>, with the calculated value <math>\cos 53</math> degrees approximately 0.6018. (3) A two-column comparison table (sine column from Lesson 2, cosine column from today) for angles 30, 45, 53, 60. (4) An arrow and question mark connecting the <math>\sin 53</math> degrees row to the <math>\cos 37</math> degrees row with the note: pattern to investigate. They also write one sentence completing the stem: In the mango-tree triangle, cosine tells us... Learners who finish early calculate the length of the sun-ray</p> | <p>Teacher says: Update your triangle diagram now. Add the adjacent label, the cosine definition box, the comparison table, and your pattern arrow. You have 4 minutes. WAIT 4 minutes. Circulate and stamp or initial three to five diagrams as a spot-check. Say to the class: Notice that your diagram now has two definition boxes — sine and cosine. Next lesson we add a third, and the three together will crack the mango-tree problem open completely. For those who finish early: use <math>\cos 53</math> degrees and the 9.0 m shadow to find the sun-ray length. Does that length match what you calculated using sine in Lesson 2? If it does not, find the error.</p> | <p>Updating the same cumulative diagram each lesson builds a physical artefact of the learners knowledge growth. Asking learners to use cosine to find the sun-ray length and cross-check with the sine result from Lesson 2 is a coherence check — both approaches must give the same hypotenuse, reinforcing that sine and cosine are complementary tools describing the same triangle. This foreshadows the Pythagorean and complementary relationships to be formalised in Lessons 6 and 7.</p> | <p>Teacher collects or photographs two or three diagrams to review after class. Criteria: (a) adjacent side correctly labelled on the 9.0 m shadow leg, (b) cosine definition box correctly written, (c) comparison table populated with reasonable 4-decimal values from tables, (d) pattern arrow present. The extension calculation should yield hypotenuse approximately equal to 9.0 divided by 0.6018 which is approximately 14.96 m, consistent with 1.6 divided by <math>\sin 53</math> degrees from Lesson 2 (approximately 14.96 m if the tree height was calculated as approximately 11.95 m and used as opposite). Teacher notes any discrepancy as a discussion seed for Lesson 4.</p> |

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|  | hypotenuse using $\cos 53^\circ$ and the 9.0 m adjacent shadow length, checking it against the value obtained from sine in Lesson 2. |  |  |  |
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### D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners correctly distinguish adjacent from opposite when setting up cosine equations, or was there persistent confusion especially in non-standard triangle orientations such as Problem 2? Note names of learners who need a brief individual conversation at the start of Lesson 4. (2) Did the side-by-side sine-cosine table reveal the complementary pattern to learners spontaneously, or did they need explicit prompting? If prompting was needed, consider whether Lesson 6 on complementary angles needs an additional bridging activity. (3) Were all five structured problems attempted by most pairs within 10 minutes, or was the pace too fast? If more than one third of pairs did not finish Problem 4 or 5, reduce the extension problem set in Lesson 4 warm-up accordingly. (4) Check the DQB sticky notes: do learners show genuine curiosity about the sine-cosine pattern, or are their questions superficial? Genuine curiosity notes indicate readiness for the tangent derivation; superficial notes suggest the pattern was not noticed and needs revisiting in the first 3 minutes of Lesson 4.

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>What did I observe?</b>                   | Learners observed that the cosine of an acute angle is a fixed ratio (adjacent divided by hypotenuse) that can be read from mathematical tables, that cosine values decrease as the angle increases, and that for each angle in their comparison table the cosine value appeared to match the sine value of a different angle — specifically the angle that adds up to 90 degrees.                                                                              |
| <b>What did I learn?</b>                     | The cosine of an acute angle $\theta$ equals the length of the adjacent side divided by the length of the hypotenuse of any right-angled triangle containing $\theta$ . Like sine, this value depends only on the angle and not on the size of the triangle. For the mango-tree scenario, $\cos 53^\circ$ is approximately 0.6018, meaning the 9.0 m shadow is approximately 0.6018 times the length of the sun-ray reaching the tree tip.                      |
| <b>How does this explain the phenomenon?</b> | Cosine, like sine, is a consistent fingerprint of an angle: it compresses the shape of any right-angled triangle with that angle into a single number. Because every right-angled triangle with a 53-degree angle has the same ratio of adjacent to hypotenuse regardless of size, a table entry is sufficient to solve for any unknown side in any such triangle anywhere in Kenya — from a hospital ramp in Nakuru to the shadow of the mango tree in Kisumu. |

## LESSON 4 (80 minutes): Tangent and the Sine–Cosine–Tangent Relationship: Solving the Mango-Tree Problem

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>Purpose</b>              | To derive the identity $\tan \theta = \sin \theta / \cos \theta$ algebraically, verify it numerically using trigonometric table values, and apply the tangent ratio to calculate the height of the mango tree from the anchoring phenomenon.                                                                                                                                                                                                                                                                                                         |
| <b>Knowledge</b>            | By the end of the lesson, the learner should be able to: (a) define the tangent ratio as Opposite/Adjacent in a right-angled triangle; (b) derive the identity $\tan \theta = \sin \theta / \cos \theta$ algebraically using the definitions of sine and cosine; (c) verify the identity numerically for $\theta = 30^\circ$ , $40^\circ$ , $60^\circ$ , and $70^\circ$ using mathematical tables; (d) apply the tangent ratio to calculate unknown heights and distances in real-life right-triangle problems.                                      |
| <b>Skills</b>               | Algebraic manipulation and symbolic reasoning; reading and interpreting trigonometric tables; numerical verification of a mathematical identity; applying SOH-CAH-TOA to a real-world measurement problem; constructing a logical mathematical argument.                                                                                                                                                                                                                                                                                             |
| <b>Attitudes</b>            | Appreciate that mathematical identities reveal deep, consistent structure in nature; show curiosity about how a single angle measurement can unlock unknown lengths; demonstrate persistence when verifying results numerically; value collaborative sense-making in mathematics.                                                                                                                                                                                                                                                                    |
| <b>Key Inquiry Question</b> | How do we apply trigonometric ratios in day-to-day activities? How are the trigonometric ratios of acute angles related to each other?                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Purpose in Storyline</b> | Lesson 4 is the pivotal evidence-gathering lesson in which the driving question is partially answered for the first time. Having established $\sin$ and $\cos$ in Lessons 1–3, learners now derive $\tan \theta = \sin \theta / \cos \theta$ , see it confirmed numerically across four angles, and — crucially — use $\tan 53^\circ$ with the 9.0 m shadow length to calculate the mango tree's height. The DQB is updated to mark this breakthrough: the class now has a working method for measuring any height using only an angle and a shadow. |
| <b>Safety Notes</b>         | No physical hazards in this lesson. If learners use printed mathematical tables, remind them to handle them carefully and not to write on shared copies. If a clinometer or protractor is passed around, ensure it is handled gently to avoid damage.                                                                                                                                                                                                                                                                                                |

### B. LESSON OVERVIEW

Learners begin by revisiting their earlier predictions about the mango-tree phenomenon and recalling the  $\sin$  and  $\cos$  definitions established in Lessons 1–3. Through guided algebraic derivation, they prove that  $\tan \theta = \sin \theta / \cos \theta$ . Working in pairs, they then verify this identity numerically for  $30^\circ$ ,  $40^\circ$ ,  $60^\circ$ , and  $70^\circ$  using mathematical tables, building confidence in the result. A whole-class sensemaking discussion consolidates understanding of why all three ratios are interconnected. The lesson reaches its climax when learners apply  $\tan 53^\circ$  and the 9.0 m shadow length to calculate the mango tree's height — the first direct, quantitative answer to the driving question. The lesson closes with a DQB update where learners record what has now been explained and what questions remain (e.g., what happens when we know the height but not the angle?).

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                | Learner Experience                                          | Teacher Moves                                                      | Sensemaking Strategy                                            | Formative Assessment Strategy                                        |
|----------------------|-------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Predict Phase</b> | Learners are shown the anchoring phenomenon scenario on the | Display the scenario on the board and read it aloud. Say: 'We have | Think-Pair-Share with public prediction display. This activates | Observable indicator: each learner has a written prediction in their |

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| <p> <b>VIDEO:</b><br/> <a href="#">Finding reciprocal trig ratios</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p>                          | <p>board: 'A student in Kisumu measures the sun angle as <math>53^\circ</math> and the mango tree's shadow as 9.0 m. She already knows <math>\sin 53^\circ \approx 0.799</math> and <math>\cos 53^\circ \approx 0.602</math> from the previous lesson. Can she calculate the tree's height — and if so, which ratio will she need?' Each learner writes a silent prediction in their exercise book (2 minutes): (a) which trigonometric ratio they think is most useful and why; (b) a numerical estimate of the tree's height. Learners then share predictions in a think-pair-share with their desk partner before three pairs are cold-called to share with the class. Predictions are pinned to the Driving Question Board under the heading 'Our Predictions — Lesson 4' without correction.</p> | <p>been gathering evidence about sin and cos for three lessons. Today we are going to find out whether there is a third ratio hiding in the triangle — one that connects the angle directly to the tree's height and shadow. But first, I want to know what YOU think.' Instruct learners to write silently for 2 minutes — no talking. After 2 minutes say: 'Now turn to your partner and compare your prediction. You have 90 seconds.' After sharing, cold-call exactly 3 pairs (choose a mix of confident and hesitant learners): 'Pair at the window bench — what ratio did you predict and why?' Allow a WAIT TIME of 12 seconds after each question before prompting. Record key words from each response on the board (e.g., 'Opposite/Adjacent', 'sin/cos', 'SOH-CAH-TOA'). Do NOT confirm or deny any prediction. Say: 'Hold those ideas — by the end of the lesson you will be able to judge your own prediction.'</p> | <p>prior knowledge of SOH-CAH-TOA and similar triangles from Lessons 1–3, creates cognitive investment in the outcome, and gives the teacher immediate formative data about whether learners are connecting ratio definitions to the physical scenario.</p>                                                                                                                                                                                                | <p>book before discussion begins. During cold-calling, listen for whether learners spontaneously name 'Opposite/Adjacent' or attempt to form sin/cos — this reveals readiness for the derivation. Flag any learner who says 'we cannot find the height' for targeted support during Phase 2.</p>                                                                                                                                                                                                                                                                                                    |
| <p><b>Observe Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Trigonometry review</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Right triangle trigonometry word problems</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>PART A — Algebraic Derivation (15 min): The teacher leads a guided derivation at the board, pausing at each step for learners to copy and annotate. Starting from <math>\sin \theta = O/H</math> and <math>\cos \theta = A/H</math>, learners are guided to form the ratio <math>\sin \theta / \cos \theta = (O/H) \div (A/H) = (O/H) \times (H/A) = O/A = \tan \theta</math>. Learners write each step in their books and annotate with arrows showing cancellation of H. They also write the result: <math>\tan \theta = \sin \theta / \cos \theta</math>, and beneath it: <math>\tan \theta = \text{Opposite} / \text{Adjacent}</math> (TAO — a memory aid introduced by the teacher).</p>                                                                                                      | <p>PART A: Write the derivation step-by-step on the board using large, clearly labelled fractions. At each cancellation step, pause and ask: 'What happens to the H in the numerator and the H in the denominator? Tell your partner in one word.' WAIT TIME 10 seconds. Cold-call 2 learners. After completing the derivation say: 'This is not a coincidence — it follows directly from the definitions we built in Lessons 1 and 2. Underline the identity in your book and write next to it: THIS IS ALWAYS TRUE FOR ANY ACUTE ANGLE.'</p>                                                                                                                                                                                                                                                                                                                                                                                    | <p>Algebraic derivation followed by numerical verification (a 'Proof then Check' strategy). The derivation builds deductive reasoning — learners see WHY the identity must be true. The numerical verification builds empirical confidence and table-reading fluency. Separating these two activities prevents learners from treating the identity as a mere formula to memorise and instead positions it as a provable, checkable mathematical truth.</p> | <p>Collect and scan 4–5 verification tables during circulation. Observable indicators of mastery: (1) all four angle rows completed with correct table readings; (2) column 4 values match column 5 to 3–4 significant figures; (3) the written conclusion uses causal language ('because', 'therefore', 'it follows that') rather than merely descriptive language ('they are the same'). Red-flag indicator: a learner who has correct sin and cos values but an incorrect ratio — this signals a division error, not a conceptual gap, and can be addressed with a brief place-value prompt.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>PART B — Numerical Verification (20 min): Learners work in pairs. Each pair receives (or opens to) a mathematical tables page. Using the table, they complete a structured verification table for <math>\theta \in \{30^\circ, 40^\circ, 60^\circ, 70^\circ\}</math>: Column 1: <math>\theta</math>; Column 2: <math>\sin \theta</math> (from tables); Column 3: <math>\cos \theta</math> (from tables); Column 4: <math>\sin \theta \div \cos \theta</math> (calculated by long division or calculator); Column 5: <math>\tan \theta</math> (from tables directly); Column 6: Do columns 4 and 5 match? (Yes/No + comment). For example, at <math>40^\circ</math>: <math>\sin 40^\circ = 0.6428</math>, <math>\cos 40^\circ = 0.7660</math>, ratio = <math>0.6428/0.7660 \approx 0.8391</math>, <math>\tan 40^\circ = 0.8391</math> — match confirmed. Learners write a one-sentence conclusion at the bottom: 'The identity <math>\tan \theta = \sin \theta / \cos \theta</math> holds for all four angles tested, confirming that the three ratios are not independent — knowing two allows you to find the third.'</p> | <p>PART B: Circulate continuously during pair work. Stop at each bench for approximately 45 seconds. Targeted prompts: (i) 'Show me how you are dividing 0.6428 by 0.7660 — walk me through it.' (ii) 'Your column 4 and column 5 are slightly different — is the difference large or small? What might cause a tiny rounding difference in tables?' (iii) 'What does a match in every row tell us about the identity?' After 18 minutes, call time: 'Pens down. I want to hear from three pairs — one for <math>30^\circ</math>, one for <math>60^\circ</math>, one for <math>70^\circ</math>.' Cold-call those three pairs. Record their numbers on the board in a class table. Say: 'Does everyone agree with these values? Raise your hand if your table matches.' Scan the room — address any mismatches immediately.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p><b>Explain Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Solving for a side in right triangles with trigonometry</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>PART A — Sensemaking Discussion (10 min): The teacher poses three discussion questions in sequence, giving learners 30 seconds of silent thinking before each open discussion. Q1: 'We now have three ratios — <math>\sin</math>, <math>\cos</math>, and <math>\tan</math> — but we just proved that <math>\tan</math> is not independent of <math>\sin</math> and <math>\cos</math>. Does that mean <math>\tan</math> is unnecessary?' Q2: 'The mango-tree shadow is 9.0 m and the sun angle is <math>53^\circ</math>. Which side of the right triangle is 9.0 m — the opposite or the adjacent? How do you know?' Q3: 'If <math>\tan \theta = \text{Opposite/Adjacent}</math>, and you know the adjacent side and the angle, how would you rearrange the formula to find the opposite side?' Learners discuss in groups of four, then representatives share. The</p>                                                                                                                                                                                                                                                     | <p>Q1 prompt: 'Think silently for 30 seconds — is <math>\tan</math> redundant? Write one word: YES or NO.' WAIT TIME 30 seconds. Then: 'Tell your group.' After 60 seconds of group talk, cold-call 2 groups. Guide toward the key insight: 'Tan is not redundant — it is CONVENIENT. When we know an angle and one side but NOT the hypotenuse, <math>\tan</math> gives us a direct route.' Write 'CONVENIENT SHORTCUT' on the board next to the identity.</p> <p>Q2: Draw the mango-tree scenario as a right triangle on the board — ground (9.0 m), tree (h), hypotenuse (sun ray). Label and ask: 'Point to the side opposite the <math>53^\circ</math> angle. Point to the side adjacent to it.' Cold-call 2 learners</p>                                                                                                 | <p>Structured Questioning with Diagram Annotation (a 'press for reasoning' strategy). By asking learners to identify sides before writing any equation, the teacher prevents the common error of confusing opposite and adjacent. The parallel calculation using similar triangles serves as a self-consistency check — a scientific practice of triangulating evidence from two independent methods, directly mirroring how Kenyan surveyors cross-check measurements.</p> | <p>Observable indicators: (1) each learner's book shows the full worked calculation <math>h = 9.0 \times \tan 53^\circ = 11.94</math> m with correct table reading; (2) learners who complete the cross-check obtain <math>h \approx 12.0</math> m and write a comparison sentence; (3) during Q1 discussion, at least 70% of groups spontaneously use the word 'convenient' or an equivalent when justifying <math>\tan</math>'s value. Flag learners who obtain <math>h = 9.0 / 1.327</math> (inverted ratio) — this is a persistent misconception requiring immediate one-to-one clarification.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>teacher scribes key algebraic steps on the board: <math>\tan 53^\circ = h / 9.0 \rightarrow h = 9.0 \times \tan 53^\circ</math>.</p> <p>PART B — Mango-Tree Calculation (10 min): Each learner independently works the calculation in their book: <math>\tan 53^\circ</math> (from tables) <math>\approx 1.3270</math>; <math>h = 9.0 \times 1.3270 = 11.943 \text{ m} \approx 11.94 \text{ m}</math>. They also perform the shadow-ratio cross-check from earlier lessons: the student's height (1.6 m) / her shadow (1.2 m) = 1.333; tree height / tree shadow = <math>h / 9.0</math>. Cross-multiplying: <math>h = 9.0 \times (1.6/1.2) = 12.0 \text{ m}</math>. The two methods agree to within rounding. Learners write a conclusion: 'The mango tree is approximately 11.94 m – 12.0 m tall. Both the similar-triangles method and the tangent ratio give consistent answers, confirming that <math>\tan \theta</math> captures the same relationship as the ratio of heights to shadows.'</p> | <p>to come to the board and label the diagram.</p> <p>Q3: Write <math>\tan 53^\circ = h / 9.0</math> on the board. Ask: 'How do I get h alone? Someone give me the next line.' WAIT TIME 12 seconds. Cold-call 1 learner. Write <math>h = 9.0 \times \tan 53^\circ</math>.</p> <p>During independent calculation: circulate and check that learners are reading <math>\tan 53^\circ</math> correctly from tables (common error: reading sin column instead of tan column). After 8 minutes, ask a learner to read their answer aloud and walk the class through the arithmetic on the board. Emphasise: 'We answered the driving question today — at least partially. Let that sink in.'</p>   |                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b> <a href="#">Intro to the trigonometric ratios</a></p> <p>Source: Khan Academy (Kiswahili)</p> <p> <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b> <a href="#">Trigonometric ratios in right triangles</a></p> <p>Source: Khan Academy (English - US curriculum)</p> <p> <a href="#">Search ARES</a></p> | <p>The teacher returns the class's attention to the Driving Question Board: 'How can the angle a sunbeam makes with the ground allow us to calculate the height of any tree, building, or hill in Kenya without measuring it directly?' Each learner writes a sticky note (or a short entry in their DQB section of their book) responding to three prompts: (1) ANSWERED TODAY: 'We can now calculate the mango tree's height because...'; (2) STILL WONDERING: one question that the lesson raised but did not resolve (e.g., 'What if we know the height but not the angle — can we work backwards?'; 'Does tan work for obtuse angles too?'; 'How do</p>                                                                                                                                                                                                                                                                                                                                            | <p>Point to the driving question written at the top of the DQB. Say: 'Three lessons ago, nobody in this room could answer this question. Today, something changed. I want you to write exactly what changed — what can you now do that you could not do before?' Allow 3 minutes of silent writing. Circulate and read over shoulders — prompt hesitant learners with: 'You just calculated 11.94 m. How did you do that? Write that down.' After sticky notes are placed, read 2 'Answered Today' notes and 2 'Still Wondering' notes aloud. Say: 'These Still Wondering notes are exactly where Lesson 5 will begin. You have just written the next lesson's starting point.' Circle the</p> | <p>Three-Column DQB Protocol (Answered / Still Wondering / Connections). This metacognitive strategy makes progress visible and tangible — learners literally see the board filling up with answered questions over the 10-lesson sequence. The 'Connections' column builds the habit of cross-lesson synthesis, which is central to NGSS Practice 6 (Constructing Explanations) and to the KICD competency of critical thinking.</p> | <p>Observable indicators: (1) 'Answered Today' notes reference <math>\tan \theta</math>, the <math>53^\circ</math> angle, and/or the 9.0 m shadow — not vague statements like 'we learned trigonometry'; (2) 'Still Wondering' notes pose genuine, answerable mathematical questions rather than personal opinions; (3) 'Connections' notes reference a specific earlier lesson by number or content (e.g., 'similar triangles from Lesson 1', 'sin definition from Lesson 2'). Collect and photograph the DQB at the end of the lesson for the teacher's formative record.</p> |

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| <p><a href="#">for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>surveyors in the Rift Valley use this for hills?'); (3) CONNECTED TO EARLIER LESSONS: 'Today's tan ratio connects to Lessons 1–3 because...' Sticky notes are placed on the DQB in three columns: Answered / Still Wondering / Connections. The teacher reads 4–5 notes aloud and uses them to preview Lesson 5 (inverse trigonometric ratios / working backwards from a known height).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>most common 'Still Wondering' question on the board and leave it there for the start of Lesson 5.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b> <a href="#">Proof of the tangent angle sum and difference identities</a></p> <p>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b> <a href="#">Trigonometric ratios in right triangles</a></p> <p>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES for similar readings</a></p> | <p>Learners open their cumulative 'Trigonometry Model' diagram — a right-angled triangle with the sun-ray scenario that they have been annotating across Lessons 1–4. Today they add three new elements: (1) Label the angle at the base as <math>\theta = 53^\circ</math>, the adjacent side as 9.0 m (shadow), and the opposite side as h (tree height). Write '<math>\tan 53^\circ = h / 9.0</math>' inside the triangle. (2) Add a 'Ratio Web' annotation beside the triangle: a small diagram showing three nodes — <math>\sin \theta</math>, <math>\cos \theta</math>, <math>\tan \theta</math> — connected by the identity <math>\tan \theta = \sin \theta / \cos \theta</math>, with arrows and labels showing the algebraic relationship. (3) Write a 'Model Statement' below the diagram: 'For any right-angled triangle, if I know one acute angle and the length of either the adjacent or opposite side, I can use <math>\tan \theta = O/A</math> to find the unknown side. This works because the ratio <math>O/A</math> is constant for a given angle, regardless of the triangle's size — just as Kisumu's sun angle gives the same ratio for every shadow at the same time of day.' Learners share their updated model with a neighbour and identify one similarity and one difference between their models</p> | <p>Distribute (or direct learners to open) their cumulative model sheet. Say: 'Your model has grown across four lessons. Today it gets three important upgrades — the angle label, the ratio web, and the model statement. These are not decoration — they are evidence that your understanding has grown.' Walk around and check that every learner has the ratio web correctly drawn (the most common omission is the arrow direction). After 8 minutes of model work, say: 'Show your model to the person next to you. Find one thing that is the same and one thing that is different. You have 2 minutes.' Cold-call 2 pairs to share a difference they noticed — use this to surface any remaining misconceptions about the direction of the identity (e.g., '<math>\tan = \sin/\cos</math>' vs '<math>\sin = \tan \times \cos</math>'). Close by saying: 'Next lesson we will ask: what if you know the height and the base but NOT the angle? Can you work backwards? That is the question on our DQB right now — and it has an answer.'</p> | <p>Cumulative Annotated Model Revision (NGSS Practice 2: Developing and Using Models). By returning to the same diagram each lesson and adding new layers, learners build a coherent, growing representation rather than isolated lesson-by-lesson notes. The 'Ratio Web' annotation makes the algebraic identity visible as a structural relationship, not just a formula — supporting KICD's emphasis on conceptual understanding over rote memorisation.</p> | <p>Observable indicators: (1) the ratio web correctly shows <math>\tan \theta = \sin \theta / \cos \theta</math> with labelled arrows; (2) the triangle diagram has all three sides and the angle correctly labelled relative to the <math>53^\circ</math> scenario; (3) the model statement uses conditional language ('if I know... then I can...') indicating procedural fluency with conceptual grounding. Exit-ticket equivalent: collect 5–6 model sheets at random as learners leave — photograph the model statement section and review before Lesson 5 to identify any persistent confusion about which side is opposite vs adjacent.</p> |

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|  | before the lesson closes. |  |  |  |
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#### D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) DERIVATION PACING — Did all learners follow the algebraic cancellation of  $H$  in the derivation, or did some copy without understanding? If more than 30% of books show errors in the derivation steps, consider opening Lesson 5 with a five-minute re-derivation using a numerical example (e.g., a 3-4-5 triangle) before moving to inverse ratios. (2) TABLE-READING ACCURACY — During circulation in Phase 2, how many pairs read the wrong column (e.g.,  $\sin$  instead of  $\tan$ )? If this was common, consider creating a colour-coded table-reading guide as a classroom resource. (3) MANGO-TREE CALCULATION — Did learners successfully obtain  $h \approx 11.94$  m, and did those who did the cross-check connect it to the similar-triangles result ( $\approx 12.0$  m)? The cross-check is a critical thinking move — if most learners skipped it, explicitly model the habit of 'checking your answer using a different method' at the start of Lesson 5. (4) DQB QUALITY — Were the 'Still Wondering' notes mathematically productive? If most notes were vague ('I want to know more about angles'), use the first 3 minutes of Lesson 5 to model a high-quality wondering note using a Kisumu land-surveying context. (5) EQUITY — Were hesitant learners included in cold-calls? Review your cold-call record: did you call on learners from all seating areas, including the back benches?

#### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>What did I observe?</b>                   | Learners observed the algebraic derivation showing that dividing $\sin \theta$ by $\cos \theta$ (each defined using the hypotenuse) causes the hypotenuse to cancel, yielding Opposite/Adjacent — the tangent ratio. They also observed numerical agreement between $\sin \theta / \cos \theta$ and $\tan \theta$ (from tables) for $\theta = 30^\circ, 40^\circ, 60^\circ$ , and $70^\circ$ , confirming the identity across four distinct angles.                                          |
| <b>What did I learn?</b>                     | Learners learned that $\tan \theta = \text{Opposite} / \text{Adjacent}$ and that this ratio is algebraically equivalent to $\sin \theta / \cos \theta$ , making all three trigonometric ratios interdependent. They learned that knowing an acute angle and the length of the adjacent side (e.g., a shadow) is sufficient to calculate the opposite side (e.g., a tree's height) using $h = \text{adjacent} \times \tan \theta$ .                                                           |
| <b>How does this explain the phenomenon?</b> | Learners explained why the mango tree in the Kisumu school compound is approximately 11.94 m tall: because the sun angle of $53^\circ$ and the 9.0 m shadow form a right triangle in which $\tan 53^\circ = \text{tree height} / \text{shadow length}$ , giving $h = 9.0 \times 1.3270 \approx 11.94$ m. This result is consistent with the similar-triangles cross-check ( $h \approx 12.0$ m), confirming that the tangent ratio is a reliable, angle-based tool for indirect measurement. |

## LESSON 5 (80 minutes): Trigonometric Ratios Using Calculators: Precision, Inverse Functions, and the Shadow Puzzle

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>Purpose</b>              | To enable learners to use scientific calculators to find sine, cosine, and tangent values of acute angles, to use inverse trigonometric functions to find angles given a ratio, and to evaluate the precision of their earlier hand-drawn triangle measurements by computing percentage error — connecting technological tools to the anchoring phenomenon of shadow-based height calculation.                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Knowledge</b>            | Learners will know: (1) how to correctly operate the sin, cos, and tan keys on a scientific calculator in degree mode; (2) how to use the inverse functions $\sin^{-1}$ , $\cos^{-1}$ , and $\tan^{-1}$ to find an unknown angle from a known ratio; (3) how to compute percentage error between a measured/estimated value and a calculator-derived value; (4) that the calculator values of trigonometric ratios are consistent with the ratios derived from hand-drawn similar triangles in earlier lessons.                                                                                                                                                                                                                                                                                      |
| <b>Skills</b>               | Learners will be able to: (1) set a scientific calculator to degree mode and accurately evaluate $\sin \theta$ , $\cos \theta$ , and $\tan \theta$ for any acute angle; (2) apply inverse trigonometric functions to recover an angle from a given ratio; (3) calculate percentage error between their Lesson 1 hand-drawn triangle ratio estimates and calculator-confirmed values; (4) use the calculator to verify that $\tan 53^\circ \approx 1.327$ and hence confirm the estimated height of the mango tree in the Kisumu phenomenon.                                                                                                                                                                                                                                                          |
| <b>Attitudes</b>            | Learners will appreciate: (1) the value of precision and integrity — honestly reporting percentage errors rather than adjusting data; (2) that technological tools (calculators) extend human capability but must be used with understanding of the underlying mathematics; (3) that the consistent, reliable values produced by the calculator reflect the deep, universal pattern hidden in similar triangles that the anchoring phenomenon pointed to.                                                                                                                                                                                                                                                                                                                                            |
| <b>Key Inquiry Question</b> | How do we apply trigonometric ratios in day-to-day activities? How are the trigonometric ratios of acute angles related to each other?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Purpose in Storyline</b> | By Lesson 5 the class has defined SOH CAH TOA, derived special-angle ratios from 30-60-90 and 45-45-90 triangles, and confirmed that similar triangles give consistent ratios. But all of this work has relied on hand-drawn diagrams and approximate measurements. Today learners bring the scientific calculator into the investigation for the first time, discovering that the calculator returns values that match — but are more precise than — their own hand-drawn results. They compute percentage error, confront their own measurement limitations with honesty, and crucially use $\tan^{-1}$ to work backwards: given the $53^\circ$ sun angle from the Kisumu clinometer, they confirm the tree height calculation, advancing the driving question answer with quantitative precision. |
| <b>Safety Notes</b>         | No physical hazards. Remind learners to handle school calculators with care (Responsibility value). Ensure calculators are returned to the storage box after the lesson. Learners who do not have access to a physical calculator may use the ARES offline calculator tool on their device.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

### B. LESSON OVERVIEW

In this 80-minute lesson, learners transition from hand-constructed geometric reasoning to calculator-assisted computation of trigonometric ratios. The lesson opens by revisiting the Kisumu mango tree phenomenon and asking learners to predict what value their calculator will give for  $\tan 53^\circ$ . Learners then conduct a structured calculator investigation worksheet, evaluating sin, cos, and tan for a range of acute angles and using inverse functions to recover angles from given ratios. A critical sensemaking activity asks each learner to retrieve their Lesson 1 hand-drawn triangle ratio estimates, compare them to calculator values, and honestly compute the percentage error — directly linking to the integrity value mandated by KICD. The lesson closes with a model revision in which learners update their cumulative right-triangle diagram to include calculator-verified values, and the Driving Question Board is annotated with a precision note: 'The calculator confirms our similar-triangle ratios — and gives us more decimal places.' The phenomenon is resolved one step further: learners confirm that  $\tan 53^\circ \times 9.0 \text{ m} \approx 11.9 \text{ m}$ , a refined estimate of the

mango tree's height.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                                                                     | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">The trig functions &amp; right triangle trig ratios</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Trigonometric ratios in right triangles</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | Learners sit in their existing groups of four. Each learner silently reads the phenomenon prompt displayed on the board or ARES screen: 'In Lesson 1 you drew right-angled triangles and estimated $\tan 53^\circ$ using your ruler. Write down: (a) the value you got, (b) what you think a scientific calculator will give for $\tan 53^\circ$ , and (c) whether you think the calculator will give exactly the same answer for every student in the class — and why.' Learners write individual predictions in their exercise books for 4 minutes before any discussion. A think-pair-share follows: each learner shares with their shoulder partner, then two pairs are cold-called to share with the class. The teacher records three or four predicted values for $\tan 53^\circ$ on the board without correcting them, creating a 'prediction wall' to be revisited after the Observe phase. | Display the prompt on the board or ARES screen before learners enter the room so it is visible immediately. Say: 'Before you touch a calculator, I want YOUR brain's best guess. Look at the triangle you drew in Lesson 1 and use it.' Allow 4 minutes of silent individual writing — enforce silence with a visible timer. After 4 minutes say: 'Turn to your shoulder partner. You have 90 seconds — share your prediction and your reason. Go.' After 90 seconds, cold-call exactly 3 pairs using the ARES random-name selector or a name stick: 'Achieng, what did you and your partner predict — and why?' WAIT 12 seconds after each cold-call before prompting. Record each predicted value on the board under the heading 'OUR PREDICTIONS FOR $\tan 53^\circ$ '. Do not evaluate or correct. Say: 'We will come back to these predictions after we investigate. Keep your Lesson 1 triangle work on your desk — you will need it.' | Individual prediction writing followed by Think-Pair-Share. This strategy activates prior knowledge from Lessons 1–4 (hand-drawn triangles, SOH CAH TOA, special angles) and surfaces misconceptions about calculator precision versus geometric estimation. Recording predictions publicly creates cognitive dissonance motivation: learners are intrinsically motivated to check whether their prediction was correct. | Observable indicator: every learner has a written numerical prediction in their book before the pair discussion begins. Teacher circulates during the 4-minute silent write and notes (without correcting) any learners who have left the prediction blank or written 'I don't know' — these learners are prioritised for targeted support in Phase 2. Listen during cold-calls for whether learners reference the ratio opposite/adjacent from their Lesson 1 triangle: this confirms retention of SOH CAH TOA from Lesson 2. |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Finding reciprocal trig ratios</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a>                                                                                                                                                                                                                                                                                                                                                                               | Learners receive the structured 'Calculator Investigation Worksheet' (printed or loaded on ARES). The worksheet has four sections: (A) Setting up — learners verify their calculator is in degree mode by checking that $\sin 90^\circ = 1.000$ ; (B) Forward ratios — learners evaluate and record $\sin$ , $\cos$ , and $\tan$ for the angles $20^\circ$ ,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Distribute or open the worksheet. Say: 'Step one — before you do anything else, check your calculator is in degree mode. Press $\sin$ , then 9, then 0, then equals. If you do not get 1.0000, raise your hand.' Circulate for 2 minutes to resolve mode errors. Once all calculators are confirmed, say: 'Complete Sections A and B.'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Structured data collection with a comparison protocol. The side-by-side table (Section B) makes the pattern across angles visible: learners can see $\sin$ increasing as the angle increases, $\cos$ decreasing, and $\tan$ growing steeply toward $90^\circ$ . The partner-comparison step builds collaborative verification habits.                                                                                    | Observable indicators: (1) All learners can confirm degree mode independently — any learner who cannot is flagged for targeted support on calculator operation. (2) In Section B, the value for $\tan 53^\circ$ should be 1.3270 (4 d.p.) — teacher spot-checks this cell specifically. (3) In Section C, learners must name the correct                                                                                                                                                                                       |

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| <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>30°, 45°, 53°, 60°, 70°, and 89°, writing answers to 4 decimal places in a table; (C) Inverse functions — learners are given four ratios (0.5000, 0.7071, 0.8192, 1.3270) and must use <math>\sin^{-1}</math>, <math>\cos^{-1}</math>, or <math>\tan^{-1}</math> to identify the angle, recording which function they used and why; (D) Phenomenon check — learners use their calculator result for <math>\tan 53^\circ</math> to recompute the height of the Kisumu mango tree (shadow = 9.0 m) and record a calculator-precision answer. After completing the worksheet individually, learners compare their tables with a partner and flag any discrepancies before the whole-class debrief.</p> | <p>You have 12 minutes. Work individually — your own data, your own table.' Set a visible 12-minute timer. Circulate using the 'listen and note' technique: crouch beside learners and silently observe their key-press sequences rather than their written answers — this reveals procedural errors (e.g., pressing <math>\sin</math> AFTER the number rather than before on some calculator models). At the 12-minute mark, say: 'Pause. Compare your Section B table with your partner. If any value differs, work out whose is correct before moving on — one minute.' Then say: 'Now Section C — inverse functions. These work backwards. Given a ratio, find the angle. Try each one, record which inverse key you used and why.' Allow 8 minutes for Section C, cold-calling 2 learners to show their key-press sequence on a class calculator projected under a document camera or shared screen. For Section D say: 'Use your calculator value for <math>\tan 53^\circ</math> and the 9.0 m shadow from our mango tree in Kisumu. Write the full calculation. This is the phenomenon we have been investigating for five lessons.' Allow 5 minutes. Cold-call one learner to read their answer aloud: 'Otieno, what height did you get for the mango tree? Walk us through your calculation.' WAIT 15 seconds.</p> | <p>Section D anchors the abstract calculator exercise back to the physical phenomenon, preventing the activity from becoming disconnected key-pressing.</p>                                                                                                  | <p>inverse function (e.g., 'I used <math>\tan^{-1}</math> because I was given opposite/adjacent') — this is the key conceptual indicator distinguishing procedural button-pressing from understanding. (4) In Section D, the accepted answer is <math>\text{height} = \tan 53^\circ \times 9.0 \approx 1.3270 \times 9.0 \approx 11.94 \text{ m}</math> — any answer outside 11.8–12.0 m prompts teacher follow-up.</p> |
| <p><b>Explain Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Intro to the trigonometric ratios</a><br/> Source: Khan Academy (Kiswahili)<br/>  <a href="#">Search ARES</a></p>               | <p>Learners now carry out the 'Percentage Error Investigation' — the integrity-linked sensemaking activity. Each learner retrieves their Lesson 1 hand-drawn triangle worksheet and finds the ratio they recorded for <math>\tan 53^\circ</math> (or the closest angle they measured). They apply</p>                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Say: 'Get out your Lesson 1 triangle worksheet. Find your measured value for <math>\tan 53^\circ</math> or the tangent of the angle closest to <math>53^\circ</math>. Write it at the top of a new page.' Give 2 minutes. Then say: 'Now I want you to be completely honest. Use this formula' — write on</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Gallery of data with class frequency distribution and collaborative explanation writing. Building the class-wide percentage error tally transforms individual errors into a shared data set, making the concept of measurement precision visible at a</p> | <p>Observable indicators: (1) Every learner has a written percentage error value — blank cells indicate the learner could not locate or did not complete Lesson 1 work (teacher flags for follow-up). (2) Group sticky-note explanations should reference measurement</p>                                                                                                                                               |

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| <p><a href="#">for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>the percentage error formula: <math>\text{Percentage Error} = \frac{ \text{Calculator Value} - \text{Hand-drawn Value} }{\text{Calculator Value}} \times 100\%</math>. Learners record their individual percentage error and then the teacher collects a class data set by asking each learner to call out their percentage error while a class frequency tally is built on the board. Groups then discuss: 'Why do different students have different percentage errors? What does this tell us about hand-drawn measurement versus the calculator?' Each group writes a two-sentence group explanation on a sticky note or ARES text card. The class reads out the sticky notes and the teacher facilitates a whole-class discussion to arrive at the shared explanation: the calculator uses a mathematically exact definition of the trigonometric function (ultimately rooted in the unit circle), whereas hand-drawn measurements are limited by the precision of the ruler and protractor. Learners then return to the driving question: 'If we used hand-drawn estimates to calculate tree heights in Kenya, how far off could we be?'</p> | <p>board: 'Percentage Error = <math>\frac{ \text{Calculator} - \text{Yours} }{\text{Calculator}} \times 100\%</math>' — 'Calculate YOUR percentage error. Do not change your Lesson 1 value. Integrity means reporting what you actually measured.' Allow 4 minutes. Then say: 'I am going to go around the room. When I point to you, call out your percentage error — just the number. I will tally on the board.' Record a frequency distribution on the board (e.g., 0–2%, 2–5%, 5–10%, &gt;10%). After collecting data from all learners say: 'Look at this spread. Why are we all different? Discuss in your group for 3 minutes and write your two-sentence explanation on your sticky note.' Cold-call 3 groups to read their sticky notes. WAIT 12 seconds after each reading. Consolidate by saying: 'So the calculator is not magic — it is applying a precise mathematical definition. Your hand-drawn measurements were limited by your tools. Both are valid ways of knowing — but they have different precision.' Connect to phenomenon: 'If Achieng used her hand-drawn <math>\tan 53^\circ</math> estimate to calculate the mango tree height and got, say, 11.4 m instead of 11.94 m, she'd be off by about half a metre. For a building in Nairobi, that error could matter. This is why engineers and surveyors use calculators.'</p> | <p>population level. The sticky-note explanation protocol (Claim-Evidence-Reasoning lite) requires learners to articulate the conceptual reason for the variation, not just observe it. Connecting the error magnitude to the real-world consequence (building height in Nairobi) grounds the abstraction in authentic stakes.</p> | <p>tools (ruler, protractor) as the source of imprecision — explanations that say only 'we made mistakes' without specifying the tool limitation are incomplete and prompt teacher probing: 'What specifically limited your measurement?' (3) During cold-calls, listen for learners who articulate that the calculator value is more reliable because it is not dependent on the quality of the diagram — this is the target understanding for this phase.</p> |
| <p><b>Driving Question Board (DQB) Creation</b><br/>  <b>VIDEO:</b><br/> <a href="#">Trigonometric ratios in right triangles</a></p>                                                                                                                                                                                      | <p>Each group returns to the class Driving Question Board (a large poster or ARES digital board organised by lesson). The board has columns: 'What we predicted', 'What we observed/found', 'What we still wonder'. Groups add two contributions: (1) a 'PRECISION</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Say: 'It is time to update our Driving Question Board. Every group needs to add at least two cards in the next 5 minutes — one precision note and one new question your investigation raised today.' Distribute sticky notes or direct learners to the ARES DQB</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Driving Question Board update with precision annotation. The DQB is a running public record of the class's growing understanding. Adding a precision note formalises the epistemological shift that occurred today: learners now understand that there are two</p>                                                              | <p>Observable indicators: (1) Every group contributes at least one precision note that correctly states a calculator value to 3 or more decimal places — groups whose note contains an incorrect value are gently probed: 'How did you get that value? Can you check it</p>                                                                                                                                                                                     |

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| <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                                     | <p>NOTE' card stating: 'Calculator values of sin, cos, tan are more precise than hand-drawn estimates. <math>\tan 53^\circ = 1.3270</math> (4 d.p.), not <math>\sim 1.25</math> from our ruler.' (2) a 'NEW QUESTION' card if any arose — for example: 'Can the calculator find angles larger than <math>90^\circ</math>?', 'What if the ground is not flat — like a hill in the Rift Valley?', or 'Why does the calculator give the same value for <math>\tan 53^\circ</math> no matter what size triangle you draw?' The teacher facilitates a 5-minute whole-class DQB review, reading out precision notes and new questions, and explicitly linking them to upcoming lessons.</p>                                                                                                                                                                                                      | <p>tool. Circulate and prompt groups who are stuck: 'What surprised you most today?' or 'What does the calculator NOT tell you that you still need to figure out?' After 5 minutes, read out each group's precision note, pausing to say: 'Do we all agree with this note? Thumbs up if yes, sideways if unsure.' For any 'sideways' thumbs, ask: 'What would you add or change?' Then read the new questions. Say: 'These are excellent questions. Look at your question — 'What if the ground is not flat, like a slope in the Rift Valley?' We will come back to that when we study angles of elevation and depression. Hold that question.' Explicitly name which future lesson addresses each new question so learners see the storyline arc.</p>                                                                                                | <p>ways to obtain trigonometric ratios (geometric construction and calculator), and that these converge but differ in precision. The 'new questions' column sustains inquiry momentum and prevents premature closure, keeping learners epistemically engaged with the phenomenon.</p>                                                                                                                                                                                                                                                                                                            | <p>on your calculator now?' (2) New questions should be scientifically or mathematically generative — surface-level questions ('Is there a <math>\tan 91^\circ</math>?) are welcomed but the teacher nudges toward application questions ('How would we use tan if the ground is sloped?') that connect to the driving question. (3) Thumbs-up consensus check on precision notes gives the teacher a rapid whole-class comprehension scan.</p>                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p><b>Model Building Phase</b></p> <p><b>VIDEO:</b><br/> <a href="#">Trigonometric ratios in right triangles - dubbed(KY)</a></p> <p>Source: Khan Academy (Kiswahili)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p> | <p>Learners open their cumulative right-triangle model — a labelled diagram that has been revised in each of the five lessons so far. Today's revision adds three elements: (1) a 'calculator value' annotation alongside every previously hand-estimated ratio, written in a different colour pen or font tag on ARES (e.g., '<math>\tan 53^\circ = 1.3270</math> ✓ calculator'); (2) an inverse-function arrow showing the reverse relationship — annotated as 'Given ratio <math>\rightarrow</math> use <math>\sin^{-1}/\cos^{-1}/\tan^{-1}</math> to find angle <math>\theta</math>'; (3) a completed numerical example using the Kisumu phenomenon: a mango tree diagram with shadow 9.0 m, angle <math>53^\circ</math>, and the computed height 11.94 m labelled. Learners write a one-paragraph 'Model Explanation' beneath their diagram explaining in their own words how the</p> | <p>Say: 'Take out your cumulative model from your ARES portfolio or exercise book. Today we make three additions — use a different colour so you can see what was added in Lesson 5.' Project a model template on the board showing where to place each of the three additions. Say: 'First — go through your ratio table and write the calculator value next to each hand-drawn estimate. Use four decimal places.' Allow 4 minutes. Then say: 'Second — add the inverse-function arrow. Label it: given ratio <math>\rightarrow</math> angle. Write one example using the Kisumu angle: <math>\tan^{-1}(1.3270) = 53^\circ</math>.' Allow 3 minutes. Then say: 'Third — draw the mango tree scenario if you haven't already, and label the height as 11.94 m, confirmed by calculator.' Allow 3 minutes. Finally say: 'Write your one-paragraph</p> | <p>Cumulative model revision with colour-coded layering and written explanation. Colour-coding the Lesson 5 additions makes the lesson's contribution to the evolving model visually distinct, helping learners track the development of their own understanding over the unit. The written paragraph is a miniature science/mathematics writing task (Claim-Evidence-Reasoning structure) that forces synthesis: learners must connect the calculator tool (today's evidence) to the overarching driving question, preventing the lesson from feeling like an isolated calculator tutorial.</p> | <p>Observable indicators: (1) The calculator-value annotations are numerically correct to 4 decimal places — teacher spot-checks <math>\tan 53^\circ = 1.3270</math> and <math>\sin 30^\circ = 0.5000</math> as anchor values. (2) The inverse-function arrow is labelled with the correct notation (<math>\sin^{-1}</math>, <math>\cos^{-1}</math>, or <math>\tan^{-1}</math>) — any learner who writes 'sin divided by' or omits the <math>^{-1}</math> notation needs targeted clarification. (3) The one-paragraph explanation explicitly connects the calculator to the phenomenon (mentions the mango tree, the sun angle, or the shadow) — paragraphs that describe only the calculator procedure without connecting to the driving question are returned with the prompt: 'How does this help Achieng find the tree height without climbing it?'</p> |

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|  | calculator confirmed what their triangles showed, and how today's lesson helps answer the driving question: 'How can the angle a sunbeam makes with the ground allow us to calculate the height of any tree, building, or hill in Kenya without measuring it directly?' | model explanation. Start with: The calculator confirmed... and end with: This helps us answer the driving question because...' Allow 5 minutes for writing. Cold-call 2 learners to read their paragraph aloud. WAIT 15 seconds after each. Affirm strong explanations: 'That paragraph shows you understand why the calculator is a tool that extends our triangle reasoning — not a replacement for it.' |  |  |
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### D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. PRECISION vs. PROCEDURE: Did learners demonstrate that they understand WHY the calculator gives a more precise value (it uses a mathematical definition, not a physical diagram), or did they simply accept the calculator as a 'magic answer machine'? What evidence from the percentage error discussion supports your answer?
2. INTEGRITY VALUE: Were learners genuinely honest in reporting their Lesson 1 percentage errors, or did you observe any tendency to retroactively adjust earlier values? How did you respond to this, and how might you reinforce the integrity value in Lesson 6?
3. INVERSE FUNCTIONS: The inverse function concept (given ratio  $\rightarrow$  find angle) is often harder for learners than the forward direction. How many learners in Section C of the worksheet correctly identified which inverse function to use AND explained their reasoning? What targeted support will you provide in Lesson 6 for those who could not?
4. PHENOMENON CONNECTION: Did the Section D 'phenomenon check' (computing the mango tree height using the calculator) feel meaningful to learners, or did it feel like a routine calculation? What could you do differently in Lesson 6 to deepen the connection between calculator tools and the real-world driving question?
5. DRIVING QUESTION PROGRESS: After five lessons, where does the class stand in answering 'How can the angle a sunbeam makes with the ground allow us to calculate the height of any tree, building, or hill in Kenya without measuring it directly?' What is the most significant gap that remains, and how will Lesson 6 address it?

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <b>What did I observe?</b>                   | Learners used scientific calculators to evaluate $\sin$ , $\cos$ , and $\tan$ for acute angles including $53^\circ$ (the Kisumu sun angle), and used inverse functions ( $\sin^{-1}$ , $\cos^{-1}$ , $\tan^{-1}$ ) to recover angles from given ratios. They also computed that $\tan 53^\circ \times 9.0 \text{ m} \approx 11.94 \text{ m}$ — a calculator-precision estimate of the mango tree height.                                                                                                                                                       |
| <b>What did I learn?</b>                     | Learners discovered that: (1) the calculator returns consistent, high-precision trigonometric ratio values that match but are more precise than hand-drawn triangle estimates; (2) inverse trigonometric functions reverse the process — given a ratio, they return the angle; (3) the percentage error between their Lesson 1 hand-drawn estimates and calculator values reflects the physical limitations of rulers and protractors, not a flaw in the mathematics; (4) integrity in mathematics means reporting errors honestly rather than adjusting data. |
| <b>How does this explain the phenomenon?</b> | The calculator does not generate new mathematics — it applies the same consistent ratio relationships between angles and side lengths in right-angled triangles that learners established geometrically in Lessons 1–4. The precision of the calculator-derived values reflects the mathematical exactness of the trigonometric function definition. When applied to the Kisumu phenomenon, $\tan$                                                                                                                                                             |

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|  | $53^\circ = 1.3270$ allows us to calculate the mango tree height as approximately 11.94 m — without climbing it — advancing the driving question answer with quantitative confidence. |
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## LESSON 6 (80 minutes): Complementary Angles and the $\sin \theta = \cos(90^\circ - \theta)$ Identity

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <b>Purpose</b>              | To enable learners to discover and prove that $\sin \theta = \cos(90^\circ - \theta)$ and apply this complementary-angle identity to solve problems involving right-angled triangles in real Kenyan contexts.                                                                                                                                                                                                                                                                                                                                       |
| <b>Knowledge</b>            | Learners will know that: (a) two angles are complementary when they sum to $90^\circ$ ; (b) in any right-angled triangle the two acute angles are always complementary; (c) $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$ for any acute angle $\theta$ ; (d) this identity follows directly from the definitions of sine and cosine using SOH CAH TOA.                                                                                                                                                        |
| <b>Skills</b>               | Learners will be able to: (a) identify complementary angle pairs; (b) complete and compare a two-column table of sin and cos values to detect the complementary identity; (c) prove $\sin \theta = \cos(90^\circ - \theta)$ geometrically using a labelled right-angled triangle diagram; (d) apply the identity to simplify expressions and solve for unknown angles; (e) relate the identity back to the shadow-and-tree phenomenon.                                                                                                              |
| <b>Attitudes</b>            | Learners appreciate that elegant mathematical relationships — such as the complementary identity — reduce the labour of real-world measurement, reflecting the beauty and efficiency of mathematics used by Kenyan engineers, architects, and surveyors.                                                                                                                                                                                                                                                                                            |
| <b>Key Inquiry Question</b> | How are the trigonometric ratios of acute angles related to each other?                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Purpose in Storyline</b> | In Lessons 1–5, learners defined SOH CAH TOA, derived special-angle ratios, and used them to calculate tree heights from shadow lengths and clinometer readings. Lesson 6 deepens that understanding: learners discover that the sine of the sun's elevation angle equals the cosine of the complementary depression angle, cutting the number of new ratios they must memorise in half. This hidden symmetry also sets up Lesson 7's tile-cutter $45^\circ$ context, where both acute angles are equal and the identity becomes trivially elegant. |
| <b>Safety Notes</b>         | No physical hazards. Ensure the short video is downloaded and cached on the ARES platform before the lesson so offline playback is seamless. Remind learners to handle protractors and rulers carefully when tracing triangle diagrams.                                                                                                                                                                                                                                                                                                             |

### B. LESSON OVERVIEW

Learners begin by predicting whether  $\sin 53^\circ$  and  $\cos 37^\circ$  are related, activating their earlier knowledge of the Kisumu mango-tree scenario ( $53^\circ$  sun-elevation angle). They then read a short structured article and watch a 4-minute offline video that introduces complementary angles. A paired table-comparison task — filling in sin and cos values for angle pairs that sum to  $90^\circ$  — leads learners to conjecture the identity  $\sin \theta = \cos(90^\circ - \theta)$ . A teacher-led geometric proof using a labelled right-angled triangle diagram consolidates understanding. A cold-call challenge (8 students, one complementary pair and identity each) checks fluency across the class. The lesson closes with the tile-cutter  $45^\circ$  supporting phenomenon as a bridge to Lesson 7, and learners update their Driving Question Board with the new insight that the sun's elevation and the complementary angle encode the same height information in two different ratios.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                           | Learner Experience                                                                         | Teacher Moves                                                                                               | Sensemaking Strategy                                                                        | Formative Assessment Strategy                                                                     |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">The trig functions</a> | Learners are shown two calculations side-by-side on the board (or ARES screen): (A) $\sin$ | Display the two side-by-side triangle diagrams — one labelled with $\theta = 53^\circ$ at ground level (the | Strategy: Prediction Journaling with Class Poll. Purpose: Activating prior knowledge of SOH | Observable indicator: Every learner has at least 2 sentences written in their book before partner |

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| <p><a href="#">&amp; right triangle trig ratios</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/><a href="#">Relate ratios in right triangles</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES for similar readings</a></p>                                                                         | <p>53° computed from the Kisumu mango-tree triangle, and (B) cos 37° computed from the same triangle but labelled from the other acute angle. They write a silent individual prediction (2–3 sentences) in their exercise books: 'Are sin 53° and cos 37° the same number? Why or why not? What does this tell us about the two acute angles in any right-angled triangle?' Learners then share predictions with a shoulder partner for 90 seconds before the teacher takes a class poll.</p>                                                                                                                                                                                                                                                                  | <p>sun-elevation angle from Lesson 5), the other labelled with the complementary angle 37° at the top vertex. Say: 'Look at these two triangles. Both come from the same mango-tree shadow scenario we explored in Kisumu. In your books, predict silently: is sin 53° equal to cos 37°? Write your reasoning — you have 2 minutes.' [WAIT — do not speak for 2 full minutes.] After writing, say: 'Turn and tell your partner your prediction and one reason.' [90-second pair talk.] Cold-call 3 students (choose one who predicted YES, one NO, one UNSURE): 'Amina, what did you predict and why?' [WAIT 10 seconds after each name before accepting the answer.] Record tally of YES / NO / UNSURE on the board without affirming correctness yet. Say: 'Hold your prediction — let's gather evidence.'</p> | <p>CAH TOA and similar triangles, and generating cognitive dissonance. By committing a prediction to paper before evidence is gathered, learners create a personal benchmark against which they will measure what they observe, making the later 'aha' moment more memorable and durable.</p>                                                                                                                                                                                                                                 | <p>talk begins. Teacher scans 4–5 books while circulating during the silent writing phase. Listen for misconceptions during cold-calls — particularly the claim that sin and cos can never be equal — to address explicitly in Phase 3.</p>                                                                                                                                                                                                                    |
| <p><b>Observe Phase</b><br/> <b>VIDEO:</b><br/><a href="#">Solving for a side in right triangles with trigonometry</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/><a href="#">Solve for an angle in right triangles</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES for similar readings</a></p> | <p>PART A — Article Reading (10 min): Learners read a one-page structured article (pre-loaded on ARES) titled 'Complementary Angles in Kenya's Construction Sites.' The article describes how a structural engineer in Nairobi's Westlands district uses a clinometer to measure both the elevation angle of the top of a building (<math>\theta</math>) and notes that the angle between the sun's ray and the building wall is (<math>90^\circ - \theta</math>). A side-bar defines complementary angles and gives three Kenyan-context examples (ramp angles in Mombasa port, roof pitch in a Kericho tea-factory, shadow angles in Kisumu). Learners annotate: underline the definition of complementary angles, circle every numeric example, and box</p> | <p>Introduce the article: 'This article is about a structural engineer called Eng. Wanjiku working in Nairobi. As you read, annotate exactly as I described — underline, circle, box. You have 10 minutes.' [Circulate, crouch to eye-level, check annotations on 6 books. Prompt quietly: 'Have you boxed the key phrase yet?']</p> <p>Before the video: 'Watch for one specific thing: what happens to the labels Opposite and Adjacent when the triangle flips perspective.' [Play video — 4 min. Do not talk over it.] After video, ask the class: 'What happened to the Opposite side when we looked from the other angle?' [WAIT 12 seconds.] Cold-call 2 students.</p>                                                                                                                                    | <p>Strategy: Data-Pattern Table with Student Conjecture Writing. Purpose: Rather than being told the identity, learners induce it from numerical evidence across five angle pairs, including the familiar 53°/37° Kisumu pair. Writing the conjecture in their own words before seeing formal notation forces active sense-making and gives the teacher a window into each pair's level of understanding. The video provides a visual-spatial complement to the numerical table, addressing multiple learning modalities.</p> | <p>Observable indicators: (1) All five rows of the table are completed with correct values (teacher checks 5 pairs during circulation). (2) The conjecture sentence, however informally worded, captures the idea that sin of an angle equals cos of its complement. (3) During cold-calls after the video, students correctly identify that Opposite and Adjacent swap — if they cannot, the teacher re-plays the 1-minute key segment before proceeding.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>the phrase 'these two angles always sum to <math>90^\circ</math>.'</p> <p>PART B — Video (4 min, offline): Learners watch a short animated video (ARES offline cache) that shows a right-angled triangle rotating so the viewer sees it from both acute-angle vertices, with the labels Opposite, Adjacent, and Hypotenuse swapping roles. The narration says: 'What was Opposite for angle <math>\theta</math> becomes Adjacent for the complementary angle.'</p> <p>PART C — Table-Comparison Task (12 min, pairs): Each pair completes a structured table with columns: <math>\theta</math>, <math>(90^\circ - \theta)</math>, <math>\sin \theta</math> (from Lesson 4 special-angles table), <math>\cos(90^\circ - \theta)</math>, and 'Are they equal?' Angle pairs used: <math>(30^\circ, 60^\circ)</math>, <math>(45^\circ, 45^\circ)</math>, <math>(53^\circ, 37^\circ)</math>, <math>(20^\circ, 70^\circ)</math>, <math>(15^\circ, 75^\circ)</math>. Learners use their special-angles tables from Lessons 4–5 for the first three rows and a calculator (or ARES tool) for the last two. They write a conjecture at the bottom of the table in their own words.</p> | <p>Distribute / display the table task. Say: 'Work in pairs. Use your special-angles tables from Lessons 4 and 5. For the <math>53^\circ</math> and <math>37^\circ</math> row, think back to our mango tree in Kisumu — you already know those values. Write your conjecture in plain English at the bottom. You have 12 minutes.' [Circulate — target 5 pairs. Ask probing questions: 'What pattern do you notice in the last column?' and 'Can you say that in one sentence starting with: <math>\sin \theta</math> equals...?'] At 12 minutes, call time. Ask 2 pairs to read their conjectures aloud. Write the best student-generated conjecture on the board verbatim before writing the formal identity below it.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Using complementary angles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan</p> | <p>PART A — Geometric Proof (individual then class): Learners draw a right-angled triangle in their books with vertices labelled P (top of tree / top vertex), Q (base of tree / right angle), R (observer / ground). They label: angle at R = <math>\theta</math> (sun's elevation), angle at P = <math>(90^\circ - \theta)</math>, hypotenuse PR = h (the sun's ray length), side PQ = tree height = opposite to <math>\theta</math>, side QR = shadow length = adjacent to <math>\theta</math>. Using SOH CAH TOA, they write <math>\sin \theta = \text{PQ/PR}</math>. They then re-label: from angle P's perspective, PQ is now</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>PART A: 'Draw the mango-tree triangle exactly as we've been using since Lesson 1. Label all three sides and both acute angles. Now write <math>\sin \theta</math> using SOH CAH TOA.' [WAIT 90 seconds.] 'Now relabel from angle P's perspective and write <math>\cos(90^\circ - \theta)</math>.' [WAIT 90 seconds.] 'What do you notice about both fractions?' [WAIT 12 seconds.] Cold-call 1 student: 'Ochieng, what is your conclusion?' Then lead the board proof step by step, saying: 'Call out the next step — what do I write?' Accept 3</p>                                                                                                                                                                      | <p>Strategy: Student-Generated Proof followed by Structured Practice Gradient. Purpose: Having learners construct the geometric proof themselves — before seeing it done on the board — ensures they understand why the identity holds, not just that it holds. The cold-call challenge (8 students, 8 unique pairs) builds fluency and public accountability, while the three-level worked examples move from pure simplification to equation-solving to reconnecting with the anchoring phenomenon,</p> | <p>Observable indicators: (1) In the individual proof, every learner's book shows two correctly labelled fractions that are visually equal. (2) All 8 cold-call students produce a valid complementary pair and a correctly written identity (teacher marks each with a tick or immediate correction on the board). (3) In Example (ii), learners correctly set up <math>\theta + 10^\circ + 2\theta - 5^\circ = 90^\circ</math> and solve to get <math>\theta = 28.33^\circ</math>; any student who writes <math>\theta + 10^\circ = 2\theta - 5^\circ</math> is flagged for targeted support. (4) In Example (iii),</p> |

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| <p>Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                              | <p>the adjacent side and QR is the opposite side. They write <math>\cos(90^\circ - \theta) = PQ/PR</math>. They box the conclusion: <math>\sin \theta = \cos(90^\circ - \theta)</math>. The teacher then leads a whole-class proof on the board, inviting learners to call out each step.</p> <p>PART B — Cold-Call Challenge (8 students): Teacher calls 8 individual students one at a time. Each must: (1) state a complementary-angle pair (not one already used by a classmate), and (2) write the corresponding identity on the board — e.g., '<math>\sin 20^\circ = \cos 70^\circ</math>'. Teacher awards a tally mark on the board for each correct response.</p> <p>PART C — Worked Examples (guided practice, 10 min): Three examples escalating in difficulty: (i) Simplify <math>\sin 72^\circ - \cos 18^\circ</math>. (ii) Find <math>\theta</math> if <math>\sin(\theta + 10^\circ) = \cos(2\theta - 5^\circ)</math>. (iii) Re-examine the Kisumu mango tree: the observer's angle is <math>53^\circ</math>; confirm that the complementary angle at the top of the tree is <math>37^\circ</math> and verify <math>\sin 53^\circ = \cos 37^\circ \approx 0.7986</math>. Learners work individually, then check with a partner before the teacher reveals solutions.</p> | <p>student volunteers.</p> <p>PART B (Cold-Call Challenge): Say: 'I need 8 volunteers — but I will choose you, not the other way around.' [Scan the room deliberately.] For each student: 'Give me a complementary pair I have not heard yet, then write the identity on the board.' [WAIT 10 seconds between naming the student and expecting an answer. If stuck after 10 seconds, offer a scaffold: 'Start with any angle between <math>1^\circ</math> and <math>89^\circ</math> — your choice.'] After all 8, say: 'Look at the board. Eight different pairs, eight identical structures. What does that tell us about this identity?'</p> <p>PART C: 'Work silently for 5 minutes on all three examples. Then check with your partner for 2 minutes.' [Circulate, watch for the common error in (ii) of setting up the equation incorrectly.] Cold-call 1 student per example to present the solution. For example (iii), explicitly say: 'This is our Kisumu tree. We've just proved that our clinometer angle and its complement are two sides of the same coin.'</p> | <p>consolidating the identity across contexts.</p>                                                                                                                                                                                                                                                                                                                                  | <p>learners confirm both values <math>\approx 0.7986</math>, linking back to the driving question.</p>                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p><b>Driving Question Board (DQB) Creation</b><br/>  <b>VIDEO:</b><br/> <a href="#">The trig functions &amp; right triangle trig ratios</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> | <p>Learners return to the class Driving Question Board — a large chart or ARES collaborative board that has been updated after each lesson. Each learner receives a sticky note (or types into the ARES DQB tool). They respond to two prompts: (1) EVIDENCE ADDED TODAY: 'Write one sentence describing the new mathematical relationship you discovered.' (2) NEW QUESTION RAISED: 'Write</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>Distribute sticky notes. Say: 'Two sentences. First: what new relationship did you discover today — be precise, use mathematical language. Second: what question does today's lesson raise for you?' [WAIT 3 minutes of silent writing.] Collect and scan quickly; select 2 evidence notes and 2 question notes that represent the range of the class. Read each aloud without naming the author. Say: 'Listen to</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Strategy: Two-Column DQB Annotation (Evidence / Wonder). Purpose: Making the DQB a living document that learners contribute to after every lesson creates a cumulative public record of the class's growing understanding. The 'wonder' column legitimises unanswered questions and motivates future lessons. Connecting the new identity explicitly to the driving question</p> | <p>Observable indicators: (1) Every learner posts both a sticky note entry (evidence and question). (2) The 'evidence' sentences use the phrase '<math>\sin \theta = \cos(90^\circ - \theta)</math>' or equivalent in the learner's own words. (3) 'Wonder' questions show genuine extension thinking (e.g., 'Does this work for angles bigger than <math>90^\circ</math>?' or 'What happens at exactly <math>45^\circ</math> — does <math>\sin</math> equal <math>\cos</math>?'), which confirms that</p> |

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| <p> <b>EXERCISE:</b><br/> <a href="#">Right triangle trigonometry word problems</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                          | <p>one question this lesson made you curious about — especially thinking about the tile-cutter's 45° angle teased at the end of class.' Learners post their notes under the columns 'What we now know' and 'What we still wonder.' The teacher reads 4 notes aloud (2 from each column) to the class and explicitly updates the class answer to the driving question: 'How can an angle tell you the height of a tree?' with the new insight: 'Because sine and cosine of complementary angles are equal, a surveyor only needs one clinometer reading — either the elevation angle OR its complement — to compute the tree height.'</p>                                                                                                                                                                                                                                                                                                                                                           | <p>this question one of you wrote...' [Read a strong 'still wonder' note.] 'Does anyone want to respond to that right now?' [WAIT 10 seconds — take 1 volunteer if available, otherwise say: 'Hold that thought — it will drive our next lesson.'] Point to the driving question banner and say: 'Update your personal answer to our driving question in your book. How does today's identity change or strengthen your answer?'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>reinforces that each lesson is a step in answering one overarching real-world puzzle, not an isolated topic.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>learners are thinking beyond the lesson rather than simply reproducing it.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>Model Building Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">The trig functions &amp; right triangle trig ratios</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Solve for a side in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners retrieve their cumulative right-angled triangle model diagram — a labelled diagram they have updated after each lesson since Lesson 1. Today's revision has two steps: (1) They add a second colour annotation to the acute angles, marking that the two angles are complementary and writing the identity <math>\sin \theta = \cos(90^\circ - \theta)</math> and <math>\cos \theta = \sin(90^\circ - \theta)</math> as annotations on the diagram. (2) They add a new 'supporting scenario' box at the bottom of their model page: the Tile-Cutter's 45° Puzzle — a brief sketch of a tile-cutter in a Nairobi hardware shop cutting a square tile diagonally, creating a 45°-45°-90° triangle, with the prompt: 'What does our new identity predict about <math>\sin 45^\circ</math> and <math>\cos 45^\circ</math>? Check against your special-angles table from Lesson 4.' Learners answer the prompt individually, then the teacher closes the lesson by previewing Lesson 7.</p> | <p>Say: 'Take out your cumulative model diagram. Use a red pen or colour for today's update. Add the complementary-angle identity to both acute angles of your triangle.' [WAIT 2 minutes.] 'Now add the tile-cutter box at the bottom. Read the prompt and answer it using your Lesson 4 table.' [WAIT 3 minutes.] Cold-call 2 students: 'Njeri, what does the identity predict <math>\sin 45^\circ</math> should equal? And does your table confirm it?' [WAIT 10 seconds.] After confirmation (<math>\sin 45^\circ = \cos 45^\circ = \sqrt{2}/2 \approx 0.7071</math>), say: 'This is not a coincidence — it is the identity being perfect. Next lesson we will explore why 45° makes both ratios identical and how a tile-cutter in Nairobi uses this every single day.' Assign exit task: 'In your book, write the updated answer to our driving question — include the complementary identity in your explanation of how the mango tree height can be calculated.'</p> | <p>Strategy: Cumulative Annotated Model Revision with Bridging Prompt. Purpose: Revising the same triangle diagram across all 10 lessons builds a tangible, growing artefact of understanding. Each addition is visually anchored to what came before, making the conceptual progression concrete. The tile-cutter bridging prompt simultaneously consolidates today's identity (learners apply it to 45°) and generates anticipation for Lesson 7, using a Kenyan artisan context that makes abstract mathematics feel locally relevant and purposeful.</p> | <p>Observable indicators: (1) The cumulative model diagram now shows complementary-angle annotations in a distinct colour, with both identities written correctly. (2) The tile-cutter box contains the correct prediction: <math>\sin 45^\circ = \cos 45^\circ</math>, verified numerically (<math>\approx 0.7071</math>). (3) The exit task answer in each learner's book explicitly mentions that either the elevation angle <math>\theta</math> or the complementary angle <math>(90^\circ - \theta)</math> can be used with the appropriate ratio because <math>\sin \theta = \cos(90^\circ - \theta)</math>, demonstrating integration of the day's learning with the driving question.</p> |

## D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record notes in your ARES teacher journal:

1. **COLD-CALL QUALITY:** During the 8-student cold-call challenge in Phase 3, which students hesitated or gave incorrect pairs? Were errors procedural (wrong arithmetic) or conceptual (not understanding what 'complement' means)? Plan a 5-minute warm-up for Lesson 7 targeting those specific students.
2. **TABLE-COMPLETION ACCURACY:** In the Phase 2 table task, did most pairs correctly complete all five rows? If the  $20^\circ/70^\circ$  and  $15^\circ/75^\circ$  rows caused difficulty (calculator use), consider providing a printed trigonometric table as a scaffold in future lessons for learners who are less confident with the ARES calculator tool.
3. **PROOF PARTICIPATION:** When you led the board proof in Phase 3A, were learners calling out steps confidently, or were you supplying most steps yourself? If participation was low, the class may not yet have fully internalised the SOH CAH TOA definitions from Lessons 2–3. Revisit those definitions as a 3-minute starter in Lesson 7.
4. **DQB WONDER QUESTIONS:** Read the 'still wonder' sticky notes after class. Do learners' questions naturally lead toward Lesson 7's  $45^\circ$  special case, or are they confused about something in today's lesson that needs re-teaching? Adjust the Lesson 7 opener accordingly.
5. **PHENOMENON CONNECTION:** Did learners explicitly connect the complementary-angle identity back to the Kisumu mango-tree scenario in their exit task answers, or did they treat today's lesson as abstract algebra? If the phenomenon connection was weak, open Lesson 7 by re-reading the original phenomenon and asking: 'Where does the complementary identity appear in the student's shadow measurement?'

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <b>What did I observe?</b>                   | Learners read an article about a Nairobi structural engineer using complementary angles, watched a 4-minute offline video showing how Opposite and Adjacent sides swap when a right-angled triangle is viewed from the complementary angle, and completed a five-row table comparing $\sin \theta$ with $\cos(90^\circ - \theta)$ for pairs including the familiar $53^\circ/37^\circ$ Kisumu mango-tree angles.                                                               |
| <b>What did I learn?</b>                     | Learners discovered and proved that $\sin \theta = \cos(90^\circ - \theta)$ for any acute angle $\theta$ , because in a right-angled triangle the two acute angles are complementary and the side that is Opposite one angle is Adjacent to the other. They applied this identity to simplify expressions, solve equations, and confirm the Kisumu tree calculation.                                                                                                           |
| <b>How does this explain the phenomenon?</b> | The complementary-angle identity means that a surveyor (or a student in Kisumu) only needs one clinometer reading — the sun's elevation angle $\theta$ — to calculate the tree height, because $\sin \theta$ and $\cos(90^\circ - \theta)$ reference the same ratio of sides in the same triangle. The identity halves the number of independent trigonometric relationships that need to be memorised, revealing an elegant symmetry at the heart of right-triangle geometry. |

## LESSON 7 (80 minutes): Special Angles (Derivation): Unlocking Exact Trigonometric Values

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>Purpose</b>              | To enable learners to derive and apply exact trigonometric values for special angles ( $0^\circ$ , $30^\circ$ , $45^\circ$ , $60^\circ$ , $90^\circ$ ) using geometric constructions, so they can solve real-world height and distance problems — like calculating the height of a mango tree in Kisumu — without a calculator.                                                                                                                                                                                                                                             |
| <b>Knowledge</b>            | Learners will know the exact trigonometric ratios (sin, cos, tan) for special angles $0^\circ$ , $30^\circ$ , $45^\circ$ , $60^\circ$ , and $90^\circ$ , derived from an equilateral triangle of side 2 (for $30^\circ/60^\circ$ ) and an isosceles right-angled triangle of leg 1 (for $45^\circ$ ), and understand why these values are exact rather than decimal approximations.                                                                                                                                                                                         |
| <b>Skills</b>               | Learners will be able to: (a) construct the two special triangles and apply Pythagoras' theorem to find missing sides; (b) read off and record exact sin, cos, and tan values for each special angle using SOH CAH TOA; (c) design a 'special angles passport card' that accurately organises these values; (d) use exact values to calculate heights and distances in context.                                                                                                                                                                                             |
| <b>Attitudes</b>            | Learners will appreciate that mathematics contains precise, elegant patterns — such as the exact values of special angles — that make computation more reliable and beautiful than decimal approximations, and will value collaborative peer learning in constructing and verifying these results.                                                                                                                                                                                                                                                                          |
| <b>Key Inquiry Question</b> | How are the trigonometric ratios of acute angles related to each other, and how do exact values for special angles make real-world calculations — like finding the height of a tree or a hill in the Rift Valley — faster and more precise?                                                                                                                                                                                                                                                                                                                                 |
| <b>Purpose in Storyline</b> | In Lesson 7 the class moves from using a calculator to approximate trig ratios (Lessons 4–6) to discovering that for certain 'special' angles the ratios are exact. The student in Kisumu measured an angle of approximately $53^\circ$ — close to $60^\circ$ — raising the question: what if the sun were at exactly $60^\circ$ ? Could we get a perfect answer? This lesson answers that by deriving exact values from geometry, deepening the class's model of the sun-angle/shadow/height relationship and preparing them to solve the driving question with precision. |
| <b>Safety Notes</b>         | No significant safety concerns. Ensure compasses used for geometric constructions are handled carefully — caps should be on when not in use. Scissors used for cutting passport cards should be safety scissors or handled with care.                                                                                                                                                                                                                                                                                                                                       |

### B. LESSON OVERVIEW

Learners derive the exact trigonometric values for the five special angles ( $0^\circ$ ,  $30^\circ$ ,  $45^\circ$ ,  $60^\circ$ ,  $90^\circ$ ) through two guided geometric constructions: (1) halving an equilateral triangle of side 2 to generate a 30-60-90 triangle, and (2) drawing an isosceles right-angled triangle with legs of 1 to generate a 45-45-90 triangle. They apply Pythagoras' theorem to find exact side lengths (including surds), then use SOH CAH TOA to read off exact ratios. A short video and a reference reading consolidate the pattern. Each group then designs a pocket-sized 'special angles passport card' — a hand-crafted reference tool — and updates the class Driving Question Board (DQB) with an exact-value table. The lesson is anchored throughout to the Kisumu shadow scenario: if the sun were at  $30^\circ$ ,  $45^\circ$ , or  $60^\circ$ , what would be the exact height of the mango tree?

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                | Learner Experience             | Teacher Moves                 | Sensemaking Strategy       | Formative Assessment Strategy      |
|----------------------|--------------------------------|-------------------------------|----------------------------|------------------------------------|
| <b>Predict Phase</b> | Learners revisit the anchoring | Write the prompt on the board | Predict–Record–Revisit: By | Observable indicator: All learners |

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| <p> <b>VIDEO:</b><br/> <a href="#">The trig functions &amp; right triangle trig ratios</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Relate ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p>                                        | <p>phenomenon. Each learner silently reads the prompt on the board: 'The clinometer in Kisumu showed the sun's rays hit the ground at about 53°. Suppose the angle were exactly 60°. What would sin 60°, cos 60°, and tan 60° be — as exact fractions or expressions, not decimals? Write your best prediction in your exercise book. You may use a diagram.' After 3 minutes of silent individual thinking, pairs share predictions. Selected pairs share with the class. Teacher records a range of predictions (decimal guesses, fraction guesses, 'I don't know') on the board without affirming or correcting any.</p>                                                               | <p>verbatim before learners enter. Say: 'We have used our calculators to get <math>\sin 53^\circ \approx 0.8</math>. But what if the angle were a rounder number — say exactly 60°? Could the answer be something cleaner than a messy decimal? Predict in your books — diagram encouraged — for 3 minutes. Silence please.' Set a visible 3-minute timer. After timer: 'Turn and share with your partner for 90 seconds.' Cold-call 4 pairs: 'Amina, what did you two predict?' 'Otieno, different or same?' 'Wanjiru, give us your reasoning.' 'Kipchoge, do you agree or disagree?' Record all predictions in a 'Prediction Wall' column on the board. Say: 'We will come back to this wall at the end of the lesson and see who was closest — and more importantly, why.'</p> | <p>committing predictions to paper before instruction, learners activate prior knowledge of ratio and similar triangles (built in Lessons 1–3) and create a cognitive 'gap' that the lesson will close. Recording publicly on the Prediction Wall creates accountability and motivates engagement with the derivation.</p>                                                                                                     | <p>have written a prediction (even 'I'm not sure because...') and drawn at least a rough triangle. Teacher scans books during pair-share. Red flag: learners who draw no triangle — prompt them: 'Can you sketch the shadow scenario from Kisumu to help you think?'</p>                                                                                                                                                                                                                                                                                                                            |
| <p><b>Observe Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">The trig functions &amp; right triangle trig ratios</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>PART A — Construction Activity (18 min): Groups of 4 receive a guided construction worksheet and the following materials: ruler, compass, protractor, pencil, coloured pens. They complete two constructions step by step:</p> <p>Construction 1 — The 30-60-90 Triangle: Draw an equilateral triangle with all sides = 2 cm. Label vertices A, B, C. Drop a perpendicular from A to BC, meeting BC at D. Measure AD using Pythagoras (<math>AD = \sqrt{3}</math>). Identify the 30°, 60°, and 90° angles. Label sides relative to each angle: Opposite, Adjacent, Hypotenuse.</p> <p>Construction 2 — The 45-45-90 Triangle: Draw a right-angled isosceles triangle with two legs</p> | <p>Distribute materials and worksheet. Say: 'You have 18 minutes for the two constructions. Read each step carefully before drawing. Every group member must do their own construction — you can compare, but each book must have both triangles.' Circulate continuously. At 5 min, check Construction 1 progress: 'Has everyone dropped the perpendicular from A? What theorem will you use to find its length?' (Wait 10 seconds.) Cold-call 2 groups for their answer before moving on. At 12 min, check Construction 2: 'What is the hypotenuse of your 45-45-90 triangle? Leave it in surd form.' (Wait 12 seconds.) Cold-call 3 groups. At 18 min, pause all groups: 'Pens down. Let's watch the video — keep your worksheet beside you and tick off each value</p>        | <p>Construct–Calculate–Confirm: The three-stage sequence (physical construction → Pythagoras calculation → video/reading confirmation) mirrors the scientific practice of 'planning and carrying out investigations' (NGSS SEP 3) and 'using mathematics' (NGSS SEP 5). Building the triangles by hand before seeing the finished table ensures learners experience the derivation as logical necessity, not memorisation.</p> | <p>Observable indicators: (1) Each learner's book has two correctly labelled triangles with Opposite/Adjacent/Hypotenuse marked relative to the angle of interest. (2) The results table shows exact surd forms (<math>\sqrt{3}</math>, <math>\sqrt{2}</math>), not decimals. Red flags: learners writing 0.866 instead of <math>\sqrt{3}/2</math> — prompt: 'Can you write that without a decimal? What did Pythagoras give you?' Learners who have <math>AD = 2</math> instead of <math>\sqrt{3}</math> — common error from forgetting to halve BC: 'What is BD after you fold the triangle?'</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>each = 1 cm. Label vertices P, Q, R with the right angle at R. Calculate the hypotenuse using Pythagoras (<math>PR = \sqrt{2}</math>). Identify the <math>45^\circ</math> angles. Label sides.</p> <p>Groups complete a results table on the worksheet, recording exact values for sin, cos, and tan of each special angle from their triangles.</p> <p>PART B — Video + Reading Consolidation (7 min): Teacher plays a 4-minute narrated animation video showing the same two constructions with a Kenyan context (a surveyor on the slopes of Mt. Kenya using exact angle values). Learners follow with a one-page reference reading (provided on ARES platform) that shows the completed exact-value table including <math>0^\circ</math> and <math>90^\circ</math> limit cases. Learners annotate the reading, circling values that match their construction results.</p> | <p>as you see it confirmed.' Play the video. After video: 'Open the reading on ARES. Circle any value in the table that your construction already found. Put a question mark next to anything new — especially <math>0^\circ</math> and <math>90^\circ</math>.' Give 3 minutes for annotation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">The trig functions &amp; right triangle trig ratios</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Trig values of <math>\pi/6</math>, <math>\pi/4</math>, and <math>\pi/3</math></a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>PART A — Sensemaking Discussion (10 min): The teacher leads a whole-class structured discussion connecting the derived values back to the Kisumu scenario. The class examines: 'If the sun in Kisumu were at exactly <math>60^\circ</math> instead of <math>53^\circ</math>, what would tan <math>60^\circ</math> be — exactly? And what height would that give the mango tree with a shadow of 9.0 m?' Learners work the calculation individually: height = <math>9.0 \times \tan 60^\circ = 9.0 \times \sqrt{3} \approx 9.0 \times 1.732 = 15.6</math> m. They compare this exact approach with the calculator approach from Lesson 6.</p> <p>PART B — Pattern Spotting (5</p>                                                                                                                                                                                              | <p>Write on the board: 'Shadow = 9.0 m. Sun angle = <math>60^\circ</math>. Height = ?' Say: 'Using only what you derived today — no calculator for the trig part — find the exact height. You may use a calculator only to multiply at the end. 3 minutes, individual work.' (Set timer.) After 3 min, cold-call 3 learners: 'Zawadi, walk us through your working.' 'Mwangi, did you get the same answer? Where did you differ?' 'Fatuma, is <math>9\sqrt{3}</math> metres an exact answer or an approximation? Explain.' (Wait 12 seconds after each question.) Then say: 'Now look at your full table. Compare sin <math>30^\circ</math> and cos <math>60^\circ</math>. What do you notice? Discuss with your partner — 90 seconds.' Cold-call 2 pairs.</p> | <p>Connecting-Back-to-Phenomenon: Every calculation in this phase is anchored to the Kisumu mango tree. By computing the tree's height using an exact value (<math>9\sqrt{3}</math> m), learners see that the special-angle derivation is not abstract — it gives a cleaner, more precise answer to the original puzzle. The complementary-angle pattern (<math>\sin \theta = \cos(90^\circ - \theta)</math>) is surfaced through guided questioning (NGSS SEP 6: Constructing Explanations), deepening the relational understanding between angles.</p> | <p>Observable indicators: (1) Learner working shows height = <math>9 \times \sqrt{3}</math> with correct application of tan = Opposite/Adjacent (SOH CAH TOA). (2) At least 3 learners can articulate why <math>\sin 30^\circ = \cos 60^\circ</math> in their own words. (3) Learners correctly identify tan <math>90^\circ</math> as undefined with a geometric reason. Red flag: learners who write <math>\tan 90^\circ = 0</math> — use targeted questioning: 'What is the Adjacent side when the angle is <math>90^\circ</math>?''</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>min): Learners examine the completed table and answer three guided questions on their worksheet: (i) What do you notice about <math>\sin 30^\circ</math> and <math>\cos 60^\circ</math>? (ii) What is the relationship between <math>\sin \theta</math> and <math>\cos(90^\circ - \theta)</math> for any special angle? (iii) Why is <math>\tan 90^\circ</math> undefined — what happens geometrically in the triangle?</p> <p>PART C — Passport Card Design (5 min planning): Groups are briefed on the Passport Card task (completed in Phase 4). Teacher models a blank template on the board: pocket-sized (A6), must include exact values for all five special angles in a neat table, a diagram of at least one construction triangle, and one real-life Kenya example using an exact value.</p>                                                                                                                                 | <p>Guide to the complementary angle relationship: '<math>\sin \theta = \cos(90^\circ - \theta)</math>'. This is not a coincidence — it comes directly from the geometry of our triangle. Why?' (Wait 15 seconds.) For <math>\tan 90^\circ</math>: 'In our equilateral triangle construction, what happens to the Opposite side as the angle approaches <math>90^\circ</math>? What about the Adjacent?' Elicit: Adjacent <math>\rightarrow 0</math>, so division is undefined. Transition: 'You have now derived these values from scratch. Your next job is to make sure you never forget them — and help a friend who missed today never forget them either. That's the Passport Card.'</p>                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">The trig functions &amp; right triangle trig ratios</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Find trig values using angle addition identities</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Each group completes their 'Special Angles Passport Card' (A6 card, coloured pens). The card must include: (1) a fully completed exact-value table for <math>\sin</math>, <math>\cos</math>, <math>\tan</math> at <math>0^\circ</math>, <math>30^\circ</math>, <math>45^\circ</math>, <math>60^\circ</math>, <math>90^\circ</math>; (2) a small labelled diagram of either the 30-60-90 or 45-45-90 triangle; (3) one Kenya-context worked example (e.g., 'A telecommunications mast in Nairobi casts a shadow of 20 m when the sun is at <math>45^\circ</math>. Height = <math>20 \times \tan 45^\circ = 20 \times 1 = 20</math> m'); (4) the group's names and a creative group title. One card per group is laminated (or covered with sticky tape) and posted on the class DQB. The DQB is then updated: the class adds a new sticky-note row titled 'Exact Values Discovered' and sticks the completed exact-value table onto</p> | <p>Say: 'You have 7 minutes to complete your group's passport card. Every value must be in exact form. Your Kenya example must be a real place and a realistic measurement.' Circulate and check cards for accuracy — particularly that <math>\sin 45^\circ = 1/\sqrt{2}</math> or <math>\sqrt{2}/2</math> (both acceptable), and that <math>\tan 30^\circ = 1/\sqrt{3}</math> or <math>\sqrt{3}/3</math>. After 7 min: 'Post your card on the DQB. Let's look at what we now know.' Point to DQB: 'In Lesson 1 we asked: how can an angle tell us the height of a tree? In Lessons 4 and 5 we found approximate ratios using calculators. Today — what is new? What can we now do that we could not do before?' Cold-call 2 learners. Summarise: 'We now have exact values. If the sun in Kisumu were at <math>30^\circ</math>, <math>45^\circ</math>, or <math>60^\circ</math>, we could give the tree's height as</p> | <p>Artefact Creation + Public Display: The Passport Card is a 'knowledge artefact' (a physical, designed object encoding what was learned). Designing it requires learners to synthesise, organise, and make choices — higher-order thinking. Posting it publicly on the DQB creates social accountability and a visual record of the class's growing understanding across the 10-lesson sequence.</p> | <p>Observable indicators: (1) Each posted passport card contains all five special angles with exact (not decimal) values. (2) The Kenya example uses correct SOH CAH TOA and a special angle. (3) DQB sticky-note is placed by learners, not the teacher — indicating ownership of the update. Red flag: decimal values on the card — teacher returns it to the group: 'Table 4, your <math>\sin 60^\circ</math> is written as 0.866. How did we derive it today? What is the exact form?'</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | the board. The teacher facilitates a 3-minute whole-class review of the DQB, pointing to which driving-question threads have now been advanced.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | a precise surd — no rounding error. This is a powerful upgrade to our model.' |            |                                     |       |       |       |     |              |                            |     |   |     |     |            |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <div>Model Building Phase</div> <div> VIDEO:</div> <div><a href="#">The trig functions &amp; right triangle trig ratios</a></div> <div>Source: Khan Academy (English - US curriculum)</div> <div> Search ARES for similar videos</div> <div> EXERCISE:</div> <div><a href="#">Right triangle trigonometry word problems</a></div> <div>Source: Khan Academy (English - US curriculum)</div> <div> Search ARES for similar readings</div> | <p>Learners return to their cumulative 'Shadow-Angle-Height' model (begun in Lesson 1 and revised in each subsequent lesson). Today's revision task: update the model to show that for special sun angles (30°, 45°, 60°), the height of any object can be expressed as an exact surd multiple of its shadow length. Specifically, learners add a new 'exact values branch' to their model diagram and complete the following table within the model:</p> <table><tr><td>  Sun Angle</td><td>  tan(angle)</td><td>  Height of mango tree (shadow = 9 m)</td></tr><tr><td> ----- </td><td> ----- </td><td> ----- </td></tr><tr><td>  30°</td><td>  <math>1/\sqrt{3}</math></td><td>  <math>9/\sqrt{3} = 3\sqrt{3}</math> m</td></tr><tr><td>  45°</td><td>  1</td><td>  9 m</td></tr><tr><td>  60°</td><td>  <math>\sqrt{3}</math></td><td>  <math>9\sqrt{3}</math> m</td></tr></table> <p>Learners also annotate their model with one new 'wondering' question to post on the DQB in the next lesson, such as: 'Does sin 53° have an exact value too, or only special angles?' or 'What happens to tan as the angle gets very close to 90°?'</p> | Sun Angle                                                                     | tan(angle) | Height of mango tree (shadow = 9 m) | ----- | ----- | ----- | 30° | $1/\sqrt{3}$ | $9/\sqrt{3} = 3\sqrt{3}$ m | 45° | 1 | 9 m | 60° | $\sqrt{3}$ | $9\sqrt{3}$ m | <p>Say: 'Open your cumulative model from Lesson 1. You have 10 minutes to add the exact-values branch. The table on the board is your guide — complete it in your model using the exact surd values you derived today.' (Write the table shell on the board with the first row completed as a model.) Set a 10-minute timer. Circulate. After 8 min: 'Two more minutes — make sure you have written at least one new wondering question on a sticky note for our DQB.' At 10 min: 'Pens down. Two volunteers to share their wondering question.' Cold-call 2 learners. Post their sticky notes on the DQB 'Questions We Still Have' column. Close: 'Every time we update this model, we get closer to fully answering our driving question. In Lesson 8, we will look at the relationship between complementary angles more deeply — your wondering questions today will guide us there.'</p> | <p>Cumulative Model Revision (NGSS SEP 2 — Developing and Using Models): Revisiting and annotating the same model across all 10 lessons externalises the learners' evolving mental models, making conceptual growth visible. Adding exact values as a new 'branch' shows learners that today's knowledge is an extension of — not a replacement for — prior knowledge. The 'wondering question' routine sustains inquiry momentum into the next lesson.</p> | <p>Observable indicators: (1) The model table shows exact surd expressions (<math>9/\sqrt{3}</math> rationalised as <math>3\sqrt{3}</math>, or left as <math>9/\sqrt{3}</math> — both accepted if exact). (2) At least one wondering question per learner is written and posted. (3) The model clearly distinguishes between 'approximate' ratios (Lessons 4–6) and 'exact' ratios (Lesson 7). Exit check: teacher collects 6 books (2 high, 2 mid, 2 emerging) to review model accuracy before Lesson 8.</p> |
| Sun Angle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | tan(angle)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Height of mango tree (shadow = 9 m)                                           |            |                                     |       |       |       |     |              |                            |     |   |     |     |            |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| 30°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $1/\sqrt{3}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $9/\sqrt{3} = 3\sqrt{3}$ m                                                    |            |                                     |       |       |       |     |              |                            |     |   |     |     |            |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 45°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 9 m                                                                           |            |                                     |       |       |       |     |              |                            |     |   |     |     |            |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 60°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\sqrt{3}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $9\sqrt{3}$ m                                                                 |            |                                     |       |       |       |     |              |                            |     |   |     |     |            |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

## D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners complete both geometric constructions correctly within the 18-minute window? If not, which construction caused the most difficulty — the equilateral triangle (30-60-90) or the isosceles right-angled triangle (45-45-90) — and what scaffolding will you provide at the start of Lesson 8? (2) How many passport cards posted on the DQB contained exact surd values rather than decimals? If fewer than 70% of groups achieved this, plan a 5-minute 'exact vs. approximate' mini-lesson at the start of Lesson 8. (3) Did the Kisumu mango-tree anchor successfully motivate the derivation, or did learners ask 'why does it matter if it's exact?' — if so, prepare a stronger real-world precision argument (e.g., engineering tolerances for a Nairobi bridge). (4) Which learners'

wondering questions showed the deepest thinking, and how can their questions shape the inquiry direction of Lesson 8 on complementary-angle relationships? (5) Were there learners who already knew the special angle values from prior tutoring — how were they challenged to go deeper (e.g., explaining the geometry to peers, deriving values for  $0^\circ$  and  $90^\circ$  as limits)?

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| What did I observe?                                          | Learners observed — through their own geometric constructions — that halving an equilateral triangle of side 2 produces a right-angled triangle with sides 1, $\sqrt{3}$ , and 2, and that an isosceles right-angled triangle with legs 1 has a hypotenuse of $\sqrt{2}$ . They observed that the trigonometric ratios read from these triangles are exact expressions involving $\sqrt{2}$ and $\sqrt{3}$ , not messy decimals, and that these values appeared consistently in the video and reference reading.                                                                                                                                                                               |
| What did I learn?                                            | Learners learned that five special angles ( $0^\circ$ , $30^\circ$ , $45^\circ$ , $60^\circ$ , $90^\circ$ ) have exact trigonometric ratios derivable from geometry alone — no calculator needed. They learned that $\sin 30^\circ = 1/2$ , $\cos 30^\circ = \sqrt{3}/2$ , $\tan 30^\circ = 1/\sqrt{3}$ ; $\sin 45^\circ = \cos 45^\circ = 1/\sqrt{2}$ ; $\sin 60^\circ = \sqrt{3}/2$ , $\cos 60^\circ = 1/2$ , $\tan 60^\circ = \sqrt{3}$ ; and the boundary values at $0^\circ$ and $90^\circ$ . They also learned that $\sin \theta = \cos(90^\circ - \theta)$ for complementary angles, and that $\tan 90^\circ$ is undefined because the Adjacent side of the triangle collapses to zero. |
| How does this explain the phenomenon?                        | Learners explained how the exact values connect back to the anchoring phenomenon: if the sun in Kisumu made an angle of exactly $60^\circ$ with the ground, the height of the mango tree (shadow = 9.0 m) would be exactly $9\sqrt{3}$ metres — a precise, unrounded answer. They explained this on their passport cards and in their cumulative models, articulating that exact values allow more reliable calculations in real-world measurement tasks such as surveying, construction, and navigation across Kenya.                                                                                                                                                                         |

## LESSON 8 (80 minutes): Special Angles: Solving Real Problems Without a Calculator

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>Purpose</b>              | Learners apply exact trigonometric values of special angles ( $0^\circ$ , $30^\circ$ , $45^\circ$ , $60^\circ$ , $90^\circ$ ) to solve real-world problems involving heights and lengths, and revise their cumulative right-triangle model to incorporate these values and the identity $\tan = \sin/\cos$ .                                                                                                                                                                                                                                  |
| <b>Knowledge</b>            | Learners know and recall the exact trigonometric ratios for special angles ( $0^\circ$ , $30^\circ$ , $45^\circ$ , $60^\circ$ , $90^\circ$ ) derived from the 30-60-90 and 45-45-90 right triangles, and recognise the identity $\tan \theta = \sin \theta / \cos \theta$ .                                                                                                                                                                                                                                                                   |
| <b>Skills</b>               | Learners calculate unknown side lengths and angles in right-angled triangles using exact special-angle values without a calculator, and annotate their cumulative model diagrams with these values.                                                                                                                                                                                                                                                                                                                                           |
| <b>Attitudes</b>            | Learners appreciate the power of exact mathematical relationships — that a builder in Eldoret, a farmer in Kericho, or an architect in Nairobi can design structures precisely without relying on electronic tools.                                                                                                                                                                                                                                                                                                                           |
| <b>Key Inquiry Question</b> | How do we apply trigonometric ratios in day-to-day activities? How are the trigonometric ratios of acute angles related to each other?                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Purpose in Storyline</b> | In Lessons 1–7 the class established SOH CAH TOA, explored similar triangles, and used calculators/tables to find ratios for arbitrary angles. Now, in Lesson 8, they discover that for five special angles the ratios are exact — no approximation needed. This unlocks the ability to solve the Eldoret builder's roof problem and to revisit the Kisumu mango-tree phenomenon with precision, advancing the driving question by showing that the angle–ratio relationship is not just consistent but, for special angles, perfectly exact. |
| <b>Safety Notes</b>         | No hazardous materials. Ensure scissor use for cutting card triangles is supervised. Learners should keep desks clear of clutter when passing card-triangle manipulatives between groups.                                                                                                                                                                                                                                                                                                                                                     |

### B. LESSON OVERVIEW

Lesson 8 of 10 in the evidence-gathering sequence. Learners begin by predicting exact ratio values for  $30^\circ$ ,  $45^\circ$ , and  $60^\circ$  using symmetry and prior knowledge of equilateral and isosceles right triangles. They then derive the exact values through a structured card-triangle activity and validate them against their calculator/table findings from earlier lessons. The Eldoret builder's roof-rafter context anchors the Explain phase: groups solve a  $60^\circ$ -pitched roof of 5 m span using exact values only. A short paired reading on special angles in Kenyan construction and engineering grounds the mathematics in authentic practice. Groups then revise their cumulative annotated right-triangle model to embed the special-angle table and the  $\tan = \sin/\cos$  identity. The lesson closes with a DQB update that maps new evidence onto the anchoring phenomenon (Kisumu mango-tree shadow and sunbeam angle).

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                | Learner Experience                                                                                                                                                                                                      | Teacher Moves                                                                                                                                            | Sensemaking Strategy                                                                                                                                                      | Formative Assessment Strategy                                                                                                                                         |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Solving for a side in right triangles with trigonometry</a> | Each learner receives a blank 3-column table with headers: Angle   $\sin$   $\cos$   $\tan$ . They individually fill in their BEST GUESS for $0^\circ$ , $30^\circ$ , $45^\circ$ , $60^\circ$ , $90^\circ$ — using only | Display the anchoring phenomenon image (student in Kisumu, mango-tree shadow, clinometer reading $53^\circ$ ). Say: 'Last lesson we used calculators and | Activation of Prior Knowledge + Prediction Journal. Learners surface informal geometric intuitions (e.g., 'at $45^\circ$ the triangle looks symmetric so $\sin = \cos$ ') | Teacher scans completed prediction tables during circulation. Observable indicator: learner has written a numerical fraction or decimal (not left blank) for at least |

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| <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>EXERCISE:</b><br/> <a href="#">Right triangle trigonometry word problems</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                               | <p>memory, symmetry reasoning, or any prior knowledge. They annotate a small sketch of a right-angled triangle to show which sides they think grow or shrink as the angle increases from <math>0^\circ</math> to <math>90^\circ</math>. They write a one-sentence prediction: 'I think <math>\sin 60^\circ</math> is ____ because ____.' Predictions are held (not shared yet) in their exercise books to be revisited after derivation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>tables. Today I want you to FORGET the calculator. Look at your blank table — what do you already know or can reason about these special angles? You have 4 minutes, working alone. Go.' Set a visible 4-minute timer. Circulate silently; do NOT confirm or correct — only observe. After 4 minutes: 'Keep your tables face-down. We will come back to them.' Cold-call 3 learners: 'Achieng, what did you write for <math>\sin 90^\circ</math>?' WAIT 12 seconds. 'Otieno, same question — agree or disagree with Achieng and say why.' WAIT 12 seconds. 'Fatuma, what about <math>\tan 45^\circ</math>? Take your time.' WAIT 15 seconds. Record the three predicted values on the board WITHOUT revealing correctness. Say: 'By the end of this lesson you will know with certainty — and you will never need a calculator for these five angles again.'</p> | <p>before formal derivation. Writing a prediction sentence forces them to articulate reasoning, creating a cognitive gap that motivates the derivation activity.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>3 of the 5 angles AND has provided a written justification sentence. Flag learners who have all cells blank — seat them next to a peer who wrote something for the pair-derivation in Phase 2.</p>                                                                                                                                                                                                                     |
| <p><b>Observe Phase</b></p> <p><b>VIDEO:</b><br/> <a href="#">The trig functions &amp; right triangle trig ratios</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>EXERCISE:</b><br/> <a href="#">Trig values of special angles</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p> | <p>Groups of 4 receive two pre-cut card triangles: (A) an equilateral triangle (side = 2 cm, labelled) and (B) a right isosceles triangle (legs = 1 cm each). Step-by-step task card instructs them to: (1) Fold Card A along its altitude to produce two 30-60-90 right triangles — measure/label all three sides using Pythagoras (<math>\frac{1}{2}</math> base = 1, hypotenuse = 2, altitude = <math>\sqrt{3}</math>). (2) Use Card B (45-45-90) — label hypotenuse = <math>\sqrt{2}</math>. (3) Complete a group special-angles table by computing <math>\sin</math>, <math>\cos</math>, <math>\tan</math> for <math>30^\circ</math>, <math>45^\circ</math>, <math>60^\circ</math> from these side lengths. (4) Use the pattern of the table to deduce <math>\sin 0^\circ</math>, <math>\cos 0^\circ</math>, <math>\sin 90^\circ</math>, <math>\cos 90^\circ</math>, and <math>\tan 90^\circ</math> (undefined). (5) Check: does <math>\tan \theta = \sin \theta \div \cos \theta</math> hold for every special angle in their table? Record 'YES/NO + working'</p> | <p>Distribute card triangles and task cards. Say: 'This is a geometry detective task. Your card triangles hold the secret of ALL special angles. Work through the steps in order — do NOT skip to the calculator yet.' Allow 18 minutes for Steps 1–5. Circulate; at the 8-minute mark cold-call one group: 'Kamau, what did your group get for the altitude of the equilateral triangle? How did you get <math>\sqrt{3}</math>?' WAIT 12 seconds. If groups stall on <math>\sqrt{3}</math>, prompt: 'What theorem connects sides of a right triangle? Use it on the 30-60-90 triangle.' At the 14-minute mark say: 'Now check your <math>\tan</math> column — does <math>\tan = \sin \div \cos</math>? Show the working for <math>60^\circ</math> right now.' WAIT 10 seconds, then ask one group to read their check aloud. After groups</p>                      | <p>Manipulative-Based Derivation (Concrete → Abstract). Physical folding of the equilateral triangle makes the origin of the 30-60-90 ratios visible and memorable — learners SEE why <math>\sin 30^\circ = 1/2</math> rather than memorising a fact. The predict–verify loop (prediction table → derivation → calculator confirmation → self-marking) creates metacognitive awareness of what was intuited correctly versus what needed formal derivation. Connecting <math>\tan \theta = \sin \theta / \cos \theta</math> algebraically confirms the identity discovered numerically in Lesson 7.</p> | <p>Collect ONE group special-angles table per group (rotate which group each lesson). Observable indicator: all 15 cells (5 angles <math>\times</math> 3 ratios) are correctly filled with exact values (fractions / surds), AND the <math>\tan = \sin/\cos</math> check is shown with working for at least 2 angles. Groups with errors in the <math>\sqrt{3}</math> row receive targeted correction before Phase 3.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>for each angle. Groups then open their calculators ONLY to verify: e.g., does <math>\sin 30^\circ</math> on calculator = 0.5? They record: 'Calculator confirms: <math>\sin 30^\circ = 0.5</math> ✓'. Finally, each learner flips over their prediction table from Phase 1 and marks each cell ✓ (correct) or ✗ (incorrect).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>complete Step 5, bring class together: 'Turn over your prediction table. Mark each cell. Hands up — who got <math>\sin 45^\circ</math> exactly right in Phase 1?' Count hands aloud. 'Interesting. Now we know the exact values. Write them in your notes in a boxed table — you will use this for the rest of secondary school.'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>Explain Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Solving for a side in right triangles with trigonometry</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Right triangle trigonometry word problems</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Part A — Eldoret Builder Problem (Guided): The teacher projects or draws on the board: 'A builder in Eldoret is constructing a roof pitched at <math>60^\circ</math>. The horizontal span of the roof is 5 m (so the half-span, from wall to ridge, is 2.5 m). Calculate the length of one roof rafter (the sloping side) using exact values only.' Learners identify the right triangle (half-span = adjacent = 2.5 m, angle = <math>60^\circ</math>, rafter = hypotenuse). They write: <math>\cos 60^\circ = \text{adjacent} / \text{hypotenuse} \rightarrow \frac{1}{2} = 2.5 / \text{rafter} \rightarrow \text{rafter} = 5</math> m. They also calculate the ridge height using <math>\tan 60^\circ = \text{height} / 2.5 \rightarrow \sqrt{3} = \text{height} / 2.5 \rightarrow \text{height} = 2.5\sqrt{3}</math> m <math>\approx 4.33</math> m. Part B — Paired Reading: Each pair reads a short 200-word article (printed on the task card) titled 'Special Angles in Kenyan Construction and Land Survey' which describes how Jua Kali artisans in Gikomba Market, Nairobi, use <math>45^\circ</math> and <math>60^\circ</math> angles in metalwork without calculators, and how surveyors in the Rift Valley use 30-60-90 geometry to mark land boundaries. After reading, pairs answer 3 questions: (1) Why are special angles useful to an artisan without a calculator? (2) A Gikomba welder makes a triangular brace at <math>30^\circ</math> to a 4 m horizontal beam — how long is the brace (hypotenuse)? (3)</p> | <p>For Part A, say: 'No calculators. This is Eldoret — the builder has only a tape measure and this table you just derived. Sketch the right triangle first, label it, THEN choose your ratio.' Allow 6 minutes. Cold-call one learner to draw the triangle on the board: 'Wanjiku, come and label the sides for us.' WAIT 10 seconds. After she labels: 'Class, do you agree with how Wanjiku has identified opposite, adjacent, hypotenuse? Show thumbs up / thumbs down.' Correct any mislabelling before proceeding. Walk through the <math>\cos 60^\circ</math> step together, then release groups for <math>\tan 60^\circ</math> step independently (3 minutes). For Part B, say: 'Read carefully — look for the connection between what the artisan does and what you just derived.' After 8 minutes of paired reading and questions, cold-call 2 pairs for Part B Q3: 'Omondi, what tree height did you get? Show your working on the board.' WAIT 12 seconds. Confirm: <math>\tan 60^\circ = h/9.0 \rightarrow \sqrt{3} = h/9.0 \rightarrow h = 9\sqrt{3} \approx 15.6</math> m. Say: 'Notice — same method as in our Kisumu phenomenon, but now exact. The angle and the shadow length are all we need.'</p> | <p>Context Transfer + Worked Example Analysis. Anchoring the exact-values skill in the Eldoret builder scenario and the Gikomba artisan reading makes the mathematics purposeful. The phenomenon reconnect question (Part B Q3) is a deliberate transfer task: learners must carry the SOH CAH TOA structure from Lessons 1–7, substitute a special angle, and produce an exact answer — demonstrating that their understanding is not context-bound. This is the NGSS Science &amp; Engineering Practice of 'Using Mathematics and Computational Thinking' applied to engineering design contexts.</p> | <p>Teacher collects paired answer sheets for Part B Q2 and Q3. Observable indicators: (1) Correct right-triangle sketch with labelled sides for Q2; (2) Working shows <math>\cos 30^\circ</math> or <math>\sin 60^\circ = \frac{1}{2}</math> used correctly to give hypotenuse = 8 m for Q2 (note: hypotenuse = <math>4/\cos 30^\circ = 4/(\sqrt{3}/2) = 8/\sqrt{3} = 8\sqrt{3}/3</math> m, OR if using <math>\sin</math>: <math>\sin 30^\circ = \text{opp}/\text{hyp}</math> so <math>\text{hyp} = 4/\sin 60^\circ</math> — accept both approaches with correct working); (3) Q3 answer of <math>9\sqrt{3}</math> m with <math>\tan 60^\circ = \sqrt{3}</math> substituted explicitly. Flag pairs with incorrect triangle orientation for Phase 4 support.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Reconnect to the phenomenon: if the Kisumu mango-tree clinometer had read exactly $60^\circ$ instead of $53^\circ$ , and the shadow was still 9.0 m, what would the tree height be using an exact value?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                            |
| <b>Driving Question Board (DQB) Creation</b><br> <b>VIDEO:</b><br><a href="#">Solving for a side in right triangles with trigonometry</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Right triangle trigonometry word problems</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | Each group reviews the class Driving Question Board displayed at the front. The DQB has two columns: 'What we have figured out' and 'What we still wonder about.' Groups receive two sticky notes each — one green (new evidence) and one orange (new wonder/question). On the GREEN note they write one specific claim from today's lesson, citing evidence: e.g., 'We figured out that $\tan 60^\circ = \sqrt{3}$ exactly, so a builder can calculate a rafter length with only arithmetic.' On the ORANGE note they write one unresolved question, e.g., 'What happens when the angle is NOT a special angle and we have no calculator — is there another method?' Groups post their notes on the DQB. The teacher reads 3–4 notes aloud and facilitates a 3-minute whole-class discussion connecting new evidence to the anchoring phenomenon: 'How does knowing exact values for special angles sharpen our answer to the driving question about the Kisumu mango tree?' | Point to the DQB and say: 'In Lessons 1 through 7 we gathered evidence that a consistent angle–ratio relationship exists and that SOH CAH TOA lets us calculate heights. Today we added something new — exact values. What should go on our board?' Allow 4 minutes for sticky-note writing. As groups post notes, read one green note aloud: 'This group says _____. Class, do you agree this is evidence that helps answer the driving question? Why?' WAIT 12 seconds. Cold-call one student who has not yet spoken this lesson: 'Njeri, where would you say we are on answering: how can an angle tell us the height of a tree?' WAIT 15 seconds. Summarise: 'We now have a complete toolkit for special angles. The driving question is nearly answered — but Lesson 9 will test whether this works beyond special angles, at any angle, for any tree in Kenya.' | Collaborative Sense-Making via Driving Question Board (Argument from Evidence). The DQB is a public, cumulative record of the class's epistemic progress. Green/orange sticky notes structure the difference between settled evidence and productive uncertainty — a direct parallel to the NGSS practice of 'Engaging in Argument from Evidence.' Requiring a cited evidence statement on the green note prevents vague claims and builds the habit of evidence-backed reasoning. | Teacher photographs the DQB after notes are posted. Observable indicator: each group's green note contains (a) a specific trigonometric claim (names a ratio and angle) AND (b) a real-world application reference. Orange notes that reveal persistent misconceptions (e.g., 'Is sin always smaller than cos?') are flagged for Lesson 9 opening address. |
| <b>Model Building Phase</b><br> <b>VIDEO:</b><br><a href="#">Solving for a side in right triangles with trigonometry</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a>                                                                                                                                                                                                                                                                                                                                                                                                | Each group retrieves their cumulative annotated right-triangle diagram begun in Lesson 1 and progressively revised in Lessons 2–7. Today's revision task (on a printed 'Model Update Card 8') asks them to: (1) Add a special-angles mini-table inside or beside the triangle, showing exact sin, cos, tan values for $30^\circ$ , $45^\circ$ , $60^\circ$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Distribute Model Update Card 8. Say: 'Your model has grown across eight lessons. Today it gains its most powerful upgrade — exact values and the tan identity. Take 12 minutes to complete all four update steps.' Circulate; at 6 minutes prompt any group still on Step 1: 'Move to Step 2 — you can refine the table later.' At 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Progressive Model Revision (Cumulative Diagrammatic Modelling). The annotated right-triangle diagram is the class's shared explanatory model — equivalent to a scientific model that is iteratively refined as new evidence is gathered. Adding the special-angles table and $\tan = \sin/\cos$ identity in Lesson 8 makes                                                                                                                                                         | Teacher collects one model per group at the end of the lesson (rotate groups each lesson). Observable indicators: (1) Special-angles mini-table present with all 9 values correct in surd/fraction form; (2) $\tan = \sin/\cos$ identity labelled with directional arrow on the triangle; (3) Both Kenyan context annotations (Eldoret rafter              |

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| <p><a href="#">for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Right triangle trigonometry word problems</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>using <math>\sqrt{\phantom{x}}</math> notation. (2) Add a label showing the identity <math>\tan \theta = \sin \theta / \cos \theta</math> with an arrow connecting the three sides of the triangle. (3) Annotate two real contexts directly on the diagram: (a) the Eldoret builder's <math>60^\circ</math> roof with rafter = 5 m labelled, and (b) the Kisumu mango tree with shadow = 9.0 m and height = <math>9\sqrt{3}</math> m labelled. (4) Write one 'Model Claim' sentence at the bottom: 'Our model now shows that for special angles, the exact ratios derived from equilateral and isosceles right triangles allow us to calculate real heights with no calculator.' Groups share their revised model with one adjacent group for a 2-minute peer critique: 'Is the special-angles table correct? Is <math>\tan = \sin/\cos</math> labelled? Are both Kenyan contexts annotated?' Peer group initials the model if satisfied.</p> | <p>minutes: 'Begin peer critique now.' During peer critique, stop one group sharing: 'Mwangi, what did your peer group say was missing from your model?' WAIT 10 seconds. After peer critique: 'Raise your hand if your model was initialled — good. If not, you have 2 minutes to add the missing element.' Close by saying: 'Your model is now a working reference tool. In Lesson 9 we will test it against angles that are NOT special — does it still hold? Bring this model to every remaining lesson.'</p> | <p>explicit the algebraic structure underlying the numerical patterns learners observed in earlier lessons. Annotating Kenyan contexts directly on the model reinforces that the model is not abstract but explanatory of real phenomena. This enacts the NGSS practice of 'Developing and Using Models.'</p> | <p>= 5 m; Kisumu tree = <math>9\sqrt{3}</math> m) present; (4) Model Claim sentence is specific and references exact values. Models missing 2 or more elements are returned with a targeted correction prompt for the start of Lesson 9.</p> |
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## D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) During the card-triangle derivation, did all groups successfully fold the equilateral triangle and apply Pythagoras to find  $\sqrt{3}$ ? If not, was the difficulty conceptual (Pythagoras) or procedural (the folding)? Plan a 5-minute Pythagoras review opener for Lesson 9 if needed. (2) In the Explain phase, how many learners correctly identified  $\cos 60^\circ$  (not  $\sin 60^\circ$ ) as the ratio linking the rafter to the half-span? Mislabelling opposite/adjacent is a persistent error — note names of learners who made this error for targeted questioning in Lesson 9. (3) Did the DQB orange notes reveal any misconceptions about the scope of special angles (e.g., learners thinking ALL angles have exact values)? If so, use this as the Lesson 9 opening hook. (4) Were the peer critique conversations during model building substantive — did peers actually check the  $\sqrt{\phantom{x}}$  notation, or did they initial without scrutiny? Consider modelling one peer critique exchange explicitly at the start of the next model-building phase. (5) Overall: Is the class ready to extend to non-special angles using tables and interpolation in Lesson 9, or does the  $\tan = \sin/\cos$  identity need another application cycle?

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                            |                                                                                                                                                                                                                                                                                                                                         |
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| <b>What did I observe?</b> | Learners folded equilateral and isosceles right-triangle card cutouts, measured and labelled all sides using Pythagoras, and read off exact sin, cos, and tan values for $30^\circ$ , $45^\circ$ , and $60^\circ$ . They verified these values against calculator outputs and self-marked their Phase 1 prediction tables.              |
| <b>What did I learn?</b>   | For special angles ( $0^\circ$ , $30^\circ$ , $45^\circ$ , $60^\circ$ , $90^\circ$ ), the trigonometric ratios are exact values derivable from the geometry of equilateral and isosceles right triangles — no calculator is required. The identity $\tan \theta = \sin \theta / \cos \theta$ holds for every special angle. These exact |

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|                                              | values can be applied directly to real problems such as calculating roof-rafter lengths for a 60°-pitched roof in Eldoret.                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>How does this explain the phenomenon?</b> | Using exact special-angle values and the relationship $\tan \theta = \sin \theta / \cos \theta$ , learners can calculate unknown side lengths in any right-angled triangle whose acute angle is a special angle — including the height of the Kisumu mango tree if the sunbeam angle were exactly 60°. This advances the driving question by showing that the angle–ratio relationship is not just consistent and calculator-dependent, but for special angles it is perfectly exact and derivable from pure geometry. |

## LESSON 9 (40 minutes): Angles of Elevation and Depression: Reading Heights from the Sky

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                           |
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| <b>Purpose</b>              | To enable learners to use angles of elevation and depression together with trigonometric ratios to calculate heights and distances in real Kenyan contexts without direct measurement                                                                                                                                                |
| <b>Knowledge</b>            | Learners will define angle of elevation and angle of depression, distinguish them from each other, and state that both are measured from the horizontal line of sight to the line of sight toward the object                                                                                                                         |
| <b>Skills</b>               | Learners will (1) draw and label diagrams showing angles of elevation and depression, (2) use a clinometer to measure an angle of elevation outdoors, (3) apply the tangent ratio to calculate an unknown height or distance from a measured baseline and angle, and (4) compare results across groups and evaluate sources of error |
| <b>Attitudes</b>            | Learners demonstrate accuracy and intellectual honesty by recording clinometer readings honestly and discussing why group results differ rather than copying a neighbour                                                                                                                                                             |
| <b>Key Inquiry Question</b> | How do we apply trigonometric ratios in day-to-day activities?                                                                                                                                                                                                                                                                       |
| <b>Purpose in Storyline</b> | Lesson 9 converts the abstract sine-cosine-tangent relationships built in Lessons 1 to 8 into a direct measurement tool: students physically measure an angle of elevation and use tangent to compute a height, bringing the driving question almost fully into view before the final consolidation in Lesson 10                     |
| <b>Safety Notes</b>         | During the outdoor activity learners must not point the straw of the clinometer toward the sun. Groups must remain within the school compound boundary. If the compound is unavailable the indoor tall-wall alternative is used.                                                                                                     |

### B. LESSON OVERVIEW

Learners open by revisiting the anchoring phenomenon: the student in Kisumu used a 53 degree sun angle and a shadow length to find the tree height using tangent. Today they ask: what if there is no shadow — can we still find the height? The lesson introduces the airport glide-path as a supporting phenomenon, establishes the formal definitions of angle of elevation and depression, and culminates in an outdoor clinometer activity at the school flagpole or water tank. Learners measure, calculate, compare group results, and update their cumulative model and the Driving Question Board with the new evidence that any height can be found from an angle and a baseline distance.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                | Learner Experience                                                                                                                                                                              | Teacher Moves                                                                                                                                                                                  | Sensemaking Strategy                                                                                                                                                                      | Formative Assessment Strategy                                                                                                                                                                  |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Solving for a side in right triangles with trigonometry</a><br>Source: Khan | Teacher projects or sketches two images side by side on the board: (A) a passenger looking UP at a Kenya Airways plane approaching Wilson Airport, and (B) the pilot looking DOWN at the runway | Teacher opens with: Recall the mango tree in Kisumu. We knew the shadow length and the sun angle, so we used tangent to get the height. Now imagine the sun is behind a cloud — no shadow. How | Eliciting prior knowledge through sketch-and-predict activates the tangent work from Lesson 4 and 8 and surfaces any confusion between looking up and looking down. Recording predictions | Teacher scans written predictions as they circulate. Look for: Does the learner reference an angle AND a length (baseline)? Do any learners already distinguish between looking up and looking |

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| <p>Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Trigonometry Word Problems: Helicopter Angle of Depression Rescue (PLIX)</a><br/> Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                               | <p>threshold. Teacher asks: In image A, what angle is the passenger seeing? In image B, what angle is the pilot using to judge the descent? Learners individually write a prediction on a half-sheet: (1) Are these two angles the same or different in size? (2) What would you need to know — besides the angle — to calculate how high the plane is above the ground at a given horizontal distance? After 90 seconds of silent writing, pairs compare predictions for 60 seconds. Two or three pairs share with the class. Teacher records key prediction phrases on the Driving Question Board column labelled Lesson 9 Predictions without affirming or correcting them yet. Teacher then asks learners to sketch a simple side-view diagram of the airport scenario showing the plane, the observer on the ground, and the runway, labelling any angle and length they think are relevant.</p> | <p>else could you find the height of something tall? Wait 10 seconds. Take a pair share now. After pairs share, teacher says: Today we meet a new scenario — a plane descending to Wilson Airport in Nairobi. I want you to predict before we study anything. Project or draw the two images. Pose the two written-prediction questions. Circulate silently for 90 seconds — do not affirm or negate. Cold-call three pairs using lolly sticks. Record their exact words on the DQB. Then say: Keep your prediction sheet. By the end of the lesson you will know whether your prediction was right.</p>                                                                                         | <p>publicly on the DQB creates accountability and gives learners a concrete claim to test.</p>                                                                                                                                                                                                                                                                                                                                                                                                            | <p>down? Flag learners who write only the angle or only a length — these need scaffolding in the Observe Phase. Checkpoint question (verbal, to the class): Thumbs up if your diagram has a horizontal line in it. Thumbs sideways if it does not. Use this to identify who already understands that both angles are measured from the horizontal.</p>                                                                                                                                                                                                                                                                                                                  |
| <p><b>Observe Phase</b><br/> <b>VIDEO:</b><br/> <a href="#">Finding reciprocal trig ratios</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Trigonometry Word Problems: Helicopter Angle of Depression Rescue (PLIX)</a><br/> Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p> | <p>PART A — VIDEO AND READING (12 minutes). Learners watch a short video (approximately 4 minutes) showing a pilot on final approach to Wilson Airport, Nairobi. The video narrates: the standard glide-path angle is 3 degrees below horizontal. When the plane is 2 000 m horizontally from the runway threshold, how high is it? After the video, learners read a one-page illustrated excerpt (provided as a printed handout or displayed on screen) that defines: (i) angle of elevation — the angle measured upward from the horizontal to the line of sight toward a higher object, (ii) angle of depression — the angle measured downward from the horizontal to</p>                                                                                                                                                                                                                          | <p>Before the video, teacher says: As you watch, listen for the glide-path angle and the horizontal distance. Sketch what you see — pause and draw if you need to. After the video, distribute the reading. Say: Read sections one and two only — you have 3 minutes. Then I will ask you to define both terms in your own words without looking at the page. After reading, cold-call 4 students (use lolly sticks) to define angle of elevation and angle of depression. For each definition, teacher says: Does anyone want to add to that? Wait 5 seconds after each addition. For Example 1, teacher says: Before I write anything, who can tell me which trigonometric ratio links the</p> | <p>The observe phase moves from real video (Wilson Airport) to formal definition to worked example to immediate application — a gradual release model. Anchoring the definitions in Kenyan contexts (Nakuru water tower, Mombasa beach, Maasai Mara watch-tower, Wilson Airport) keeps meaning local and concrete. The glide-path problem connects directly to the Lesson 9 supporting phenomenon and provides the first numerical evidence that angles of depression solve real navigation problems.</p> | <p>After the four cold-calls on definitions, teacher asks the class: Show me on your mini-whiteboard or notebook — draw the angle of depression from a point. Scan for common error: learners who draw the angle from the vertical rather than the horizontal. Correct immediately with a sketch. After the glide-path pair task, teacher cold-calls two pairs to write working on the board. Check: (1) Is the diagram present and labelled? (2) Is tan used correctly? (3) Is the answer reasonable — about 105 m for a plane 2 km out? If a pair uses sin or cos instead of tan, ask: Which sides do you know? Which ratio uses those two sides? Wait 6 seconds.</p> |

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|  | <p>the line of sight toward a lower object, and (iii) that these two angles are alternate interior angles and therefore equal in size when the observer and the object are at the same base level. The reading includes two labelled diagrams: a person looking up at a water tower in Nakuru (elevation) and a lifeguard at Mombasa beach looking down at a swimmer (depression). PART B — GUIDED WORKED EXAMPLES (8 minutes). Teacher works two examples on the board with full learner participation via cold-call: Example 1 (Elevation) — A student stands 15 m from the base of the school water tank. The angle of elevation to the top of the tank is 62 degrees. Find the height of the tank. Draw, label, write <math>\tan 62^\circ = \frac{\text{height}}{15}</math>, solve: height equals 15 times <math>\tan 62^\circ</math> equals 15 times 1.8807 equals 28.2 m. Example 2 (Depression) — A ranger at the top of a 40 m watch-tower at Maasai Mara spots a vehicle on the flat plain. The angle of depression is 28 degrees. How far is the vehicle from the base of the tower? Draw, label, write <math>\tan 28^\circ = \frac{40}{\text{distance}}</math>, solve: distance equals 40 divided by <math>\tan 28^\circ</math> equals 40 divided by 0.5317 equals 75.2 m. Learners copy diagrams and working into their exercise books. PART C — AIRPORT GLIDE-PATH PROBLEM (5 minutes, pairs). Each pair solves: The Wilson Airport glide-path angle is 3 degrees. When the plane is 2 000 m from the runway threshold horizontally, calculate the height of the plane above the ground. They use <math>\tan 3^\circ = \frac{\text{height}}{2000}</math></p> | <p>opposite side and the adjacent side? Wait 8 seconds. Cold-call. After the ratio is named, teacher says: Good — we are going to use tangent today because we always know the horizontal distance and want the vertical height — or vice versa. Pose the glide-path pair task and say: You have 5 minutes. Show every step. I will cold-call two pairs to present on the board.</p> |  |  |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | from tables or Lesson 5 calculator skills. Expected answer: height equals 2 000 times 0.0524 equals 104.8 m.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Solving for a side in right triangles with trigonometry</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Trigonometry Word Problems: Helicopter Angle of Depression Rescue (PLIX)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p><b>OUTDOOR CLINOMETER ACTIVITY</b> (approximately 12 minutes total, including transition). Groups of four receive a pre-assembled clinometer (a protractor taped to a 30 cm cardboard strip, with a straw taped along the straight edge and a plumb-line — a piece of string with a small stone — hanging from the centre hole of the protractor). Each group also receives a 20 m measuring tape and a data-recording sheet. Procedure: (1) One member holds the clinometer and looks through the straw at the top of the flagpole or water tank. (2) A second member reads the angle between the plumb-line and the 90-degree mark on the protractor — this is the angle of elevation. (3) A third member measures the horizontal distance from the observer to the base of the object. (4) All four members record the data independently on their sheets. The group calculates the height: height equals distance times <math>\tan(\text{angle of elevation})</math>. Groups are told to take two readings from two different distances and calculate the height both times to check consistency. Back in class, each group writes its two height estimates on the board. If the outdoor activity is not feasible due to rain or space, an indoor alternative is used: teacher has pre-measured the height of one wall or a mark near the ceiling. Groups use the clinometer from a measured distance inside the room and compare their calculated height to the actual measured</p> | <p>Before going outside, teacher says: You are now going to do exactly what the student in Kisumu did — but instead of a shadow angle, you will measure the angle of elevation directly with a clinometer. The clinometer is your protractor turned into a field instrument. Demonstrate the clinometer technique for 60 seconds indoors: Hold it flat, look along the straw, let the plumb-line hang freely, read the angle. Then say: Two rules: do not point the straw at the sun. Stay within the school boundary. You have 8 minutes outdoors. When you return, every group writes both height estimates on the board. Transition takes 1 to 2 minutes. While outdoors, circulate to each group. Ask: What angle are you reading? Which side is the opposite? Which is the adjacent? Back indoors, facilitating the error discussion, teacher says: Look at the values on the board. They range from X to Y. Should they all be the same? Wait 5 seconds. What could cause the spread? After learners suggest reasons, teacher says: This is what scientists and engineers call measurement uncertainty — it is not failure, it is honesty about the limits of our tools. How might a professional surveyor reduce this error? Wait 8 seconds. Cold-call two students.</p> | <p>The outdoor clinometer activity is the kinesthetic peak of the entire ten-lesson sequence — learners physically enact the very process that the Kisumu student used conceptually in Lesson 1. Comparing group results and discussing error turns the activity into genuine scientific inquiry rather than a recipe exercise. The eye-height refinement introduces the idea of systematic error correction, deepening understanding of the measurement-calculation chain.</p> | <p>Teacher checks each group data-recording sheet during the outdoor phase. Confirm: angle is recorded in degrees, distance in metres, calculation shows <math>\tan</math> ratio set up correctly, units are stated. Back indoors, assess the board values: if one group is an outlier (e.g., all others get 8 to 10 m for a flagpole and one group gets 20 m), invite that group to re-examine their diagram — did they read the complementary angle by mistake? Exit check: each learner writes one sentence stating one source of error in their clinometer reading and how it could be minimised.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>value. After results are on the board, teacher facilitates a whole-class discussion: Why do the groups get slightly different answers? What are the sources of error? Learners suggest: parallax in reading the clinometer, measuring tape not held horizontally, not accounting for the observer eye height above the ground. Teacher notes that the observer eye height (approximately 1.5 m) should be added to the calculated height for a more accurate answer — this is a refinement moment.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Solving for a side in right triangles with trigonometry</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Trigonometry Word Problems: Helicopter Angle of Depression Rescue (PLIX)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Each group receives two sticky notes of different colours. On the first (green), they write new evidence added to our answer of the driving question: We can find the height of any object — tree, tank, flagpole, or aircraft — using the angle of elevation from a known horizontal distance and the tangent ratio. On the second (orange), they write one remaining question or limitation: e.g., What if the ground is not flat? What if we cannot measure the baseline directly? What if we are on a hill? Groups post their sticky notes on the DQB in the Lesson 9 column. Teacher reads three green notes aloud and asks the class: Compare this new evidence with what we added in Lesson 4 when we first used tangent on the mango tree. What is new today? Expected responses: today we measured the angle ourselves rather than being given it; today we used the angle of elevation rather than the sun angle; today we also met the angle of depression. Teacher then reads two orange notes and says: These</p> | <p>After groups post their sticky notes, teacher says: Let us look at how far we have come. Point to the DQB Lesson 1 column. In Lesson 1 you predicted an angle could reveal a tree height. In Lesson 4 you used tangent on the mango tree for the first time. In Lesson 9 you actually went outside, measured an angle yourself, and computed a height that you can verify by measuring the real flagpole. How confident are you now that the driving question is answered? Show me with fingers — 1 is not at all, 5 is completely answered. Scan the room. Most should show 4. Say: One lesson remains because real problems in Kenya — surveying, construction, aviation — involve complications. Your orange sticky notes just identified some of them. Well done.</p> | <p>The two-colour sticky-note protocol separates evidence from remaining questions, teaching learners to distinguish what is known from what is still open — a core scientific literacy practice. Reading selected notes aloud and connecting them to earlier DQB entries makes the cumulative knowledge-building arc visible to every learner.</p> | <p>Teacher photographs the DQB after sticky notes are posted. At a glance, check: Do the green notes contain the key idea (tangent, angle of elevation, baseline)? Do the orange notes show genuine conceptual reach (non-flat ground, inaccessible baselines) rather than trivial questions? If green notes are vague (e.g., we learned about angles), prompt the group: Can you name the specific ratio and the specific measurement you need?</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | are exactly the kinds of questions professional surveyors in Kenya ask. We will begin to address them in Lesson 10.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Solving for a side in right triangles with trigonometry</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Trigonometry Word Problems: Helicopter Angle of Depression Rescue (PLIX)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners open their cumulative annotated right-angled triangle diagram that has been built across Lessons 4, 7, and 8. They add two new labelled elements: (1) An eye symbol at one vertex with an arrow going up to the opposite vertex labelled Angle of Elevation and a second arrow going down labelled Angle of Depression, with a note that both are measured from the horizontal. (2) The generalised formula: height equals baseline times <math>\tan(\text{angle of elevation})</math> or horizontal distance equals height divided by <math>\tan(\text{angle of depression})</math>. Below the diagram, each learner writes a two-sentence connection back to the anchoring phenomenon: In the Kisumu mango-tree scenario, the sun angle of 53 degrees acted as an angle of elevation from the tip of the shadow to the top of the tree. Today I used the same relationship — tangent of the elevation angle times the baseline equals the height — with a clinometer instead of a shadow. One learner from each group shares their two sentences with the class. Teacher collects or photographs the updated models for formative records.</p> | <p>Teacher says: Take out your cumulative model. You are going to add today's learning in the same way you added the special-angle values in Lesson 8 — as permanent features of the diagram, not on a separate page. Give learners 4 minutes to annotate. Circulate and look for three things: Is the horizontal line drawn at the observer point? Is elevation labelled as upward from horizontal and depression as downward from horizontal? Does the formula use <math>\tan</math> (not <math>\sin</math> or <math>\cos</math>)? After 4 minutes, cold-call one learner per group to read their two-sentence phenomenon connection aloud. After each reading, teacher asks the class: Does this statement correctly connect the angle to the tangent ratio and to a real height? Thumbs up or sideways. If thumbs are sideways, ask: What needs to change? Wait 5 seconds before accepting suggestions. Close by saying: In Lesson 10 we bring all of this together — sine, cosine, tangent, special angles, elevation, and depression — across three Kenyan contexts. Bring your full cumulative model.</p> | <p>Adding to the same cumulative diagram across multiple lessons externalises the conceptual structure of the unit and forces learners to see each lesson as building on rather than replacing previous knowledge. The two-sentence phenomenon connection is a micro-explanation task that assesses whether the learner can link formal mathematics to the real-world context — the highest level of the intended learning.</p> | <p>Collect or photograph the updated cumulative models. Evaluate against three criteria: (1) Angle of elevation and depression both labelled correctly relative to the horizontal — not the vertical. (2) Formula height equals baseline times <math>\tan(\text{angle})</math> present. (3) Two-sentence connection mentions tangent, an angle, and the mango-tree or clinometer context. Use a simple three-point rubric (3 — all criteria met, 2 — two criteria met, 1 — one or zero criteria met) to identify learners who need additional support before Lesson 10.</p> |

## D. TEACHER REFLECTION

After the lesson, consider: Did the outdoor clinometer activity run within the time allocation? If it overran, which phase was shortened and did that affect the model-building quality? Were learners able to articulate the difference between angle of elevation and angle of depression consistently, or is confusion remaining? Did the error-analysis discussion generate genuine reasoning about measurement uncertainty, or did learners simply list errors without explaining their effect on the calculated height? Review the photographed cumulative models: are the diagrams becoming richer and more connected across lessons, or are learners treating each lesson as

isolated notes? Plan for Lesson 10: identify the two or three learners whose models show the weakest phenomenon connections — give them a targeted prompt at the start of Lesson 10 before the integrated problem set begins.

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) |                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| What did I observe?                                          | We observed that when a student looks up at the top of the school flagpole and measures the angle with a clinometer, the horizontal distance to the base and that angle are enough information to calculate the height — without climbing the pole or using a shadow.                                                                                                                                                       |
| What did I learn?                                            | We learned that the angle of elevation is measured upward from the horizontal to the line of sight toward a higher object, the angle of depression is measured downward from the horizontal to the line of sight toward a lower object, and in both cases the tangent ratio connects the angle to the opposite side (height or depth) and the adjacent side (horizontal distance).                                          |
| How does this explain the phenomenon?                        | We explained these observations using the tangent ratio: height equals baseline times $\tan(\text{angle of elevation})$ . This is the same relationship used in Lesson 4 to find the mango-tree height from the sun angle, confirming that the driving question can be answered for any tall object — tree, flagpole, water tank, or aircraft on a glide path — as long as we can measure an angle and a baseline distance. |

## LESSON 10 (80 minutes): Real-Life Applications, DQB Closure & Final Model Revision

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>Purpose</b>              | To consolidate and apply trigonometric ratios of acute angles in integrated Kenyan real-life contexts, close the Driving Question Board, and produce a final annotated model that connects all unit concepts back to the anchoring mango-tree phenomenon.                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Knowledge</b>            | Learners demonstrate that trigonometric ratios (sin, cos, tan and their reciprocals) for acute angles define consistent relationships between angles and side lengths in right-angled triangles using SOH CAH TOA, that special angles ( $0^\circ$ , $30^\circ$ , $45^\circ$ , $60^\circ$ , $90^\circ$ ) yield exact, memorable ratios derived from 30-60-90 and 45-45-90 triangles, and that complementary-angle identities ( $\sin \theta = \cos(90^\circ - \theta)$ ) link all six ratios into a unified framework applicable at any scale.                                                                                                                        |
| <b>Skills</b>               | Learners select appropriate trigonometric ratios to solve multi-step problems involving heights, distances, and angles in surveying, telecommunications, and aviation contexts; interpret clinometer and shadow data; present mathematical reasoning to peers; and annotate a cumulative model diagram showing concept-to-phenomenon links.                                                                                                                                                                                                                                                                                                                           |
| <b>Attitudes</b>            | Learners appreciate the power and elegance of trigonometric ratios as practical tools used by Kenyan surveyors, engineers, and pilots; show confidence in presenting solutions; and value collaborative sense-making as a means of resolving uncertainty.                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Key Inquiry Question</b> | How do we apply trigonometric ratios in day-to-day activities? How are the trigonometric ratios of acute angles related to each other?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Purpose in Storyline</b> | This is the culminating lesson of the unit. Learners bring together every concept — SOH CAH TOA, special angles, reciprocal ratios, complementary relationships, and multi-step problem solving — to tackle three integrated Kenyan problems (a plot boundary in Kisumu, a telecommunications mast in Kericho, and a glide path at Wilson Airport). They then return to the original mango-tree phenomenon to give a complete, evidence-based Final Explanation, close every card on the Driving Question Board, and revise their individual models into a polished, annotated final diagram that shows exactly how trigonometric ratios answer the driving question. |
| <b>Safety Notes</b>         | No physical hazards in this lesson. Ensure learners handle printed problem cards and rulers carefully. If the session moves outdoors for a shadow-measurement extension, remind learners to avoid staring directly at the sun and to stay within the school compound boundary.                                                                                                                                                                                                                                                                                                                                                                                        |

### B. LESSON OVERVIEW

Lesson 10 is the unit's Final Explanation lesson. In the Predict phase learners revisit their very first shadow-and-tree predictions from Lesson 1 and articulate what they now believe — and why. In the Observe phase groups rotate through three rich Kenyan application stations (Kisumu plot survey, Kericho telecom mast, Wilson Airport glide path), gathering evidence by solving and cross-checking multi-step trigonometry problems. In the Explain phase each group presents one station solution and the class co-constructs the complete SOH CAH TOA → special angles → reciprocals → complementary-identities chain that answers the driving question. The DQB phase closes every prediction and question card posted across all ten lessons. Finally, in the Model Building phase each learner submits an individually annotated final model diagram — a right-angled triangle labelled with all six ratios, the original mango-tree data, and arrows connecting each ratio to the Kenyan real-life context in which it was most powerfully used.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| <p><b>Predict Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Finding reciprocal trig ratios</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Each learner retrieves their Lesson 1 prediction card (kept in their science journal or returned by the teacher) on which they wrote how a shadow could reveal a tree's height. Working individually for 3 minutes, they write a 'Then vs Now' comparison: what they predicted in Lesson 1 and what they now know, using precise mathematical language (ratio, opposite, adjacent, hypotenuse, sin, cos, tan, angle of elevation). They share with a shoulder partner, then three pairs are cold-called to read both halves aloud. The anchoring phenomenon is re-displayed: Achieng's 1.6 m height casts a 1.2 m shadow; the mango tree's shadow is 9.0 m; the sun's angle is approximately <math>53^\circ</math>. The class is asked: 'Using everything you now know, could you calculate the mango tree's height in more than one way? Predict how many distinct methods you can use — write the number and name each method.' Learners record predictions silently (2 minutes).</p> | <p>Display the original phenomenon image (shadow diagram from Lesson 1) on the board. Say: 'Before we solve anything new today, I want you to travel back to Lesson 1. Pull out your prediction card.' Distribute cards if collected. Allow 3 minutes of silent individual writing. Then say: 'Turn to your shoulder partner — you have 90 seconds. Partner A shares Lesson 1 prediction; Partner B shares what they now know. Then swap.' After 90 seconds call TIME. Cold-call 3 pairs (choose one high-confidence, one mid-confidence, one who has been quieter across the unit). Use the sentence frame on the board: 'In Lesson 1 I predicted _____. Now I know _____ because _____.' After each pair, ask the class: 'Does anyone want to add or respectfully challenge?' Allow 10 seconds WAIT TIME after each challenge prompt. Then say: 'Now, eyes on the board. The same mango tree, the same shadow, the same angle. How many distinct mathematical methods can you now use to find the tree's height? Write a number and list the method names. You have 2 minutes — go.' Circulate and note the range of numbers students write (expect 2–4 methods: shadow ratio / similar triangles, tan ratio directly, sin with hypotenuse, cos with hypotenuse after finding hypotenuse). After 2 minutes cold-call 4 learners. Record their method counts on the board without evaluating. Say: 'We will return to this question at the end of the lesson to see who was right.'</p> | <p>Strategy: 'Then vs Now' Reflective Prediction. By forcing learners to articulate the gap between their Lesson 1 intuition and their current understanding in precise mathematical language, this strategy activates the full unit's schema, surfaces any residual misconceptions before the final explanation, and creates cognitive readiness for synthesis. Naming multiple solution methods primes learners to see trigonometric ratios as a connected system rather than isolated procedures.</p> | <p>Observable indicator: Each learner's 'Then vs Now' card uses at least two of the following terms correctly — ratio, opposite, adjacent, hypotenuse, sin, cos, tan, angle of elevation, complementary. Teacher scans cards during circulation and flags any learner who cannot write a 'Now' statement with mathematical vocabulary for targeted support during the Observe phase stations.</p> |
| <p><b>Observe Phase</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>Groups of four rotate through three problem stations (12 minutes)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Before releasing groups say: 'Each station is a real Kenyan context.'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Strategy: Rotating Evidence Stations with Kenyan Contexts.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Observable indicators: (1) Group Answer Sheets show a labelled</p>                                                                                                                                                                                                                                                                                                                             |

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| <p> <b>VIDEO:</b><br/> <a href="#">Finding reciprocal trig ratios</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Relate ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>each, 1-minute transition). Each station card is printed on coloured paper and contains a diagram, all given measurements, and three scaffolded sub-questions progressing from identify → calculate → verify.</p> <p>Station A — Kisumu Plot Survey (Green card): A land surveyor in Kisumu needs to find the width of a rectangular plot whose diagonal boundary fence runs at <math>37^\circ</math> to the longer side. The diagonal measures 48 m. Sub-questions: (i) Identify which trigonometric ratio links the <math>37^\circ</math> angle, the diagonal (hypotenuse), and the shorter side (opposite). (ii) Calculate the shorter side to 1 d.p. (iii) Verify using Pythagoras' theorem.</p> <p>Station B — Kericho Telecom Mast (Yellow card): A Safaricom engineer needs to know the height of a new mast in Kericho. She stands 30 m from the base on flat ground and measures the angle of elevation to the top as <math>60^\circ</math>. Sub-questions: (i) Draw and label the right-angled triangle. (ii) Use the exact value of <math>\tan 60^\circ = \sqrt{3}</math> to calculate the mast height exactly, then to 2 d.p. (iii) If a support wire runs from the top of the mast to her feet, calculate the wire length using the exact value of <math>\cos 60^\circ</math>.</p> <p>Station C — Wilson Airport Glide Path (Blue card): Aircraft approaching Wilson Airport in Nairobi descend on a glide path of <math>3^\circ</math>. A pilot is at an altitude of 300 m when she begins her final approach. Sub-questions: (i) Use <math>\sin 3^\circ</math> (tables or calculator) to find</p> | <p>Your job is not just to get a number — it is to show the reasoning chain: identify the ratio, set up the equation, solve, and verify or interpret. All four group members must be able to explain the solution.' Assign groups to starting stations (use pre-assigned colour cards). Set a visible 12-minute timer. At the 11-minute mark say: 'One minute — make sure every member can explain your solution.' Sound a soft signal for transitions. During rotations, circulate and listen for: (a) correct ratio identification, (b) correct substitution, (c) use of exact values at Station B, (d) complementary angle reasoning at Station C. At Station B, probe with: 'Why can we use <math>\sqrt{3}</math> exactly instead of a decimal? What does that tell us about special angles?' Allow 10 seconds WAIT TIME before accepting responses. At Station C, probe: 'The pilot's angle is <math>3^\circ</math> and the co-pilot says <math>87^\circ</math>. Where did <math>87^\circ</math> come from? Convince me with an identity.' After all three rotations, say: 'Before we move to presentations — quickly, without looking at your sheet — can the mango-tree problem be solved the same way as Station A or Station B? Thumbs up for A, sideways for B, down for both, fingers crossed for both.' Take a quick visual poll and say: 'Hold that thought.'</p> | <p>Rotating stations ensure every group encounters all three application domains (surveying, telecommunications, aviation), building the evidence base that trigonometric ratios are universally applicable — not just a classroom procedure. The progression from identify → calculate → verify within each station mirrors the scientific practice of 'Planning and Carrying Out Investigations' (NGSS SEP 3) and 'Using Mathematics' (NGSS SEP 5). Requiring all group members to be presentation-ready instills individual accountability within collaborative structures.</p> | <p>diagram, a correctly written ratio equation (e.g., <math>\tan 60^\circ = h/30</math>), and a numerical answer with units at each station. (2) At Station B, learners write '<math>30\sqrt{3}</math> m' or '<math>51.96</math> m' — either is acceptable but exact form is preferred. (3) At Station C, learners explicitly write '<math>\sin 3^\circ = \cos 87^\circ</math> because <math>3^\circ + 87^\circ = 90^\circ</math>' — any group that writes only a numerical answer without the identity explanation is flagged for the Explain phase discussion. Teacher records which groups show the complementary-identity reasoning unprompted.</p> |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>the slant distance the aircraft travels from that altitude to touchdown. (ii) Use <math>\cos 3^\circ</math> to find the horizontal ground distance. (iii) The co-pilot says the angle of elevation from the threshold of the runway to the aircraft is <math>87^\circ</math>. Explain why this makes sense using the complementary angle identity <math>\sin \theta = \cos(90^\circ - \theta)</math>.</p> <p>All groups record solutions on a shared Group Answer Sheet. Groups are told they must be ready to present any one station — the teacher will randomly assign at the start of the Explain phase.</p>                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Finding reciprocal trig ratios</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Each group presents one assigned station solution (5 minutes per group, 15 minutes total). Presenters write their triangle diagram and calculation on the board; the class listens and completes a 'Gallery Critique' slip — one thing correct, one question or extension. After all three presentations, the teacher facilitates a whole-class synthesis: learners co-construct a 'Trigonometry Concept Map' on the board that shows SOH CAH TOA at the centre, branching to special angles (<math>30^\circ</math>, <math>45^\circ</math>, <math>60^\circ</math>), reciprocal ratios (cosec, sec, cot), complementary identities, and arrows pointing to each of the three Kenyan stations. A final arrow points back to the mango-tree diagram on which learners demonstrate all applicable methods. The class then formally answers the driving question in writing: 'The angle a sunbeam makes with the ground allows us to calculate height because...' (minimum three sentences, must</p> | <p>Randomly draw station assignments from a hat. Say: 'Group ____, you are presenting Station A — Kisumu. You have 30 seconds to choose your presenter and your diagram-drawer. Go.' While groups prepare, distribute Gallery Critique slips. During each presentation, stand to the side and listen. After each presentation, ask the class: 'On your slip — write one thing they did correctly and one question or extension you have. You have 60 seconds.' Cold-call two learners per presentation to share their slips (target different learners for each presentation — aim for at least 8 different voices across the three presentations). Respond to each slip with: 'Can the presenting group answer that question?' Allow 10 seconds WAIT TIME. After all three presentations, say: 'Now we build the map together. I'll draw the trunk — SOH CAH TOA. Who can give me the first branch? What concept connects directly to SOH CAH TOA?' Wait 10 seconds, then cold-</p> | <p>Strategy: Gallery Critique + Collaborative Concept Mapping. Gallery Critique (one correct, one question) trains learners to evaluate mathematical reasoning rather than just absorb it, embodying 'Engaging in Argument from Evidence' (NGSS SEP 7). The co-constructed concept map externalises the unit's conceptual architecture, making the connections between ratio types, special angles, and real-life applications visible and permanent — directly building the sensemaking capacity needed for the final model. Writing the driving-question answer forces every learner to synthesise individually before the model phase.</p> | <p>Observable indicators: (1) Each learner's Gallery Critique slip names a specific mathematical step (not 'they drew a triangle' but 'they correctly identified <math>\tan = \text{opposite/adjacent}</math>'). (2) The written driving-question answer contains all three required terms (ratio, similar triangles, angle of elevation) in a logically connected argument. (3) Volunteers drawing arrows on the mango-tree diagram can justify each arrow verbally — teacher asks 'Why this ratio for this triangle?' and accepts answers that name the given sides.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | include ratio, similar triangles, and angle of elevation).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | call. Build the concept map collaboratively, asking 'Where does this branch connect?' for each new concept. When the map is complete, point to the mango-tree diagram: 'How many of these branches can we use to find the tree's height? Come and draw an arrow for each one.' Invite 4 volunteers. Finally, say: 'In your journals, write the answer to our driving question. You have 4 minutes. Use ratio, similar triangles, and angle of elevation — all three terms must appear.' After 4 minutes cold-call 3 learners to read aloud. Post exemplary responses on the board.                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Finding reciprocal trig ratios</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>The full Driving Question Board — accumulated across all 10 lessons — is displayed. Cards are organised into four colour-coded columns: Predictions (Lesson 1), Questions (Lessons 1–9), Evidence (Lessons 2–9), and Claims (ongoing). Working in pairs, learners receive a printed DQB Summary Sheet listing every card's text. They spend 8 minutes marking each card with one of three stamps: ✓ Fully Answered, ~ Partially Answered, ? Still Uncertain. Pairs then share their stamps with another pair to form a group of four and reach consensus on any disagreements. The teacher leads a whole-class DQB review: each original Lesson 1 prediction card is read aloud, the class votes on its status, and the teacher moves it to a 'Resolved' column. Key unresolved '?' cards (if any) are acknowledged as productive open questions — extensions beyond Grade 10 scope (e.g., 'What happens with angles greater than 90°?' —</p> | <p>Uncover the full DQB (all cards visible). Say: 'This board tells the story of our learning across 10 lessons. Every question you ever asked, every prediction you ever made — it is all here.' Give pairs the DQB Summary Sheet. Say: 'In 8 minutes, stamp every card. Be honest — if something is still unclear to you, mark it with a question mark. That is not failure; that is intellectual honesty.' After 8 minutes, say: 'Find a second pair and compare your stamps. If you disagree, you have 2 minutes to convince each other using evidence from your journals.' After 2 minutes, begin whole-class review. Read each Lesson 1 prediction card aloud. Ask: 'Who predicted this? Raise your hand.' Acknowledge the original predictor by name. Then ask: 'Class — fully answered, partially answered, or still uncertain? Show me fingers: one, two, or three.' Record the majority vote and physically move or annotate the card. For any '?' cards, say: 'This is a beautiful</p> | <p>Strategy: DQB Closure with Consensus Stamping. Returning to the full DQB makes the unit's learning trajectory visible and honours the epistemic practice of tracking claims against evidence over time — mirroring how scientists and engineers revise understanding iteratively (NGSS SEP 6: Constructing Explanations). Consensus stamping in pairs then fours builds the collaborative reasoning skill mandated by the Core Competency of Communication and Collaboration. Acknowledging unresolved questions as productive extensions models intellectual honesty and intrinsic motivation for further inquiry.</p> | <p>Observable indicators: (1) Every pair's DQB Summary Sheet is fully stamped — no blank cards. (2) Disagreements between pairs are resolved by citing a specific lesson's evidence (e.g., 'In Lesson 4 we found that <math>\sin 30^\circ = 0.5</math> exactly, so this prediction is fully answered'). (3) During choral response, the teacher listens for the presence of the words 'ratio', 'right-angled triangle', and 'angle of elevation' — if absent, probes with 'Can someone add the mathematical language?'</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>pointing forward to Trigonometry II). The driving question is read aloud one final time: 'How can the angle a sunbeam makes with the ground allow us to calculate the height of any tree, building, or hill in Kenya without measuring it directly?' The class answers chorally using the sentence frame drafted in the Explain phase.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>question — it belongs to Trigonometry II, Grade 11. Write it in your journal as a question to carry forward.' End by saying: 'Let us answer our driving question together — everyone, on three: one, two, three—' and allow the choral response. Then say: 'Achieng in Kisumu started with a shadow and a protractor. You have just shown that a simple angle connects her to every tree, mast, and runway in Kenya. That is the power of mathematics.'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Finding reciprocal trig ratios</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Trigonometric ratios in right triangles</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Each learner works individually to produce their Final Annotated Model — a revision and completion of the model they have updated across all 10 lessons. The model must include: (1) A large right-angled triangle labelled with <math>\theta</math>, Opposite, Adjacent, Hypotenuse. (2) All six trigonometric ratios written as fractions beside the triangle. (3) A special-angles table (<math>0^\circ</math>, <math>30^\circ</math>, <math>45^\circ</math>, <math>60^\circ</math>, <math>90^\circ</math>) for sin, cos, tan. (4) The complementary identity <math>\sin \theta = \cos(90^\circ - \theta)</math> written and illustrated with one example. (5) The original mango-tree data (Achieng: height 1.6 m, shadow 1.2 m; tree shadow 9.0 m; angle <math>53^\circ</math>) worked through using at least two distinct methods, with answers circled. (6) Three labelled arrows — one pointing to the Kisumu plot, one to the Kericho mast, one to the Wilson Airport glide path — each annotated with the ratio used and why. (7) A 'Final Explanation' box at the bottom answering the driving question in 2–3 sentences. Learners spend 15 minutes on their models. They then do a 'Lesson 1 vs Lesson 10' comparison: they hold their original</p> | <p>Distribute blank A3 paper (or learners use their last journal page). Say: 'This is your final model — the most important piece of mathematics you will produce in this unit. It should be something you could show a friend in Mombasa or a farmer in the Rift Valley and use it to explain how trigonometry works in the real world.' Display the seven required components as a checklist on the board. Say: 'You have 15 minutes. Work in silence — this is individual. I will be circulating and I may ask you to explain any part of your model to me verbally.' Circulate every 2 minutes. When you see a learner struggling with the mango-tree calculation, crouch to their level and ask quietly: 'Which sides do you know? Which ratio connects them?' — do not give the answer. When you see a strong model, ask the learner: 'Can you add one more thing that would make this model useful to someone who has never studied trigonometry?' At the 13-minute mark say: 'Two minutes — make sure your Final Explanation box is complete and your mango-tree answers are circled.' At 15 minutes</p> | <p>Strategy: Cumulative Model Revision with L1 vs L10 Comparison. The final annotated model is the unit's capstone sensemaking artefact — it requires learners to represent, connect, and explain the entire conceptual structure in a single diagram, embodying 'Developing and Using Models' (NGSS SEP 2) at its highest level. The L1 vs L10 sticky-note comparison makes cognitive growth concrete and visible, activating metacognitive reflection. Requiring two distinct methods for the mango-tree calculation confirms conceptual flexibility rather than procedural memorisation.</p> | <p>Observable indicators: (1) The final model contains all seven required components — teacher uses the checklist to score each submission (0–7 points) and returns with written feedback within one lesson. (2) The 'Final Explanation' box uses the words 'ratio', 'right-angled triangle', 'angle', and 'height' in a coherent argument — vague answers ('trigonometry helps us') are flagged for verbal follow-up. (3) The sticky note names three specific mathematical differences (e.g., 'In Lesson 1 I did not know what tan meant; now I know <math>\tan \theta = \text{opposite/adjacent}</math>') — non-specific differences ('I know more now') are returned with the prompt 'Give me the mathematics.' (4) The mango-tree calculation shows at least two correctly labelled methods with matching numerical answers (tree height <math>\approx 12.0</math> m from shadow ratio; <math>\approx 12.0</math> m from <math>\tan 53^\circ \times 9.0</math> m).</p> |

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|  | Lesson 1 model sketch next to the final model and write three differences on a sticky note, which they attach to the final model before submission. | say: 'Pause. Take out your Lesson 1 model sketch. Hold them side by side. On a sticky note, write exactly three differences — be specific and mathematical. Attach it to your final model.' Collect all final models. Close the lesson by asking: 'One word — how do you feel about trigonometry now compared to Lesson 1? Whisper it to yourself, then someone brave, share it with the class.' Accept 4–5 responses. End with: 'The mango tree in Kisumu is still standing. Now you have the tools to measure it — and every tree, mast, and mountain in Kenya — without leaving the ground.' |  |  |
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#### D. TEACHER REFLECTION

After collecting final models, use the 7-component checklist to score each submission before the next lesson and identify the three most common gaps — likely candidates are: (a) incomplete special-angles tables (missing  $0^\circ$  or  $90^\circ$  entries), (b) the complementary identity stated but not illustrated with a numerical example, and (c) 'Final Explanation' boxes that are procedurally correct but do not reference the anchoring phenomenon. Return models with written feedback and a single starred 'growth question' personalised to each learner's gap. Reflect on: Did the DQB closure reveal any questions that the lesson sequence did not adequately address — for example, questions about obtuse or reflex angles? If so, note these as explicit 'bridge questions' to introduce at the start of Trigonometry II. Consider whether the three Kenyan application stations were equally accessible — Station C (Wilson Airport glide path) requires reading comprehension about aviation that may have disadvantaged learners unfamiliar with airports; in a future iteration, offer an alternative third station set at Jomo Kenyatta International Airport with a more scaffolded diagram. Finally, audit gender participation across all 10 lessons: were female learners (like the phenomenon's protagonist Achieng) equally represented in cold-call responses and group presentations? Adjust cold-call lists in future units to ensure equity.

#### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

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| <b>What did I observe?</b> | Students observed that the same trigonometric ratios that describe Achieng's shadow-and-tree relationship in Kisumu also govern the angle of a plot boundary fence in Kisumu (Station A), the height of a Safaricom telecommunications mast in Kericho calculated exactly using $\tan 60^\circ = \sqrt{3}$ (Station B), and the glide-path distance of an aircraft approaching Wilson Airport where the complementary-angle identity $\sin 3^\circ = \cos 87^\circ$ was verified numerically (Station C). They also observed, by holding their Lesson 1 and Lesson 10 models side by side, the measurable growth in their own understanding across the unit.                                                                             |
| <b>What did I learn?</b>   | Students learned that trigonometric ratios ( $\sin$ , $\cos$ , $\tan$ and their reciprocals $\operatorname{cosec}$ , $\sec$ , $\cot$ ) form a unified, interconnected system — not isolated formulas — in which SOH CAH TOA, special-angle exact values, and the complementary identity $\sin \theta = \cos(90^\circ - \theta)$ are all expressions of the same underlying geometric relationship between angles and side-length ratios in right-angled triangles. They learned that this relationship is scale-invariant: it works identically whether the triangle is the small one formed by Achieng's shadow or the large one formed by an aircraft's glide path, because similar triangles preserve ratios. They confirmed that the |

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|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | mango tree's height ( $\approx 12.0$ m) can be found by at least two independent trigonometric methods, validating the consistency of the ratio system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>How does this explain the phenomenon?</b> | Students explained that the angle a sunbeam makes with the ground (the angle of elevation) creates a right-angled triangle whose sides are related by fixed trigonometric ratios regardless of the triangle's size, because all triangles with the same angle are similar and therefore preserve the ratio of any two corresponding sides. Therefore, once you know the angle and one side length (a shadow, a measured distance, or an altitude), the trigonometric ratio — sin, cos, or tan — acts as a conversion factor that scales any known length into the unknown height or distance. This is why Achieng, a land surveyor in Kisumu, a Safaricom engineer in Kericho, and a pilot approaching Wilson Airport all use the same mathematical tool: the consistent, angle-determined ratios encoded in SOH CAH TOA. |

| DIFFERENTIATION AND INCLUSION       |                                                                                       |                                                                                  |
|-------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Learner Need                        | Support Strategies                                                                    | Extension Strategies                                                             |
| English Language Learners           | Provide bilingual glossaries; allow use of first language for initial model drawing.  | Write extended explanations of observations; lead group discussions in English.  |
| Students with learning difficulties | Provide structured note frames; pair with a supportive partner during investigations. | Mentor peers; create additional visual models to explain concepts.               |
| Gifted and advanced learners        | Access to additional reading materials; challenge questions during DQB.               | Lead model-building discussions; research and present extension topics to class. |